

Determining the Structure of Dynamic Factor Models

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Abstract

We propose a simple and computationally efficient *alternating least squares* (ALS) algorithm for estimating dynamic factor models with finite filter length. We demonstrate that the factors and factor loadings are consistent up to an invertible matrix whose dimension equals the number of dynamic factors. Building on this result, we introduce two procedures for selecting the number of dynamic factors and the filter length. We illustrate the practical usefulness of the proposed framework through two empirical applications. First, we estimate impulse responses to monetary policy shocks identified using a large-dimensional system. Second, we construct directional connectedness measures for U.S. state-level and county-level COVID-19 positive cases.

1 Introduction

Factor models are a central tool in empirical macroeconomics and finance for summarizing co-movements in large-dimensional time series. Their appeal lies in the idea that a small number of latent factors can explain the dynamics of many observed variables.

A fundamental distinction is between *static* and *dynamic* factor models. In a static factor model, the observables x_t depend contemporaneously on a low-dimensional vector of factors f_t ,

$$x_t = \lambda f_t + \varepsilon_t,$$

whereas a dynamic factor model allows leads and lags of the factors to affect the observables,

$$x_t = \sum_{k=-m_1}^{m_2} \lambda_k f_{t-k} + \varepsilon_t.$$

The dynamic specification is more general and economically informative, as it captures delayed responses and propagation mechanisms that are central to macroeconomic and financial dynamics.

There are many settings in which dynamic factor structures yield more interpretable models. Early work by Chamberlain and Rothschild [1983] employs a static factor structure for asset returns motivated by arbitrage pricing theory. However, empirical evidence suggests that factor loadings may be dynamic. For example, Carhart [1997] introduces a momentum factor constructed from lagged monthly returns, implying delayed responses of asset prices to factor innovations.

Dynamic factor models are also essential for structural analysis. Forni et al. [2009] exploits the dynamic structure to identify structural shocks, performing a second-step dimension reduction on the residuals of a VAR estimated using static factors. This step

reduces the number of identifying restrictions from $r^2 = (mq)^2$ to q^2 , where r denotes the number of static factors, q the number of dynamic factors, and $m = m_1 + m_2 + 1$ the filter length. If dynamic factors can be estimated directly, this second-step reduction is unnecessary. Estimation and inference can then be conducted in a substantially lower-dimensional space, with autocovariance matrices of dimension q^2 rather than r^2 .

Dynamic factor models also provide a natural framework for modeling second-moment properties of large-dimensional time series. The factor structure imposes strong restrictions on the autocovariance matrices, yielding a manageable number of parameters even in large systems. While Fan et al. [2013] exploits static factor structures to estimate large covariance matrices, estimation of large-dimensional autocovariance matrices has received far less attention. Accurate estimates of these objects can be used for large-dimensional forecasting, impulse response analysis, and estimating network connectedness measures. We provide empirical illustrations of estimating impulse responses and network connectedness measures with large systems in Section 6.

Despite these advantages, most empirical applications rely on static factor models due to their computational and theoretical tractability. Dynamic structures are typically approximated by extracting a large number of static factors, implicitly capturing leads and lags of the underlying dynamic factors. This practice obscures the distinction between static and dynamic components and complicates structural interpretation. Dynamic factors correspond more closely to primitive economic shocks, making their identification—and the choice of filter length—central to structural analysis.

This paper proposes *alternating least squares (ALS)* framework for estimating dynamic factor models. The method iteratively updates factors and loadings through simple least squares steps, making it computationally efficient and scalable to large datasets. We establish consistency under double asymptotics, with the estimator identified up to an invertible transformation of dimension equal to the number of dynamic factors. Building on this framework, we develop procedures for selecting both the number of dynamic factors and the filter length. Together, these contributions provide a unified, practical, and theoretically grounded approach to uncovering the dynamic structure of large-dimensional time series.

1.1 Related Literature

There is a large literature on dynamic factor models, which can be broadly divided into two classes. The first class comprises general dynamic factor models with infinite filter length, which do not admit a finite-dimensional static representation. The second class consists of models with a finite filter length, for which an equivalent static representation exists. Much of the empirical and theoretical work has focused on these static representations rather than on the underlying dynamic factors themselves. Our work falls into the second category: the dynamic factor models with a finite filter length. What distinguishes our work from many existing ones is that we work directly with the dynamic factors themselves, rather than the static representations.

The general dynamic factor model with infinite filter length was introduced by Forni et al. [2000], who proposed estimation based on frequency-domain principal components. Subsequent work has developed one-sided representations, estimation methods, and asymptotic theory for these models: Forni et al. [2015], Forni et al. [2017], and Barigozzi et al. [2024]. Tests for the number of dynamic factors in the general dynamic factor framework have been proposed by Hallin and Liška [2007] and Onatski [2010].

A separate strand of the literature studies the static representations of dynamic factor

models. Bai [2003] and Bai and Li [2016] provide inferential theory for static factors estimated by principal components and quasi-maximum likelihood, respectively. Procedures for determining the number of static factors include information criteria [Bai and Ng, 2002] and eigenvalue-ratio tests [Ahn and Horenstein, 2013], among others. Despite this extensive literature, comparatively little attention has been paid to the direct estimation of dynamic factors. An exception is Bai and Wang [2015], who studies Bayesian estimation of dynamic factor models.

This paper is most closely related to Bai and Ng [2007] and Amengual and Watson [2007], who study tests for the number of dynamic factors under VAR specifications for the static representations with low-rank innovations. In both papers, estimation proceeds in two steps: static factors are first estimated, after which additional regressions are used to recover the number of dynamic factors. In contrast, our approach estimates the dynamic factors directly and uses these estimates to determine both the number of dynamic factors and the filter length. We propose two testing procedures, which can be viewed as dynamic generalizations of the information-criterion and eigenvalue-ratio tests of Bai and Ng [2002] and Ahn and Horenstein [2013]. By operating directly on the dynamic factor representation, our procedures avoid the inefficiencies and loss of information inherent in static approximations.

1.2 Notation

We denote by $\|\cdot\|$ the Euclidean norm when applied to vectors and the corresponding spectral norm when applied to matrices, and by $\|\cdot\|$ the Frobenius norm. The row-major vectorization operator is denoted by $\text{vec}(\cdot)$.

2 Model, Assumptions, and Identification

2.1 Model

We consider the dynamic factor model

$$x_{it} = \sum_{k=-m_1}^{m_2} \lambda'_{ik} f_{t-k} + \varepsilon_{it}, \quad i = 1, \dots, N, \quad t = 1, \dots, T, \quad (1)$$

where $f_t \in \mathbb{R}^q$ is a q -dimensional vector of dynamic factors and $\lambda_{ik} \in \mathbb{R}^q$ are factor loadings. The nonnegative integers m_1 and m_2 denote the number of leads and lags of f_t entering the measurement equation. We define the *filter length* as $m = m_1 + m_2 + 1$. Several familiar models are nested within this framework. When $m_1 = m_2 = 0$ ($m = 1$), the model reduces to a static factor model. When $m_1 = -\infty$ and $m_2 = \infty$, it corresponds to the general dynamic factor model of Forni et al. [2000]. We restrict attention to the finite-filter case $m < \infty$, which can be viewed either as a finite-dimensional specialization of Forni et al. [2000] or as a dynamic generalization of the conventional static factor model of Bai [2003] and Stock and Watson [2002]. The model admits an equivalent static representation,

$$x_{it} = \gamma'_i g_t + \varepsilon_{it}, \quad g_t = \begin{pmatrix} f_{t+m_1} \\ \vdots \\ f_{t-m_2} \end{pmatrix}, \quad \gamma_i = \begin{pmatrix} \lambda_{i,-m_1} \\ \vdots \\ \lambda_{i,m_2} \end{pmatrix}, \quad (2)$$

where $g_t \in \mathbb{R}^{qm}$ collects the relevant leads and lags of the dynamic factors. Define

$$x_t = (x_{1t}, \dots, x_{Nt})', \quad \varepsilon_t = (\varepsilon_{1t}, \dots, \varepsilon_{Nt})',$$

and stack observations over time as

$$X = (x_1, \dots, x_T)', \quad E = (\varepsilon_1, \dots, \varepsilon_T)', \quad G = (g_1, \dots, g_T)'.$$

The model can then be written compactly as

$$x_t = \Gamma g_t + \varepsilon_t, \quad X = G\Gamma' + E, \quad (3)$$

where $\Gamma = (\lambda_{-m_1}, \dots, \lambda_{m_2})$ is the $N \times qm$ matrix of factor loadings. Let $F_{-k} = (f_{1-k}, \dots, f_{T-k})'$ collect the k -th lag of the factors. Then, we may write

$$G = (F_{m_1}, \dots, F_{-m_2}), \quad X = \sum_{k=-m_1}^{m_2} F_{-k} \lambda'_k + E. \quad (4)$$

We let $F = F_0$ for the $T \times q$ matrix collecting $(f_t)_{t=1}^T$.

As shown by Bai and Ng [2007] and Forni et al. [2009], if the dynamic factors (f_t) follow a $\text{VAR}(p)$ process, then the static representation (g_t) follows a $\text{VAR}(\tilde{p})$ process, where $\tilde{p} = 1$ if $m \geq p$ and $\tilde{p} = p - m + 1$ if $m < p$. When $m \geq p$, estimation based on the static representation requires fitting a $\text{VAR}(1)$ in dimension qm , involving $q^2 m^2$ parameters. In contrast, direct estimation of (f_t) requires fitting a $\text{VAR}(p)$ in dimension q , with only pq^2 parameters. Since $m > 1$ in a genuine dynamic factor model, the reduction in dimensionality is substantial. When $m < p$, estimation of the static representation requires $(p - m + 1)q^2 m^2$ parameters, which again strictly exceeds pq^2 whenever $m > 1$. Thus, direct estimation of dynamic factors is always more parsimonious than estimation based on static representations.

2.2 Assumptions

We impose the following two assumptions, which are required to establish the consistency of the estimated factors and factor loadings. We'll introduce two additional assumptions in Section 4, which are needed to establish consistency for tests of the number of dynamic factors and filter length.

Assumption 1. *The factors and factor loadings satisfy*

$$\frac{1}{T} G' G \rightarrow_p M_g > 0, \quad \frac{1}{N} \Gamma' \Gamma \rightarrow_p M_\gamma > 0$$

as $T \rightarrow \infty$ and $N \rightarrow \infty$, where M_g and M_γ are $qm \times qm$ positive definite matrices. Furthermore, let $G^{(m+1)}$ denote the $T \times q(m+1)$ matrix

$$G^{(m+1)} = (F_{m_1} \quad \dots \quad F_{-m_2} \quad F_{-m_2-1}),$$

i.e., G augmented with an additional lag term. Then, $\frac{1}{T} G^{(m+1)'} G^{(m+1)}$ is also positive definite in the limit.

Assumption 2. *The idiosyncratic errors satisfy*

$$\|E\| = O_p(\sqrt{\max(N, T)}).$$

Assumption 1 requires that both the factors and factor loadings are *pervasive*, meaning that their influence remains non-negligible as $T \rightarrow \infty$ and $N \rightarrow \infty$. Specifically, $\frac{1}{T} G' G \rightarrow_p$

$M_g > 0$ ensures that the dynamic factors g_t maintain sufficient temporal variation; no linear combination of factors becomes degenerate. Also, $\frac{1}{N}\Gamma'\Gamma \rightarrow_p M_\gamma > 0$ ensures that each factor affects a non-negligible fraction of the cross-sectional units. Intuitively, this assumption holds as long as the factors are not perfectly predictable from a finite number of lags and the loadings are sufficiently “spread out” across the cross-section. Assumption 2 is standard in the factor models literature, e.g., Bai and Ng [2023].

Example. Let $f_t \in \mathbb{R}^q$ follow a VAR(1) process

$$f_t = Af_{t-1} + u_t, \quad u_t \sim \text{i.i.d. } \mathbb{N}(0, I_q),$$

and let the factor loadings λ_{ik} be independent draws from a distribution with finite second moments and full-rank covariance. Then, for large T and N ,

$$\frac{1}{T}G'G \approx \mathbb{E}[g_t g_t'] > 0, \quad \frac{1}{N}\Gamma'\Gamma \approx \mathbb{E}[\gamma_i \gamma_i'] > 0,$$

so Assumption 1 is satisfied. This setup corresponds to the standard “pervasive factor” framework commonly used in static factor models.

2.3 Identification

Consider the static factor model of dimension r :

$$x_t = \Gamma g_t + \varepsilon_t,$$

where $g_t \in \mathbb{R}^r$ and $\Gamma \in \mathbb{R}^{N \times r}$. Suppose the common component Γg_t is asymptotically identified. In the absence of dynamic structure, the factors and loadings are identified only up to an invertible $r \times r$ matrix H_r , i.e., any transformation

$$g_t \mapsto H_r g_t, \quad \Gamma \mapsto \Gamma H_r^{-1}$$

yields the same common component.

Now, suppose the factors have a dynamic structure such that g_t consists of m blocks of dimension q , corresponding to leads and lags of the underlying factors f_t :

$$x_t = \Gamma g_t + \varepsilon_t = \sum_{k=-m_1}^{m_2} \lambda_k f_{t-k} + \varepsilon_t,$$

$$g_t = \begin{pmatrix} f_{t+m_1} \\ \vdots \\ f_{t-m_2} \end{pmatrix}, \quad r = qm.$$

In this case, a transformation $g_t \mapsto H_r g_t$ preserves the lead-lag structure if and only if

$$H_r = I_m \otimes H_q,$$

where H_q is a $q \times q$ invertible matrix. Under Assumptions 1 and 2, Bai and Wang [2014] show that these are the only transformations that leave the common component unchanged. Therefore, in dynamic factor models, the factors and loadings are identified up to an invertible $q \times q$ matrix, rather than an $r \times r$ matrix as in the static representation.

Finally, the dynamic factor model is subject to *timing indeterminacy*. That is, the common component

$$\sum_{k=-m_1}^{m_2} \lambda_k f_{t-k}$$

can equivalently be written as a one-sided filter

$$\sum_{k=1}^m \tilde{\lambda}_k \tilde{f}_{t-k}$$

by redefining the latent factors and loadings:

$$f_{t-k} \mapsto \tilde{f}_{t-k-m_1-1}, \quad \lambda_k \mapsto \tilde{\lambda}_{k+m_1+1}.$$

Thus, without loss of generality, we can assume the filter is one-sided.

3 Estimation

Suppose we observe $(x_t)_{t=1}^T$. Given the dynamic factor model

$$x_t = \sum_{k=-m_1}^{m_2} \lambda_k f_{t-k} + \varepsilon_t,$$

the dynamic factors f_t are indexed from $t = 1 - m_2$ to $t = T + m_1$ to account for the lags and leads. We jointly estimate $(f_t)_{t=1-m_2}^{T+m_1}$ and $(\lambda_k)_{k=-m_1}^{m_2}$ by minimizing the least squares objective

$$\frac{1}{NT} \left\| X - \sum_{k=-m_1}^{m_2} F_{-k} \lambda'_k \right\|^2 \quad (5)$$

over (f_t) and (λ_k) .

The first-order conditions for the factor loadings λ_j are

$$\sum_{k=-m_1}^{m_2} \hat{\lambda}_k \hat{F}'_{-k} \hat{F}_{-j} = X' \hat{F}_{-j}, \quad j = -m_1, \dots, m_2. \quad (6)$$

We define P as the $T + m_1 + m_2$ dimensional one-step upward cyclic permutation that acts as

$$(Pa)_t = a_{t+1 \pmod{T+m_1+m_2}},$$

i.e., P sends $(a_1, \dots, a_{T+m_1+m_2})' \in \mathbb{R}^{T+m_1+m_2}$ to $(a_2, \dots, a_{T+m_1+m_2-1}, a_1)' \in \mathbb{R}^{T+m_1+m_2}$. More concretely, P is $(T + m_1 + m_2) \times (T + m_1 + m_2)$ dimensional zero matrix with ones on 1-th diagonal and $-(T + m_1 + m_2)$ -th diagonal. Let $P_j = P^j$. To handle boundary effects, we impose $f_t = 0$ for $t = 1 - m_2, \dots, m_1$ and $t = T - m_2 + 1, \dots, T + m_1$. Then, the first-order conditions for the factors $(f_t)_{t=1}^T$ are

$$\sum_{k,j=-m_1}^{m_2} P_{j-k} \hat{F} \hat{\lambda}'_k \hat{\lambda}_j = \sum_{j=-m_1}^{m_2} P_j \tilde{X} \hat{\lambda}_j, \quad (7)$$

where the augmented matrices

$$\tilde{X} = \begin{pmatrix} 0_{m_2 \times N} \\ X \\ 0_{m_1 \times N} \end{pmatrix}, \quad \tilde{F}_{-k} = \begin{pmatrix} 0_{m_2 \times q} \\ F \\ 0_{m_1 \times q} \end{pmatrix},$$

include m_2 zero rows on top and m_1 zero rows on bottom to account for the leads and lags.

We solve (5) using an alternating least squares (ALS) procedure:

1. Start with an initial guess for $(\hat{\lambda}_k)$.
2. Given $(\hat{\lambda}_k)$, update (f_t) by solving (7).
3. Given $(\hat{f}_t)$, update (λ_k) by solving (6).
4. Repeat steps 2–3 until convergence.

The first-order conditions (6) defines a multivariate regression with rank qm . The first-order condition (7) can be efficiently solved using the discrete Fourier transform, requiring inversion of only $q \times q$ matrices. Since each step minimizes a least squares objective, the algorithm guarantees a monotone decrease of the objective function and convergence.

3.1 Consistency

Proposition 1. *Let Assumptions 1 and 2 hold. Suppose both (f_t) and (λ_k) are unknown, and we estimate them by minimizing the objective function (5). Then*

$$\begin{aligned} \frac{1}{\sqrt{N}} \left\| \hat{\Gamma} - \Gamma H_\gamma' \right\| &= O_p \left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}} \right), \\ \frac{1}{\sqrt{T}} \left\| \hat{G} - G H_\gamma^{-1} \right\| &= O_p \left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}} \right), \end{aligned}$$

where $H_\gamma \in \mathbb{R}^{mq \times mq}$ is defined by

$$H_\gamma = (\hat{G}' \hat{G})^{-1} \hat{G}' G.$$

Consequently, the estimated common component satisfies

$$\frac{1}{\sqrt{NT}} \left\| \sum_{k=-m_1}^{m_2} \hat{F}_{-k} \hat{\lambda}'_k - \sum_{k=-m_1}^{m_2} F_{-k} \lambda'_k \right\| = \frac{1}{\sqrt{NT}} \left\| \hat{G} \hat{\Gamma}' - G \Gamma' \right\| = O_p \left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}} \right).$$

Proposition 1 implies that if a dynamic factor structure exists, minimizing the corresponding least squares objective yields consistent estimates of both factors and factor loadings. Note that we do not require the factors or factor loadings to be orthogonal to the idiosyncratic errors to obtain consistency. Since the dynamic factor model generalizes the static factor model, this result also implies consistency of the static factors and loadings under the same conditions. However, imposing orthogonality between the factors and idiosyncratic errors can improve the convergence rates. For static factor models, orthogonality improves the rates to

$$O_p \left(\frac{1}{\sqrt{N}} + \frac{1}{T} \right) \quad \text{for factors,} \quad O_p \left(\frac{1}{\sqrt{T}} + \frac{1}{N} \right) \quad \text{for loadings.}$$

For dynamic factor models, orthogonality improves the rate of convergence for the factor loadings Γ to

$$O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{N}\right),$$

but it does not improve the rate for the dynamic factors. This limitation arises because the boundary values, i.e., f_t for $t = 1, \dots, m_1$ and $t = T - m_2 + 1, \dots, T$, cannot be consistently estimated for each finite T .

Theorem 1. *Let Assumptions 1 and 2 hold. Suppose both (f_t) and (λ_k) are unknown, and we estimate them by minimizing the objective function (5). Then, for $k = -m_1, \dots, m_2$, we have*

$$\begin{aligned} \frac{1}{\sqrt{T}} \left\| \hat{F} - F H_f' \right\| &= O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right) \\ \frac{1}{\sqrt{N}} \left\| \hat{\lambda}_k - \lambda_k H_f^{-1} \right\| &= O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right), \end{aligned}$$

where $H_f \in \mathbb{R}^{q \times q}$ is defined as

$$H_f = \left(\sum_{k=-m_1}^{m_2} \lambda_k' \hat{\lambda}_k \right) \left(\sum_{k=-m_1}^{m_2} \hat{\lambda}_k' \hat{\lambda}_k \right)^{-1}.$$

Consequently, each component of the common component satisfies

$$\frac{1}{\sqrt{NT}} \left\| \hat{F}_{-k} \hat{\lambda}_k' - F_{-k} \lambda_k' \right\| = O_p\left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}}\right), \quad k = -m_1, \dots, m_2.$$

Theorem 1 extends Proposition 1 by establishing the consistency of the dynamic factors up to a $q \times q$ invertible transformation, reflecting the rotational indeterminacy inherent in dynamic factor models.

4 Testing for Number of Dynamic Factors

4.1 Test Based on Dynamic Singular Value Ratio

The dynamic factor model $x_t = \sum_{k=-m_1}^{m_2} \lambda_k f_{t-k} + \varepsilon_t$ is observationally equivalent to a one-sided representation $x_t = \sum_{k=1}^m \lambda_k f_{t-k} + \varepsilon_t$, because we can always redefine the unobserved factors via a shift, e.g., $f_t \mapsto f_{t+m_1+1}$. For notational simplicity, we adopt this one-sided representation, so that

$$X = \sum_{k=1}^m F_{-k} \lambda_k' + E.$$

Let (q_0, m_0) denote the true number of dynamic factors and the true filter length, both unknown, and let (q, m) denote candidate values used in estimation. It is well known that the static representation with $(q, m) = (q_0 m_0, 1)$ also correctly specifies a dynamic factor model with (q_0, m_0) . Dynamic factor model may also admit multiple equivalent representations

other than the static representation. In other words, there may exist multiple (q, m) pairs that can also correctly specify the original model with (q_0, m_0) . For example, consider

$$x_t = \sum_{k=1}^3 \lambda_k f_{t-k} + \varepsilon_t, \quad f_t \in \mathbb{R}^2.$$

This model can equivalently be written as

$$\begin{aligned} x_t &= \lambda_1 f_{t-1} + \lambda_2 f_{t-2} + \lambda_3 f_{t-3} + \varepsilon_t \\ &= (\lambda_{1,1} \quad \lambda_{1,2} \quad \lambda_{2,2} - \lambda_{3,1}) g_{t-1}^{(3)} + (\lambda_{2,1} - \lambda_{1,2} \quad \lambda_{3,1} \quad \lambda_{3,2}) g_{t-2}^{(3)} + \varepsilon_t \\ &= (\lambda_1 \quad \lambda_2 \quad \lambda_3) g_{t-1}^{(6)} + \varepsilon_t, \end{aligned}$$

where $g_{t-1}^{(3)} = \begin{pmatrix} f_{t-1,1} \\ f_{t-1,2} + f_{t-2,1} \\ f_{t-2,2} \end{pmatrix}$ and $g_{t-1}^{(6)} = \begin{pmatrix} f_{t-1} \\ f_{t-2} \\ f_{t-3} \end{pmatrix}$. Here, $f_{t-k,j}$ and $\lambda_{k,j}$ denote the j -th element of f_{t-k} and the j -th column of λ_k , respectively.

This shows that a model with $(q_0, m_0) = (2, 3)$ also admits representations with $(q, m) = (3, 2)$ or $(q, m) = (6, 1)$. More generally, if $m < m_0$, more factors are needed to achieve correct specification, and if $q > q_0$, a smaller filter length may suffice to preserve correct specification. By convention, we denote (q_0, m_0) as the representation with the smallest number of factors. We formalize this observation in the following proposition.

Proposition 2. *Let (q_0, m_0) denote the true number of dynamic factors and the true filter length. Then the dynamic factor model with (q_0, m_0) also admits equivalent representations with*

$$(q, m) = \left(\left\lceil \frac{q_0 m_0}{m} \right\rceil, m \right), \quad 1 \leq m < m_0,$$

or

$$(q, m) = \left(q, \left\lceil \frac{q_0 m_0}{q} \right\rceil \right), \quad q > q_0.$$

In other words, the common component $\sum_{k=1}^{m_0} \lambda_k f_{t-k}$, with $f_t \in \mathbb{R}^{q_0}$, can also be represented as $\sum_{k=1}^m \Gamma_k g_{t-k}$ with the factor number and filter length given above.

To construct a test for the pair (q, m) corresponding to the correctly specified dynamic factor model, we employ a notion of *dynamic singular values*, which generalizes the classical singular value decomposition to incorporate temporal structure. Formally, let $\delta_{q,m}(X)$ denote the q -th dynamic singular value of a matrix X with filter length m , defined as

$$\delta_{q,m}(X) = \min_{(f_t) \in \mathbb{R}^{q-1}, (\lambda_k) \in \mathbb{R}^{q-1}} \left\| X - \sum_{k=1}^m F_{-k} \lambda'_k \right\|,$$

where $q \geq 1$ and $m \geq 0$. Note that we let (f_t) and (λ_k) be optimization arguments, not the true factors and factor loadings, whenever they appear as arguments within a minimization problem. Intuitively, $\delta_{q,m}(X)$ measures the residual spectral norm of X after removing $q-1$ dynamic factors with filter length m .

Several special cases illustrate the connection with standard singular values. First, if $m = 0$, or if $q = 1$, the dynamic singular value reduces to the spectral norm of X . Second, when $m = 1$, $\delta_{q,1}(X)$ coincides with the usual q -th singular value of X :

$$\delta_{q,1}(X) = \min_{F \in \mathbb{R}^{T \times (q-1)}, \lambda \in \mathbb{R}^{N \times (q-1)}} \|X - F\lambda'\|.$$

Note that $\delta_{q+1,m}(X)$ can be interpreted as the minimization problem

$$\delta_{qm+1,1}(X) = \min_{G \in \mathbb{R}^{T \times qm}, \Gamma \in \mathbb{R}^{T \times qm}} \|X - G\Gamma'\|$$

with the additional constraints that m blocks of $G \in \mathbb{R}^{T \times qm}$ with q columns are given as leads and lags of each other. This implies that we have $\delta_{qm+1,1}(X) \leq \delta_{q+1,m}(X)$ for any q and m . More generally, we have

$$\delta_{\tilde{q}+1,\tilde{m}}(X) \leq \delta_{q+1,m}(X)$$

for any integer $q, \tilde{q}, m, \tilde{m}$ that satisfies $qm \leq \tilde{q}\tilde{m}$ and $m \geq \tilde{m}$ because $\delta_{\tilde{q}+1,\tilde{m}}(X)$ minimizes the objective function over a larger space due to Proposition 2.

Assumption 3. For any $\epsilon > 0$, there exists a constant $c_0 > 0$ such that

$$\mathbb{P}\left(\frac{1}{\sqrt{N+T}} \delta_{k,1}(E) < c_0\right) \leq \epsilon$$

for all $k = 1, \dots, k_{\max} + 1$ and for all sufficiently large $N + T$.

Assumption 4. For any $\epsilon > 0$, there exists a constant $c_2 > 0$ such that

$$\mathbb{P}\left(\frac{1}{\sqrt{NT}} \delta_{q,m}\left(\sum_{k=1}^{m_0} F_{-k} \lambda'_k\right) < c_2\right) \leq \epsilon$$

for all $q = 1, \dots, q_0$ and $m = 1, \dots, m_{\max}$, and for all sufficiently large NT .

Assumption 3 requires that the leading singular values of the idiosyncratic component E be bounded away from zero in probability after appropriate normalization. This property is known to hold under very mild conditions on the cross-sectional and temporal dependence of E . In particular, under general conditions, the number of singular values that remain bounded away from zero grows at rate $\min(N, T)$; see, for example, Ahn and Horenstein [2013]. As a direct implication of Assumption 3, for any $\epsilon > 0$ there exists $c_0 > 0$ such that

$$\mathbb{P}\left(\frac{1}{\sqrt{N+T}} \delta_{q+1,m}(E) < c_0\right) \leq \epsilon$$

for all pairs (q, m) satisfying $qm \leq k_{\max}$, and for all sufficiently large $N + T$.

Assumption 4 imposes a lower bound on the dynamic singular values of the common component. Note that

$$\begin{aligned} \frac{1}{\sqrt{NT}} \delta_{q_0, m_0} \left(\sum_{k=1}^{m_0} F_{-k} \lambda'_k \right) &\geq \frac{1}{\sqrt{NT}} \delta_{(q_0-1)m_0+1, 1} \left(\sum_{k=1}^{m_0} F_{-k} \lambda'_k \right) \\ &= \frac{1}{\sqrt{NT}} \delta_{q_0 m_0 - m_0 + 1, 1}(G\Gamma'), \end{aligned}$$

where the inequality follows from the additional restrictions imposed on the minimization problem on the left hand side. The term $\frac{1}{\sqrt{NT}} \delta_{q_0 m_0 - m_0 + 1, 1}(G\Gamma')$ is strictly positive in the limit provided that $m_0 \geq 1$. Indeed, the k -th singular value of $(NT)^{-1/2} G\Gamma'$ equals the square root of the k -th eigenvalue of

$$\left(\frac{1}{N} G'G \right) \left(\frac{1}{T} \Gamma' \Gamma \right),$$

which is bounded away from zero for all $k = 1, \dots, q_0 m_0$ by Assumption 1. Consequently,

$$\frac{1}{\sqrt{NT}} \delta_{q,m} \left(\sum_{k=1}^{m_0} F_{-k} \lambda'_k \right)$$

is bounded away from zero in probability for all $q = 1, \dots, q_0$ and $m = 1, \dots, m_0$.

Assumption 4 further requires this lower bound to persist for $m = m_0 + 1, \dots, m_{\max}$. This restriction rules out the possibility that omitted dynamic factors can be fully recovered by increasing the filter length of the remaining factors. In other words, when fewer than q_0 dynamic factors are extracted, enlarging the lag polynomial alone is insufficient to span the factor space associated with the omitted components.

This requirement is satisfied whenever the spectral density matrix of the common component has rank q_0 almost everywhere, which is standard in the frequency-domain literature on dynamic factor models (see, e.g., Forni et al., 2000, 2015). To see the connection, let $(\tilde{F}_{-k}, \tilde{\lambda}_k)$ denote the minimizer in the definition of

$$\delta_{q,m} \left(\sum_{k=1}^{m_0} F_{-k} \lambda'_k \right), \quad q < q_0.$$

The squared spectral norm of the approximation error

$$\sum_{k=1}^{m_0} F_{-k} \lambda'_k - \sum_{k=1}^m \tilde{F}_{-k} \tilde{\lambda}'_k$$

is bounded below by the squared Frobenius norm divided by the finite rank $(q_0 m_0 + qm)$. The population counterpart to the Frobenius norm corresponds to the population mean-squared error of a rank- q , finite-degree lag-polynomial approximation to the common component. Under the assumption that the spectral density of the common component has rank q_0 on a set of frequencies of positive measure, this population approximation error is bounded below by

$$\int_{-\pi}^{\pi} \mu_{q_0}(\omega) d\omega,$$

where $\mu_{q_0}(\omega)$ denotes the q_0 -th largest eigenvalue of the spectral density matrix of the common component at frequency ω . Since this integral is strictly positive, the dynamic singular values remain bounded away from zero even when $m > m_0$, thereby establishing Assumption 4.

We construct a test based on estimated dynamic singular values. Let $\hat{\delta}_{q+1,m}(X)$ denote the estimated $(q+1)$ -th dynamic singular value, defined as the spectral norm of the residual matrix obtained after fitting a dynamic factor model with q factors and filter length m . Specifically,

$$\hat{\delta}_{q+1,m}(X) = \left\| X - \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)'} \right\|,$$

where $\{\hat{F}_{-k}^{(q)}, \hat{\lambda}_k^{(q)}\}_{k=1}^m$ are the least-squares estimators obtained by minimizing the Frobenius norm of the residual matrix under q dynamic factors, i.e., the objective function (5).

It is important to note that the alternating least squares algorithm yields the minimizer of the Frobenius norm, whereas the dynamic singular value is defined as the minimum of

the spectral norm. The two minimization problems coincide in the static case ($m = 1$), in which case $\hat{\delta}_{q+1,1}(X)$ reduces to the usual $(q + 1)$ -th singular value of X . For $m > 1$, the two minimizers generally differ. While dynamic singular values can in principle be computed by directly minimizing the spectral norm, we rely on the Frobenius-norm solution for computational tractability. This approximation does not affect the asymptotic validity of the proposed test.

Fixing m , we estimate the number of dynamic factors using a ratio-type statistic based on adjacent estimated dynamic singular values.

Theorem 2. *Suppose Assumptions 1, 2, 3, and 4 hold. Let $q_{\max}$ and $m_{\max}$ be integers satisfying $(q_{\max} + q_0 + 1)m_{\max} \leq k_{\max}$. Fix $m \geq 1$ and define*

$$\hat{q} = \operatorname{argmax}_{1 \leq q \leq q_{\max}} DR_{q,m}(X),$$

where $DR_{q,m}(X) = \frac{\hat{\delta}_{q,m}(X)}{\hat{\delta}_{q+1,m}(X)}$. If $m_0 \leq m \leq m_{\max}$, then

$$\mathbb{P}\{\hat{q} = q_0\} \rightarrow 1 \quad \text{as } N, T \rightarrow \infty.$$

If instead $m < m_0$, then

$$\mathbb{P}\left\{\hat{q} = \left\lceil \frac{q_0 m_0}{m} \right\rceil\right\} \rightarrow 1,$$

where $\lceil \cdot \rceil$ denotes the ceiling function.

The theorem shows that, when the filter length is at least as large as the true value, the ratio statistic consistently identifies the true number of dynamic factors. When the filter length is misspecified downward, the procedure selects the minimal number of factors required to span the true common component, reflecting the observational equivalence between (q_0, m_0) and alternative (q, m) representations discussed in Proposition 2. When $m = 1$, the DR statistic coincides with the eigenvalue ratio (ER) test of Ahn and Horenstein [2013] and consistently estimates the number of static factors, $r = q_0 m_0$. Theorem 2 thus generalizes the ER test to dynamic factor models by allowing the filter length m to exceed one.

Theorem 3. *Suppose Assumptions 1, 2, 3, and 4 hold. Let $q_{\max}$ and $m_{\max}$ satisfy $(q_{\max} + q_0 + 1)m_{\max} \leq k_{\max}$. Fix q and define*

$$\hat{m} = \operatorname{argmax}_{1 \leq m \leq m_{\max}} DR_{q,m}(X),$$

where $DR_{q,m}(X) = \frac{\hat{\delta}_{q+1,m-1}(X)}{\hat{\delta}_{q+1,m}(X)}$. If $q = q_0 \geq 1$, then

$$\mathbb{P}\{\hat{m} = m_0\} \rightarrow 1$$

as $N, T \rightarrow \infty$. If $q > q_0$, then

$$\mathbb{P}\left\{\hat{m} = \left\lceil \frac{q_0 m_0}{q} \right\rceil\right\} \rightarrow 1,$$

where $\lceil \cdot \rceil$ denotes the ceiling operator.

Theorem 3 provides a consistent estimator of the true filter length in dynamic factor models. If $q = q_0$, the test uncovers the true filter length. If $q > q_0$, the test select the smallest filter length required to span the true common component. The case $q < q_0$ is excluded. Under Assumption (4), $\delta_{q+1,m}(X) = O_p(\sqrt{NT})$ for all $m \leq m_{\max}$ when $q < q_0$. As a result, $DR_{q,m}(X)$ may not diverge at $m = m_0$, and ratio-based tests for the filter length fail under underspecification of the number of dynamic factors.

4.2 Test Based on Information Criteria

While the ratio-based test succeeds in uncovering all the dynamic specifications of the model discussed in Proposition 2, in many cases, the most useful test would be a test that directly finds the most parsimonious specification (q_0, m_0) . We consider a test based on information criteria to estimate (q_0, m_0) . Let

$$V_{q,m}(X) = \frac{1}{NT} \left\| X - \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)'} \right\|^2,$$

where $\hat{F}_{-k}^{(q)}$ and $\hat{\lambda}_k^{(q)'}$ are least squares estimates with q dynamic factors. Following Bai and Ng [2002], "The crucial element in consistent estimation of the number of factors is a penalty factor that vanishes but still dominates the difference in the sum of squared residuals between the true and the overparametrized model." In this spirit, we define two information criteria:

$$\begin{aligned} PC(q, m) &= V_{q,m}(X) + q(m+1)g(N, T), \\ DC(q, m) &= \frac{1}{NT} \hat{\delta}_{q+1,m}^2(X) + q(m+1)g(N, T), \end{aligned}$$

where $g(N, T)$ is a vanishing penalty function. Compared to Bai and Ng [2002], the key differences are: (i) model fit is evaluated after estimating the dynamic factor model, not its static representation, and (ii) the penalty accounts for both the number of dynamic factors q and the number of static factors qm . This ensures the selection of the most parsimonious dynamic representation.

The DC criterion is motivated naturally from the static case. For static models, the estimated factors and loadings minimize both the Frobenius norm and the spectral norm. Hence, the squared spectral norm can serve as an alternative objective. While this argument does not trivially extend to dynamic factors, the DC criterion also succeeds in finding the correct number of dynamic factors.

Theorem 4. Suppose Assumptions 1, 2, 3, and 4 hold. Let $\hat{q}$ and $\hat{m}$ be given by

$$\hat{q}, \hat{m} = \underset{0 \leq q \leq q_{\max}, 0 \leq m \leq m_{\max}}{\operatorname{argmin}} PC(q, m)$$

or

$$\hat{q}, \hat{m} = \underset{0 \leq q \leq q_{\max}, 0 \leq m \leq m_{\max}}{\operatorname{argmin}} DC(q, m).$$

If $g(N, T) \rightarrow 0$ and $\left(\frac{N+T}{NT}\right) g(N, T) \rightarrow \infty$ as $N, T \rightarrow \infty$, we have

$$P(\hat{q} = q_0) \rightarrow 1, \quad \text{and} \quad P(\hat{m} = m_0) \rightarrow 1$$

as $N, T \rightarrow \infty$.

The key to consistency is penalizing both the number of static factors qm and the number of dynamic factors q . Penalizing only qm would select the static representation of the dynamic factor model, since among tuples with the same static factors $r = qm$, $(qm, 1)$ minimizes the squared error. Including q as a separate term in the penalty ensures recovery of the most parsimonious dynamic structure.

Corollary 1. *Suppose Assumptions 1, 2, 3, and 4 hold. Let*

$$(\hat{q}, \hat{m}) = \underset{0 \leq q \leq q_{\max}, 0 \leq m \leq m_{\max}}{\operatorname{argmin}} IC(q, m),$$

where

$$IC(q, m) = \log V_{q,m}(X) + q(m+1)g(N, T).$$

If $g(N, T) \rightarrow 0$ and $(\frac{N+T}{NT})g(N, T) \rightarrow \infty$, then

$$\mathbb{P}(\hat{q} = q_0) \rightarrow 1, \quad \mathbb{P}(\hat{m} = m_0) \rightarrow 1.$$

The proof is omitted, as it follows directly from Corollary 1 of Bai and Ng [2002]. The IC criterion is convenient because it does not require scaling the penalty term, so that the test becomes independent of $q_{\max}$ or $m_{\max}$. As in Bai and Ng [2002], we use three candidate penalty terms:

$$\begin{aligned} g_1(N, T) &= \left(\frac{N+T}{NT} \right) \log \left(\frac{NT}{N+T} \right) \\ g_2(N, T) &= \left(\frac{N+T}{NT} \right) \log (\min(N, T)) \\ g_3(N, T) &= \left(\frac{1}{\min(N, T)} \right) \log (\min(N, T)). \end{aligned}$$

We also scale the penalty term with $\hat{\sigma}^2 = V_{q_{\max}, 1}(X)$ for PC criterion and $\hat{\sigma}^2 = \frac{1}{NT} \hat{\delta}_{q_{\max}+1, 1}(X)$ for DC criteria¹.

Both PC and IC can be viewed as dynamic extensions of the original criteria in Bai and Ng [2002]: if $m = 1$, they reduce numerically to Bai and Ng's criteria, up to a multiplicative factor of two in the penalty term.

5 Simulations

We conduct Monte Carlo simulations of dynamic factor models to evaluate the finite-sample performance of tests for the number of dynamic factors and the filter length. Specifically, we simulate the model

$$x_t = \sum_{k=-m_1}^{m_2} \lambda_k f_{t-k} + \varepsilon_t,$$

where the idiosyncratic errors satisfy

$$\varepsilon_{it} = \sqrt{\frac{\theta(1-\rho^2)}{1+2J\beta^2}} e_{it}, \quad e_{it} = \rho e_{i,t-1} + v_{it} + \sum_{1 \leq |j| \leq J} \beta v_{i-j,t}.$$

¹ $\hat{\sigma}^2 = V_{q_{\max}, m_{\max}}(X)$ or $\frac{1}{NT} \hat{\delta}_{q_{\max}+1, m_{\max}}(X)$ may also be considered.

The innovations v_{it} and the factor loadings λ_k are drawn independently from a standard normal distribution. This data-generating process is identical to those considered in Bai and Ng [2002] and Ahn and Horenstein [2013], except that we allow the filter length $m = m_1 + m_2 + 1$ to exceed one.

The common factors f_t follow a VARMA(1, 1) process,

$$f_t = Af_{t-1} + u_t + \Theta u_{t-1},$$

under the following four specifications:

1. $\rho = \beta = J = 0$ and $A = \Theta = 0$.
2. $\rho = 0.5$, $\beta = 0.2$, $J = 10$, and $A = \Theta = 0$.
3. $\rho = \beta = J = 0$, $A = \text{Diag}(0.7, 0.5, 0.3)$, and $\Theta = 0$.
4. $\rho = \beta = J = 0$, $A = 0$, and $\Theta = \text{Diag}(0.7, 0.5, 0.3)$.

By construction, the variance of ε_{it} equals θ . The variance of the common component $\sum_{k=-m_1}^{m_2} \lambda_k f_{t-k}$ is given by $m \cdot \text{tr}(\Sigma_f)$, where $\Sigma_f = \mathbb{E}[f_t f_t']$. We therefore set $\theta = m \cdot \text{tr}(\Sigma_f)$, ensuring that the signal-to-noise ratio equals one in all specifications.

The first specification corresponds to an *exact* dynamic factor model, in which both the factors and idiosyncratic errors are serially uncorrelated. The second specification represents an *approximate* dynamic factor model, allowing for heteroskedasticity as well as serial and cross-sectional dependence in the idiosyncratic errors. The third and fourth specifications examine settings in which the factors follow VAR(1) and VMA(1) processes, respectively.

5.1 Test Based on Dynamic Singular Value Ratio

We use the four specifications described in the previous section. We use $DR_{q,m}(X)$ to estimate the number of q given m .

		$m = 1$		$m = 2$		$m = 3$		$m = 4$		$m = 5$		$m = 6$	
N	T	$< \tilde{q}$	$= \tilde{q}$	$< \tilde{q}$	$= \tilde{q}$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$
$N = 50$	$T = 50$	1.00	0.00	0.18	0.82	0.00	0.98	0.10	0.90	0.41	0.55	0.00	0.00
$N = 50$	$T = 100$	1.00	0.00	0.06	0.94	0.00	0.97	0.00	1.00	0.06	0.94	0.15	0.85
$N = 50$	$T = 150$	1.00	0.00	0.02	0.98	0.00	0.98	0.00	1.00	0.00	1.00	0.03	0.97
$N = 50$	$T = 200$	1.00	0.00	0.02	0.98	0.00	0.98	0.00	1.00	0.00	1.00	0.00	1.00
$N = 100$	$T = 50$	1.00	0.00	0.01	0.99	0.00	0.99	0.04	0.96	0.36	0.64	0.22	0.14
$N = 100$	$T = 100$	1.00	0.00	0.00	1.00	0.00	0.97	0.00	1.00	0.00	1.00	0.03	0.97
$N = 100$	$T = 150$	1.00	0.00	0.00	1.00	0.00	0.99	0.00	1.00	0.00	1.00	0.00	1.00
$N = 100$	$T = 200$	1.00	0.00	0.00	1.00	0.00	0.99	0.00	1.00	0.00	1.00	0.00	1.00
$N = 150$	$T = 50$	1.00	0.00	0.00	1.00	0.00	0.99	0.02	0.98	0.30	0.70	0.64	0.33
$N = 150$	$T = 100$	1.00	0.00	0.00	1.00	0.00	0.98	0.00	1.00	0.00	1.00	0.05	0.95
$N = 150$	$T = 150$	1.00	0.00	0.00	1.00	0.00	0.99	0.00	1.00	0.00	1.00	0.00	1.00
$N = 150$	$T = 200$	1.00	0.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
$N = 200$	$T = 50$	1.00	0.00	0.02	0.98	0.00	1.00	0.00	1.00	0.18	0.82	0.54	0.46
$N = 200$	$T = 100$	1.00	0.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.02	0.98
$N = 200$	$T = 150$	1.00	0.00	0.00	1.00	0.00	0.99	0.00	1.00	0.00	1.00	0.00	1.00
$N = 200$	$T = 200$	1.00	0.00	0.00	1.00	0.00	0.99	0.00	1.00	0.00	1.00	0.00	1.00

Table 1: where $\tilde{q} = \lceil \frac{q_0 m_0}{m} \rceil$, First Specification

		$m = 1$		$m = 2$		$m = 3$		$m = 4$		$m = 5$		$m = 6$	
N	T	$< \tilde{q}$	$= \tilde{q}$	$< \tilde{q}$	$= \tilde{q}$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$
$N = 50$	$T = 50$	1.00	0.00	0.04	0.16	0.03	0.13	0.05	0.34	0.09	0.30	0.03	0.09
$N = 50$	$T = 100$	1.00	0.00	0.00	0.10	0.00	0.11	0.04	0.29	0.10	0.28	0.15	0.30
$N = 50$	$T = 150$	1.00	0.00	0.01	0.03	0.00	0.09	0.02	0.15	0.07	0.12	0.10	0.07
$N = 50$	$T = 200$	1.00	0.00	0.00	0.02	0.01	0.14	0.03	0.20	0.04	0.16	0.09	0.06
$N = 100$	$T = 50$	1.00	0.00	0.15	0.40	0.03	0.47	0.05	0.61	0.20	0.51	0.06	0.13
$N = 100$	$T = 100$	1.00	0.00	0.08	0.56	0.00	0.83	0.04	0.75	0.34	0.46	0.52	0.29
$N = 100$	$T = 150$	1.00	0.00	0.09	0.57	0.00	0.89	0.10	0.78	0.23	0.61	0.49	0.33
$N = 100$	$T = 200$	1.00	0.00	0.08	0.70	0.00	0.99	0.03	0.89	0.18	0.67	0.33	0.47
$N = 150$	$T = 50$	1.00	0.00	0.21	0.66	0.06	0.82	0.18	0.73	0.46	0.44	0.33	0.16
$N = 150$	$T = 100$	1.00	0.00	0.05	0.93	0.00	0.98	0.08	0.91	0.38	0.57	0.61	0.34
$N = 150$	$T = 150$	1.00	0.00	0.00	1.00	0.00	1.00	0.02	0.98	0.18	0.82	0.42	0.58
$N = 150$	$T = 200$	1.00	0.00	0.01	0.99	0.00	1.00	0.00	1.00	0.10	0.90	0.28	0.72
$N = 200$	$T = 50$	1.00	0.00	0.15	0.80	0.01	0.92	0.13	0.85	0.46	0.49	0.38	0.11
$N = 200$	$T = 100$	1.00	0.00	0.01	0.99	0.00	1.00	0.01	0.98	0.14	0.86	0.48	0.51
$N = 200$	$T = 150$	1.00	0.00	0.00	1.00	0.00	0.99	0.00	1.00	0.11	0.89	0.28	0.72
$N = 200$	$T = 200$	1.00	0.00	0.00	1.00	0.00	0.99	0.00	1.00	0.01	0.99	0.14	0.86

Table 2: where $\tilde{q} = \lceil \frac{q_0 m_0}{m} \rceil$, Second Specification

		$m = 1$		$m = 2$		$m = 3$		$m = 4$		$m = 5$		$m = 6$	
N	T	$< \tilde{q}$	$= \tilde{q}$	$< \tilde{q}$	$= \tilde{q}$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$
$N = 50$	$T = 50$	1.00	0.00	0.67	0.07	0.63	0.16	0.77	0.15	0.71	0.15	0.52	0.07
$N = 50$	$T = 100$	1.00	0.00	0.70	0.01	0.64	0.16	0.79	0.10	0.84	0.10	0.81	0.09
$N = 50$	$T = 150$	1.00	0.00	0.66	0.00	0.66	0.25	0.86	0.06	0.89	0.03	0.89	0.03
$N = 50$	$T = 200$	1.00	0.00	0.73	0.01	0.72	0.21	0.84	0.09	0.88	0.07	0.92	0.06
$N = 100$	$T = 50$	1.00	0.00	0.93	0.03	0.75	0.23	0.85	0.12	0.85	0.12	0.73	0.05
$N = 100$	$T = 100$	1.00	0.00	0.98	0.00	0.54	0.45	0.78	0.22	0.90	0.10	0.89	0.11
$N = 100$	$T = 150$	1.00	0.00	0.99	0.00	0.44	0.56	0.67	0.33	0.78	0.22	0.83	0.17
$N = 100$	$T = 200$	1.00	0.00	0.99	0.00	0.25	0.75	0.55	0.45	0.73	0.27	0.79	0.21
$N = 150$	$T = 50$	1.00	0.00	0.98	0.02	0.70	0.30	0.84	0.16	0.87	0.13	0.90	0.04
$N = 150$	$T = 100$	1.00	0.00	0.99	0.01	0.35	0.65	0.65	0.35	0.83	0.17	0.80	0.20
$N = 150$	$T = 150$	1.00	0.00	1.00	0.00	0.17	0.83	0.49	0.51	0.66	0.34	0.68	0.32
$N = 150$	$T = 200$	1.00	0.00	1.00	0.00	0.12	0.88	0.35	0.65	0.53	0.47	0.55	0.45
$N = 200$	$T = 50$	1.00	0.00	0.96	0.04	0.57	0.43	0.80	0.20	0.91	0.09	0.92	0.03
$N = 200$	$T = 100$	1.00	0.00	0.95	0.05	0.20	0.80	0.46	0.54	0.70	0.30	0.77	0.23
$N = 200$	$T = 150$	1.00	0.00	0.94	0.06	0.04	0.96	0.29	0.71	0.47	0.53	0.53	0.47
$N = 200$	$T = 200$	1.00	0.00	0.93	0.07	0.03	0.97	0.15	0.85	0.29	0.71	0.37	0.63

Table 3: where $\tilde{q} = \lceil \frac{q_0 m_0}{m} \rceil$, Third Specification

		$m = 1$		$m = 2$		$m = 3$		$m = 4$		$m = 5$		$m = 6$	
N	T	$< \tilde{q}$	$= \tilde{q}$	$< \tilde{q}$	$= \tilde{q}$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$
$N = 50$	$T = 50$	1.00	0.00	0.15	0.09	0.09	0.11	0.21	0.33	0.30	0.28	0.16	0.13
$N = 50$	$T = 100$	1.00	0.00	0.13	0.07	0.06	0.33	0.12	0.30	0.26	0.29	0.33	0.20
$N = 50$	$T = 150$	1.00	0.00	0.10	0.04	0.04	0.47	0.07	0.31	0.20	0.24	0.23	0.21
$N = 50$	$T = 200$	1.00	0.00	0.09	0.01	0.02	0.52	0.12	0.40	0.17	0.24	0.31	0.14
$N = 100$	$T = 50$	1.00	0.00	0.64	0.10	0.16	0.61	0.40	0.47	0.67	0.25	0.42	0.13
$N = 100$	$T = 100$	1.00	0.00	0.76	0.10	0.05	0.94	0.15	0.82	0.39	0.57	0.55	0.38
$N = 100$	$T = 150$	1.00	0.00	0.80	0.07	0.01	0.99	0.09	0.91	0.28	0.68	0.46	0.51
$N = 100$	$T = 200$	1.00	0.00	0.81	0.08	0.01	0.99	0.03	0.97	0.14	0.86	0.39	0.60
$N = 150$	$T = 50$	1.00	0.00	0.70	0.18	0.18	0.77	0.38	0.61	0.58	0.36	0.58	0.16
$N = 150$	$T = 100$	1.00	0.00	0.69	0.29	0.00	1.00	0.01	0.99	0.30	0.70	0.48	0.52
$N = 150$	$T = 150$	1.00	0.00	0.55	0.45	0.00	1.00	0.02	0.98	0.09	0.91	0.24	0.76
$N = 150$	$T = 200$	1.00	0.00	0.57	0.43	0.00	1.00	0.00	1.00	0.07	0.93	0.14	0.86
$N = 200$	$T = 50$	1.00	0.00	0.66	0.32	0.11	0.89	0.33	0.66	0.67	0.32	0.73	0.11
$N = 200$	$T = 100$	1.00	0.00	0.56	0.44	0.00	1.00	0.05	0.95	0.22	0.78	0.42	0.58
$N = 200$	$T = 150$	1.00	0.00	0.39	0.61	0.00	1.00	0.01	0.99	0.12	0.88	0.24	0.76
$N = 200$	$T = 200$	1.00	0.00	0.36	0.64	0.00	1.00	0.00	1.00	0.02	0.98	0.07	0.93

Table 4: where $\tilde{q} = \lceil \frac{q_0 m_0}{m} \rceil$, Fourth Specification

Now, we use $DR_{q,m}(X)$ to estimate the number of m given q .

		$q = 3$		$q = 4$		$q = 5$		$q = 6$		$q = 7$		$q = 8$	
N	T	$< m_0$	$= m_0$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$
$N = 50$	$T = 50$	0.01	0.97	0.43	0.57	0.02	0.98	0.10	0.90	0.50	0.50	0.02	0.00
$N = 50$	$T = 100$	0.00	0.97	0.14	0.86	0.00	1.00	0.01	0.99	0.08	0.92	0.51	0.49
$N = 50$	$T = 150$	0.00	0.98	0.10	0.90	0.00	1.00	0.00	1.00	0.01	0.99	0.30	0.70
$N = 50$	$T = 200$	0.00	0.98	0.03	0.97	0.00	1.00	0.00	1.00	0.00	1.00	0.26	0.74
$N = 100$	$T = 50$	0.00	0.99	0.12	0.88	0.00	1.00	0.02	0.98	0.10	0.90	0.42	0.50
$N = 100$	$T = 100$	0.00	0.97	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.02	0.98
$N = 100$	$T = 150$	0.00	0.99	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
$N = 100$	$T = 200$	0.00	0.99	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
$N = 150$	$T = 50$	0.00	0.99	0.02	0.98	0.00	1.00	0.01	0.99	0.01	0.99	0.29	0.71
$N = 150$	$T = 100$	0.00	0.98	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
$N = 150$	$T = 150$	0.00	0.99	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
$N = 150$	$T = 200$	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
$N = 200$	$T = 50$	0.00	1.00	0.03	0.97	0.00	1.00	0.00	1.00	0.01	0.99	0.19	0.81
$N = 200$	$T = 100$	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
$N = 200$	$T = 150$	0.00	0.99	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
$N = 200$	$T = 200$	0.00	0.99	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00

Table 5: where $\tilde{m} = \lceil \frac{q_0 m_0}{q} \rceil$, First Specification

		$q = 3$		$q = 4$		$q = 5$		$q = 6$		$q = 7$		$q = 8$	
N	T	$< m_0$	$= m_0$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$
$N = 50$	$T = 50$	0.48	0.22	0.35	0.56	0.13	0.50	0.05	0.95	0.01	0.99	0.02	0.93
$N = 50$	$T = 100$	0.47	0.33	0.49	0.46	0.06	0.44	0.00	1.00	0.01	0.99	0.00	1.00
$N = 50$	$T = 150$	0.69	0.25	0.57	0.40	0.15	0.44	0.00	1.00	0.00	1.00	0.00	1.00
$N = 50$	$T = 200$	0.68	0.28	0.61	0.37	0.15	0.52	0.00	0.99	0.00	1.00	0.00	1.00
$N = 100$	$T = 50$	0.24	0.56	0.35	0.62	0.11	0.68	0.07	0.91	0.03	0.97	0.04	0.95
$N = 100$	$T = 100$	0.09	0.86	0.29	0.70	0.02	0.94	0.02	0.97	0.00	1.00	0.00	1.00
$N = 100$	$T = 150$	0.02	0.97	0.31	0.69	0.02	0.98	0.01	0.99	0.00	1.00	0.03	0.97
$N = 100$	$T = 200$	0.01	0.99	0.20	0.80	0.00	1.00	0.00	1.00	0.01	0.99	0.00	1.00
$N = 150$	$T = 50$	0.16	0.78	0.33	0.67	0.04	0.95	0.05	0.94	0.09	0.91	0.18	0.81
$N = 150$	$T = 100$	0.02	0.98	0.09	0.91	0.00	1.00	0.00	1.00	0.00	1.00	0.02	0.98
$N = 150$	$T = 150$	0.00	1.00	0.01	0.99	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
$N = 150$	$T = 200$	0.00	1.00	0.01	0.99	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
$N = 200$	$T = 50$	0.05	0.94	0.29	0.70	0.02	0.98	0.03	0.97	0.07	0.93	0.26	0.73
$N = 200$	$T = 100$	0.00	1.00	0.02	0.98	0.00	1.00	0.00	1.00	0.00	1.00	0.01	0.99
$N = 200$	$T = 150$	0.00	0.99	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
$N = 200$	$T = 200$	0.00	0.99	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00

Table 6: where $\tilde{m} = \lceil \frac{q_0 m_0}{q} \rceil$, Second Specification

		$q = 3$		$q = 4$		$q = 5$		$q = 6$		$q = 7$		$q = 8$	
N	T	$< m_0$	$= m_0$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$
$N = 50$	$T = 50$	1.00	0.00	0.99	0.01	0.97	0.03	0.89	0.11	0.90	0.10	0.91	0.08
$N = 50$	$T = 100$	1.00	0.00	1.00	0.00	1.00	0.00	0.90	0.10	0.88	0.12	0.97	0.03
$N = 50$	$T = 150$	1.00	0.00	1.00	0.00	1.00	0.00	0.96	0.04	0.95	0.05	0.99	0.01
$N = 50$	$T = 200$	1.00	0.00	1.00	0.00	1.00	0.00	0.99	0.01	0.97	0.03	1.00	0.00
$N = 100$	$T = 50$	0.98	0.02	1.00	0.00	0.97	0.03	0.95	0.05	0.99	0.01	0.99	0.01
$N = 100$	$T = 100$	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
$N = 100$	$T = 150$	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
$N = 100$	$T = 200$	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00
$N = 150$	$T = 50$	0.99	0.01	1.00	0.00	0.96	0.04	0.99	0.01	1.00	0.00	1.00	0.00
$N = 150$	$T = 100$	1.00	0.00	1.00	0.00	0.97	0.03	1.00	0.00	0.99	0.01	1.00	0.00
$N = 150$	$T = 150$	1.00	0.00	1.00	0.00	0.99	0.01	1.00	0.00	1.00	0.00	1.00	0.00
$N = 150$	$T = 200$	1.00	0.00	1.00	0.00	0.99	0.01	1.00	0.00	1.00	0.00	1.00	0.00
$N = 200$	$T = 50$	0.95	0.05	0.99	0.01	0.93	0.07	0.97	0.03	0.99	0.01	0.99	0.01
$N = 200$	$T = 100$	0.97	0.03	1.00	0.00	0.97	0.03	1.00	0.00	1.00	0.00	1.00	0.00
$N = 200$	$T = 150$	0.98	0.02	1.00	0.00	0.98	0.02	1.00	0.00	1.00	0.00	1.00	0.00
$N = 200$	$T = 200$	0.99	0.01	1.00	0.00	0.97	0.03	1.00	0.00	1.00	0.00	1.00	0.00

Table 7: where $\tilde{m} = \lceil \frac{q_0 m_0}{q} \rceil$, Third Specification

		$q = 3$		$q = 4$		$q = 5$		$q = 6$		$q = 7$		$q = 8$	
N	T	$< m_0$	$= m_0$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$	$< \tilde{m}$	$= \tilde{m}$
$N = 50$	$T = 50$	0.77	0.04	0.81	0.18	0.61	0.29	0.35	0.65	0.29	0.71	0.36	0.64
$N = 50$	$T = 100$	0.93	0.05	0.94	0.06	0.57	0.38	0.28	0.72	0.22	0.78	0.35	0.65
$N = 50$	$T = 150$	0.97	0.03	0.98	0.02	0.59	0.38	0.20	0.80	0.24	0.76	0.34	0.66
$N = 50$	$T = 200$	0.95	0.05	1.00	0.00	0.60	0.39	0.32	0.68	0.24	0.76	0.32	0.68
$N = 100$	$T = 50$	0.69	0.27	0.92	0.08	0.52	0.47	0.61	0.39	0.60	0.40	0.77	0.23
$N = 100$	$T = 100$	0.64	0.36	0.98	0.02	0.50	0.50	0.50	0.50	0.74	0.26	0.83	0.17
$N = 100$	$T = 150$	0.73	0.27	0.99	0.01	0.37	0.63	0.66	0.34	0.80	0.20	0.84	0.16
$N = 100$	$T = 200$	0.71	0.29	0.98	0.02	0.30	0.70	0.75	0.25	0.83	0.17	0.89	0.11
$N = 150$	$T = 50$	0.45	0.55	0.90	0.10	0.37	0.63	0.42	0.58	0.71	0.29	0.86	0.14
$N = 150$	$T = 100$	0.32	0.68	0.91	0.09	0.19	0.81	0.53	0.47	0.82	0.18	0.91	0.09
$N = 150$	$T = 150$	0.24	0.76	0.88	0.12	0.14	0.86	0.50	0.50	0.79	0.21	0.96	0.04
$N = 150$	$T = 200$	0.18	0.82	0.85	0.15	0.08	0.92	0.47	0.53	0.84	0.16	0.95	0.05
$N = 200$	$T = 50$	0.41	0.58	0.87	0.13	0.36	0.64	0.48	0.52	0.76	0.24	0.91	0.09
$N = 200$	$T = 100$	0.13	0.87	0.83	0.17	0.10	0.90	0.40	0.60	0.66	0.34	0.96	0.04
$N = 200$	$T = 150$	0.09	0.90	0.75	0.25	0.03	0.97	0.33	0.67	0.75	0.25	0.94	0.06
$N = 200$	$T = 200$	0.06	0.94	0.77	0.23	0.03	0.97	0.32	0.68	0.74	0.26	0.95	0.05

Table 8: where $\tilde{m} = \lceil \frac{q_0 m_0}{q} \rceil$, Fourth Specification

5.2 Test Based on Information Criteria

We see finite sample performance of the *DC*, *PC*, *IC* criteria in estimating q . We compare the results with the two existing methods given by Bai and Ng [2007] and Amengual and Watson [2007]. In Bai and Ng [2007], we tried

		<i>DC</i>		<i>PC</i>		<i>IC</i>		<i>BN</i>		<i>AW</i>	
N	T	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$
$N = 50$	$T = 50$	0.01	0.97	0.20	0.79	0.17	0.00	0.01	0.96	0.07	0.92
$N = 50$	$T = 100$	0.00	0.97	0.00	0.98	0.00	0.98	0.00	1.00	0.00	1.00
$N = 50$	$T = 150$	0.00	0.99	0.00	0.98	0.00	0.98	0.00	1.00	0.00	1.00
$N = 50$	$T = 200$	0.00	0.98	0.00	0.98	0.00	0.98	0.00	1.00	0.00	0.98
$N = 100$	$T = 50$	0.00	1.00	0.00	1.00	0.00	0.00	0.01	0.99	0.00	1.00
$N = 100$	$T = 100$	0.00	0.98	0.00	0.97	0.00	0.97	0.00	1.00	0.00	1.00
$N = 100$	$T = 150$	0.00	0.99	0.00	0.99	0.00	0.99	0.00	1.00	0.00	1.00
$N = 100$	$T = 200$	0.00	0.99	0.00	0.99	0.00	0.99	0.00	1.00	0.00	1.00
$N = 150$	$T = 50$	0.00	0.99	0.00	1.00	0.00	0.00	0.00	1.00	0.00	1.00
$N = 150$	$T = 100$	0.00	0.98	0.00	0.99	0.00	0.99	0.00	1.00	0.00	1.00
$N = 150$	$T = 150$	0.00	0.99	0.00	1.00	0.00	0.99	0.00	1.00	0.00	1.00
$N = 150$	$T = 200$	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
$N = 200$	$T = 50$	0.00	0.99	0.00	1.00	0.00	0.00	0.00	1.00	0.00	1.00
$N = 200$	$T = 100$	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00	0.00	1.00
$N = 200$	$T = 150$	0.00	0.99	0.00	0.99	0.00	0.99	0.00	1.00	0.00	1.00
$N = 200$	$T = 200$	0.00	0.99	0.00	0.99	0.00	0.99	0.00	1.00	0.00	1.00

Table 9: $\hat{q}$, First Specification

		<i>DC</i>		<i>PC</i>		<i>IC</i>		<i>BN</i>		<i>AW</i>	
N	T	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$
$N = 50$	$T = 50$	0.50	0.37	0.01	0.96	0.04	0.02	0.02	0.49	0.00	0.00
$N = 50$	$T = 100$	0.11	0.44	0.00	0.93	0.00	0.00	0.00	0.39	0.00	0.00
$N = 50$	$T = 150$	0.01	0.18	0.00	0.88	0.00	0.00	0.00	0.21	0.00	0.00
$N = 50$	$T = 200$	0.00	0.09	0.00	0.92	0.00	0.00	0.00	0.20	0.00	0.00
$N = 100$	$T = 50$	0.19	0.76	0.00	1.00	0.00	0.00	0.01	0.83	0.00	0.00
$N = 100$	$T = 100$	0.00	0.92	0.00	0.99	0.00	0.54	0.00	0.51	0.00	0.00
$N = 100$	$T = 150$	0.00	0.89	0.00	1.00	0.00	0.02	0.00	0.44	0.00	0.00
$N = 100$	$T = 200$	0.00	0.93	0.00	1.00	0.00	0.00	0.00	0.55	0.00	0.00
$N = 150$	$T = 50$	0.08	0.89	0.00	1.00	0.00	0.00	0.01	0.92	0.00	0.00
$N = 150$	$T = 100$	0.00	0.97	0.00	0.99	0.00	0.94	0.00	0.96	0.00	0.00
$N = 150$	$T = 150$	0.00	0.97	0.00	1.00	0.00	0.78	0.00	0.83	0.00	0.00
$N = 150$	$T = 200$	0.00	1.00	0.00	1.00	0.00	0.20	0.00	0.91	0.00	0.00
$N = 200$	$T = 50$	0.02	0.96	0.00	1.00	0.00	0.00	0.04	0.95	0.00	0.00
$N = 200$	$T = 100$	0.00	0.98	0.00	1.00	0.00	1.00	0.00	1.00	0.00	0.00
$N = 200$	$T = 150$	0.00	0.99	0.00	0.99	0.00	0.99	0.00	1.00	0.00	0.00
$N = 200$	$T = 200$	0.00	0.99	0.00	0.99	0.00	0.97	0.00	1.00	0.00	0.00

Table 10: $\hat{q}$, Second Specification

		<i>DC</i>		<i>PC</i>		<i>IC</i>		<i>BN</i>		<i>AW</i>	
N	T	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$
$N = 50$	$T = 50$	0.22	0.34	0.00	0.50	0.00	0.00	0.01	0.48	0.00	0.00
$N = 50$	$T = 100$	0.02	0.16	0.00	0.47	0.00	0.00	0.00	0.25	0.00	0.00
$N = 50$	$T = 150$	0.00	0.03	0.00	0.48	0.00	0.00	0.00	0.26	0.00	0.00
$N = 50$	$T = 200$	0.00	0.05	0.00	0.32	0.00	0.00	0.00	0.23	0.00	0.00
$N = 100$	$T = 50$	0.10	0.50	0.00	0.89	0.00	0.00	0.01	0.77	0.00	0.00
$N = 100$	$T = 100$	0.00	0.59	0.00	0.99	0.00	0.45	0.00	0.31	0.00	0.00
$N = 100$	$T = 150$	0.00	0.53	0.00	1.00	0.00	0.03	0.00	0.37	0.00	0.00
$N = 100$	$T = 200$	0.00	0.48	0.00	1.00	0.00	0.03	0.00	0.43	0.00	0.00
$N = 150$	$T = 50$	0.03	0.67	0.00	0.90	0.00	0.00	0.04	0.90	0.00	0.00
$N = 150$	$T = 100$	0.00	0.78	0.00	0.99	0.00	0.78	0.00	0.93	0.00	0.00
$N = 150$	$T = 150$	0.00	0.83	0.00	1.00	0.00	0.78	0.00	0.74	0.00	0.00
$N = 150$	$T = 200$	0.00	0.83	0.00	1.00	0.00	0.27	0.00	0.77	0.00	0.00
$N = 200$	$T = 50$	0.02	0.80	0.00	0.88	0.00	0.00	0.06	0.92	0.00	0.00
$N = 200$	$T = 100$	0.00	0.93	0.00	1.00	0.00	0.94	0.00	1.00	0.00	0.00
$N = 200$	$T = 150$	0.00	0.97	0.00	1.00	0.00	0.97	0.00	0.99	0.00	0.00
$N = 200$	$T = 200$	0.00	0.99	0.00	1.00	0.00	0.96	0.00	0.97	0.00	0.00

Table 11: $\hat{q}$, Third Specification

		<i>DC</i>		<i>PC</i>		<i>IC</i>		<i>BN</i>		<i>AW</i>	
N	T	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$	$< q_0$	$= q_0$
$N = 50$	$T = 50$	0.17	0.46	0.00	0.73	0.00	0.02	0.14	0.60	0.00	0.00
$N = 50$	$T = 100$	0.00	0.24	0.00	0.87	0.00	0.01	0.00	0.36	0.00	0.00
$N = 50$	$T = 150$	0.00	0.08	0.00	0.74	0.00	0.00	0.00	0.32	0.00	0.00
$N = 50$	$T = 200$	0.00	0.08	0.00	0.57	0.00	0.00	0.00	0.28	0.00	0.00
$N = 100$	$T = 50$	0.07	0.61	0.00	1.00	0.00	0.00	0.46	0.43	0.00	0.00
$N = 100$	$T = 100$	0.00	0.70	0.00	1.00	0.00	0.47	0.00	0.49	0.00	0.00
$N = 100$	$T = 150$	0.00	0.73	0.00	1.00	0.00	0.02	0.00	0.48	0.00	0.00
$N = 100$	$T = 200$	0.00	0.73	0.00	0.98	0.00	0.00	0.00	0.68	0.00	0.00
$N = 150$	$T = 50$	0.03	0.83	0.00	1.00	0.00	0.00	0.65	0.32	0.00	0.00
$N = 150$	$T = 100$	0.00	0.97	0.00	1.00	0.00	0.94	0.00	0.92	0.00	0.00
$N = 150$	$T = 150$	0.00	0.97	0.00	1.00	0.00	0.83	0.00	0.85	0.00	0.00
$N = 150$	$T = 200$	0.00	0.97	0.00	1.00	0.00	0.22	0.00	0.95	0.00	0.00
$N = 200$	$T = 50$	0.00	0.92	0.00	1.00	0.00	0.00	0.51	0.43	0.00	0.00
$N = 200$	$T = 100$	0.00	0.96	0.00	1.00	0.00	0.99	0.00	0.98	0.00	0.00
$N = 200$	$T = 150$	0.00	0.96	0.00	0.99	0.00	0.99	0.00	1.00	0.00	0.00
$N = 200$	$T = 200$	0.00	0.99	0.00	1.00	0.00	0.94	0.00	0.98	0.00	0.00

Table 12: $\hat{q}$, Fourth Specification

		<i>DC</i>		<i>PC</i>		<i>IC</i>	
N	T	$< m_0$	$= m_0$	$< m_0$	$= m_0$	$< m_0$	$= m_0$
$N = 50$	$T = 50$	0.02	0.97	0.01	0.99	0.92	0.08
$N = 50$	$T = 100$	0.03	0.97	0.02	0.98	0.02	0.98
$N = 50$	$T = 150$	0.01	0.98	0.02	0.98	0.02	0.98
$N = 50$	$T = 200$	0.02	0.98	0.02	0.98	0.02	0.98
$N = 100$	$T = 50$	0.00	1.00	0.00	1.00	0.02	0.98
$N = 100$	$T = 100$	0.02	0.97	0.03	0.97	0.03	0.97
$N = 100$	$T = 150$	0.01	0.99	0.01	0.99	0.01	0.99
$N = 100$	$T = 200$	0.01	0.99	0.01	0.99	0.01	0.99
$N = 150$	$T = 50$	0.01	0.99	0.00	1.00	0.00	1.00
$N = 150$	$T = 100$	0.02	0.98	0.01	0.99	0.01	0.99
$N = 150$	$T = 150$	0.01	0.99	0.00	1.00	0.01	0.99
$N = 150$	$T = 200$	0.00	1.00	0.00	1.00	0.00	1.00
$N = 200$	$T = 50$	0.01	0.99	0.00	1.00	0.00	1.00
$N = 200$	$T = 100$	0.00	1.00	0.00	1.00	0.00	1.00
$N = 200$	$T = 150$	0.01	0.99	0.01	0.99	0.01	0.99
$N = 200$	$T = 200$	0.01	0.99	0.01	0.99	0.01	0.99

Table 13: $\hat{m}$, First Specification

		<i>DC</i>		<i>PC</i>		<i>IC</i>	
N	T	$< m_0$	$= m_0$	$< m_0$	$= m_0$	$< m_0$	$= m_0$
$N = 50$	$T = 50$	0.40	0.39	0.03	0.97	0.72	0.28
$N = 50$	$T = 100$	0.47	0.43	0.07	0.93	1.00	0.00
$N = 50$	$T = 150$	0.76	0.24	0.12	0.88	1.00	0.00
$N = 50$	$T = 200$	0.85	0.15	0.08	0.92	1.00	0.00
$N = 100$	$T = 50$	0.06	0.73	0.00	1.00	0.13	0.82
$N = 100$	$T = 100$	0.05	0.93	0.01	0.99	0.46	0.54
$N = 100$	$T = 150$	0.06	0.92	0.00	1.00	0.98	0.02
$N = 100$	$T = 200$	0.04	0.96	0.00	1.00	1.00	0.00
$N = 150$	$T = 50$	0.03	0.86	0.00	1.00	0.01	0.96
$N = 150$	$T = 100$	0.02	0.97	0.01	0.99	0.05	0.95
$N = 150$	$T = 150$	0.03	0.96	0.00	1.00	0.22	0.78
$N = 150$	$T = 200$	0.00	1.00	0.00	1.00	0.80	0.20
$N = 200$	$T = 50$	0.01	0.94	0.00	1.00	0.00	0.95
$N = 200$	$T = 100$	0.02	0.98	0.00	1.00	0.00	1.00
$N = 200$	$T = 150$	0.01	0.99	0.01	0.99	0.01	0.99
$N = 200$	$T = 200$	0.01	0.99	0.01	0.99	0.03	0.97

Table 14: $\hat{m}$, Second Specification

		<i>DC</i>		<i>PC</i>		<i>IC</i>	
N	T	$< m_0$	$= m_0$	$< m_0$	$= m_0$	$< m_0$	$= m_0$
$N = 50$	$T = 50$	0.63	0.27	0.50	0.50	0.79	0.21
$N = 50$	$T = 100$	0.86	0.12	0.53	0.47	1.00	0.00
$N = 50$	$T = 150$	0.93	0.07	0.52	0.48	1.00	0.00
$N = 50$	$T = 200$	0.91	0.09	0.68	0.32	1.00	0.00
$N = 100$	$T = 50$	0.39	0.44	0.11	0.89	0.46	0.54
$N = 100$	$T = 100$	0.34	0.63	0.01	0.99	0.55	0.45
$N = 100$	$T = 150$	0.39	0.61	0.00	1.00	0.97	0.03
$N = 100$	$T = 200$	0.45	0.55	0.00	1.00	0.97	0.03
$N = 150$	$T = 50$	0.26	0.67	0.10	0.90	0.25	0.72
$N = 150$	$T = 100$	0.16	0.83	0.01	0.99	0.22	0.78
$N = 150$	$T = 150$	0.12	0.88	0.00	1.00	0.22	0.78
$N = 150$	$T = 200$	0.13	0.87	0.00	1.00	0.73	0.27
$N = 200$	$T = 50$	0.17	0.79	0.12	0.88	0.13	0.84
$N = 200$	$T = 100$	0.07	0.93	0.00	1.00	0.06	0.94
$N = 200$	$T = 150$	0.03	0.97	0.00	1.00	0.03	0.97
$N = 200$	$T = 200$	0.01	0.99	0.00	1.00	0.04	0.96

Table 15: $\hat{m}$, Third Specification

		<i>DC</i>		<i>PC</i>		<i>IC</i>	
N	T	$< m_0$	$= m_0$	$< m_0$	$= m_0$	$< m_0$	$= m_0$
$N = 50$	$T = 50$	0.66	0.23	0.27	0.73	0.76	0.24
$N = 50$	$T = 100$	0.80	0.20	0.13	0.87	0.99	0.01
$N = 50$	$T = 150$	0.90	0.10	0.26	0.74	1.00	0.00
$N = 50$	$T = 200$	0.92	0.08	0.43	0.57	1.00	0.00
$N = 100$	$T = 50$	0.28	0.56	0.00	1.00	0.22	0.78
$N = 100$	$T = 100$	0.23	0.76	0.00	1.00	0.53	0.47
$N = 100$	$T = 150$	0.22	0.77	0.00	1.00	0.98	0.02
$N = 100$	$T = 200$	0.24	0.76	0.02	0.98	1.00	0.00
$N = 150$	$T = 50$	0.12	0.79	0.00	1.00	0.03	0.93
$N = 150$	$T = 100$	0.03	0.97	0.00	1.00	0.06	0.94
$N = 150$	$T = 150$	0.01	0.99	0.00	1.00	0.17	0.83
$N = 150$	$T = 200$	0.03	0.97	0.00	1.00	0.78	0.22
$N = 200$	$T = 50$	0.07	0.89	0.00	1.00	0.00	0.93
$N = 200$	$T = 100$	0.02	0.98	0.00	1.00	0.00	1.00
$N = 200$	$T = 150$	0.04	0.95	0.01	0.99	0.01	0.99
$N = 200$	$T = 200$	0.01	0.99	0.00	1.00	0.06	0.94

Table 16: $\hat{m}$, Fourth Specification

6 Empirical Illustration

We give two empirical illustrations of dynamic factor models. In Section 6.1, we estimate impulse responses of US macroeconomic variables to the identified monetary policy shock using a dynamic factor model. In Section 6.2, we estimate directional connectedness measures of COVID-19 positive cases as in Diebold and Yilmaz [2014]. We estimate the measure at both the state-level and county-level using a dynamic factor structure for regularization.

6.1 Structural Shock Identification using Dynamic Factor Model

Let (x_t) denote an N -dimensional time series of macroeconomic variables generated by the dynamic factor model

$$x_t = \sum_{k=0}^{m-1} \lambda_k f_{t-k} + \varepsilon_t,$$

where q -dimensional factor process (f_t) follows a VAR(p)

$$f_t = \sum_{l=1}^p A_l f_{t-l} + u_t, \quad u_t = f_t - \mathbb{E}[f_t \mid (f_s)_{s < t}],$$

with innovation covariance matrix $\Sigma_u = \mathbb{E}(u_t u_t')$. Let $\Sigma_u = QQ'$ be a Cholesky decomposition. Then, for any invertible $q \times q$ matrix H , the factor dynamics can be written as

$$f_t = \sum_{l=1}^p A_l f_{t-l} + QH e_t, \quad e_t = H^{-1} Q^{-1} u_t.$$

Substituting into the measurement equation yields the moving-average representation

$$x_t = \sum_{k=-m_1}^{m_2} \lambda_k L^k \left(I - \sum_{l=1}^p A_l L^l \right)^{-1} Q H e_t + \varepsilon_t,$$

where L is a lag operator so that the impulse response of x_t to the lagged structural shocks e_{t-h} is given by

$$\sum_{k=-m_1}^{m_2} \lambda_k L^k \left(I - \sum_{l=1}^p A_l L^l \right)^{-1} Q H.$$

The matrix H can be identified using standard identification schemes from the SVAR literature.

For comparison, we also estimate the model following as in Forni and Gambetti [2010]. In this approach, we first estimate the static representation

$$x_t = \Gamma g_t + \varepsilon_t,$$

and then fit a VAR($\tilde{p}$) to the static factors,

$$g_t = \sum_{l=1}^{\tilde{p}} \tilde{A}_l g_{t-l} + \tilde{u}_t, \quad \tilde{u}_t = g_t - \mathbb{E}[g_t \mid (g_s)_{s < t}].$$

Note that $\tilde{u}_t = (u'_t, 0, \dots, 0)'$ so that the innovation covariance matrix $\tilde{\Sigma}_u = \mathbb{E}(\tilde{u}_t \tilde{u}'_t)$ is r dimensional with rank q . Let R be $r \times q$ dimensional matrix satisfying $\tilde{\Sigma}_u = R R'$. As before, for any invertible $q \times q$ matrix H , the factor dynamics can be written as

$$g_t = \sum_{l=1}^{\tilde{p}} \tilde{A}_l g_{t-l} + R H e_t, \quad e_t = H^{-1} (R' R)^{-1} R' \tilde{u}_t.$$

Structural shocks e_t are again identified through restrictions on H .

We estimate the dynamic effects of monetary policy shocks using both approaches, where identification follows Eichenbaum and Evans [1995]. Industrial production and CPI are assumed not to respond contemporaneously to a monetary policy shock. All other variables, including exchange rates, are allowed to respond contemporaneously to innovations in the federal funds rate².

We use the FRED-MD data set of McCracken and Ng [2015] from March 1973 to November 2007. The sample excludes the fixed exchange rate regime and the global financial crisis, following Forni and Gambetti [2010]³. The sample contains $T = 417$ observations on $N = 124$ macroeconomic variables. Stationary transformations are kept minimal: prices enter in log differences, while interest rates are in levels.

The estimated dynamic structure (q_0, m_0) equals $(2, 3)$, $(3, 2)$, and $(3, 2)$ under IC_2 , DC_2 , and PC_2 , respectively. We set $(q, m) = (3, 2)$ for the subsequent analysis. VAR lag

²Additional zero restrictions, other than on contemporaneous response of industrial production and CPI, can be imposed. In that case, the restrictions become over-identifying and must be enforced during estimation. This is straightforward in our ALS framework. For example, in the factor loading update step, one may impose the restrictions via constrained least squares on $(\hat{f}_t)$.

³Results for the extended sample (March 1973–December 2019) are reported in the Appendix 8.6.

orders are selected by BIC for both the estimated dynamic factors ($\hat{f}_t$) and static factors ($\hat{g}_t$), yielding $(p, \tilde{p}) = (3, 2)$.

The fraction of variance explained by the model with $(q, m) = (3, 2)$ is 0.51. This exceeds the 0.41 obtained from a static factor model with the same number of factors, $(q, m) = (3, 1)$, but is below the 0.54 achieved by the static representation of the original model with $(q, m) = (6, 1)$. However, estimation of the $\text{VAR}(\tilde{p})$ for (g_t) requires a second-step dimension reduction of the residuals ($\tilde{u}_t$), reducing the dimension from $r = 6$ to $q = 3$. The explained variation in this second step is 0.89. Accounting for this loss, the effective explained variation of the model $(q, m) = (6, 1)$ is 0.48, which is lower than the explained variation 0.51 of the model $(q, m) = (3, 2)$.

Figure 1 reports impulse responses of log output, log inflation, the federal funds rate, and the log exchange rate. Confidence bands are constructed using a residual bootstrap at the 90% level. The point estimates are qualitatively similar across methods. However, confidence bands based on the dynamic factor model are narrower, indicating efficiency gains from direct estimation of dynamic factors.

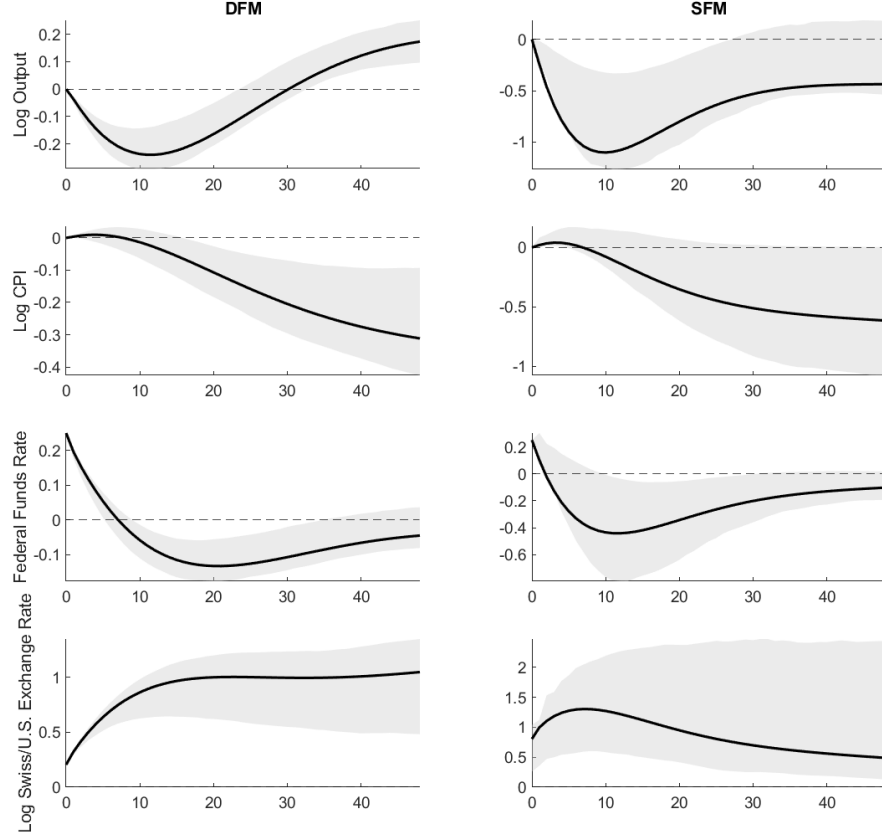


Figure 1: Estimated impulse responses of log output, log CPI, the federal funds rate, and the log exchange rate to a monetary policy shock, based on the dynamic factor model (left) and the static factor model (right).

6.2 Network Connectedness of the US State-Level COVID Cases

Let (x_t) denote an N -dimensional time series of weekly COVID positive cases across US states generated by the dynamic factor model

$$x_t = \sum_{k=0}^{m-1} \lambda_k f_{t-k} + \varepsilon_t,$$

where q -dimensional factor process (f_t) follows a $\text{VAR}(p)$ process. We additionally assume that (i) (f_t) and (ε_t) are jointly Gaussian, (ii) (f_t) and (ε_t) are uncorrelated at all leads and lags, and (iii) the idiosyncratic errors (ε_t) are serially and cross-sectionally uncorrelated, with possible heteroskedasticity. Under these conditions, all cross-sectional and serial correlations in (x_t) are explained by (f_t) .

Let $\Sigma_x(h) = \mathbb{E}[x_t x'_{t-h}]$, $\Sigma_f(h) = \mathbb{E}[f_t f'_{t-h}]$, and $\Sigma_\varepsilon = \mathbb{E}[\varepsilon_t \varepsilon'_t]$. The model-implied autocovariance matrices are

$$\Sigma_x(h) = \begin{cases} \sum_{k,j=-m_1}^{m_2} \lambda_k \Sigma_f(j-k) \lambda'_j + \Sigma_\varepsilon, & h = 0, \\ \sum_{k,j=-m_1}^{m_2} \lambda_k \Sigma_f(h+j-k) \lambda'_j, & h \neq 0. \end{cases}$$

After estimating the dynamic factor model, the diagonal elements of Σ_ε are estimated using the sample residuals ($\hat{\varepsilon}_t$), while off-diagonal elements are set to zero.⁴ Given the estimated factors ($\hat{f}_t$), we fit a VAR(p) to obtain $\hat{\Sigma}_f(h)$ and construct the corresponding model-implied autocovariances. Combined with the estimated factor loadings ($\hat{\lambda}_k$), this yields plug-in estimates of $\Sigma_x(h)$ for all lags. In this analysis, we focus on computing network connectedness measures defined by Diebold and Yilmaz [2014] based on estimated autocovariances. Let the generalized impulse response of state i at horizon h to a shock in state j be given as

$$s_{ij}^h = \mathbb{E}[x_{i,t+h} \mid e_{j,t} = \delta_j, \Omega_{t-1}] - \mathbb{E}[x_{i,t+h} \mid e_{j,t} = 0, \Omega_{t-1}],$$

where $\Omega_t = \text{span}\{x_s : s \leq t\}$ and $e_{j,t}$ is the j -th element of the projection residual $e_t = x_t - \mathbb{E}[x_t \mid \Omega_{t-1}]$. This coincides with the generalized impulse response of Pesaran and Shin [1998] when applied to a VAR model. Under our assumptions of linearity and Gaussianity, s_{ij}^h can be computed via the Durbin-Levinson or innovations algorithm using the model-implied autocovariances.⁵ The Gaussianity assumption can be discarded, at the cost of replacing the conditional expectation with linear projections.

Following Pesaran and Shin [1998], we define the h -step generalized forecast error variance decomposition as

$$d_{ij}^h = \frac{\sum_{l=0}^{h-1} (s_{ij}^l)^2}{\mathbb{E}[(x_{i,t+h} - \mathbb{E}[x_{i,t+h} \mid \Omega_t])^2]}.$$

Also, following Diebold and Yilmaz [2014], we normalize the variance decomposition to add up to one

$$\tilde{d}_{ij}^h = \frac{d_{ij}^h}{\sum_{j=1}^N d_{ij}^h} = \frac{\sum_{l=0}^{h-1} (s_{ij}^l)^2}{\sum_{j=1}^N \sum_{l=0}^{h-1} (s_{ij}^l)^2},$$

and define the directional connectedness from state j to i at lag h as $C_{i \leftarrow j}^h = \tilde{d}_{ij}^h$. The total directional connectedness from all other states to state i and from state j to all other states are defined as

$$C_{i \leftarrow \bullet}^h = \sum_{j \neq i} C_{i \leftarrow j}^h, \quad C_{\bullet \leftarrow j}^h = \sum_{i \neq j} C_{i \leftarrow j}^h,$$

and the net total directional connectedness is defined as $C_i^h = C_{\bullet \leftarrow i}^h - C_{i \leftarrow \bullet}^h$.

Since the dynamic factor structure allows us to estimate the autocovariance matrices $\Sigma_x(h)$ for all lags h , the connectedness measures can be computed in a straightforward manner. The estimation of large-dimensional connectedness measures has also been studied by Demiret et al. [2018], who estimate connectedness using high-dimensional VARs with

⁴More sophisticated estimators can be used, e.g., Fan et al. [2013].

⁵The projection coefficients $(\Phi_k)_{k=1}^\infty$ in $\mathbb{E}[x_t \mid \Omega_{t-1}] = \sum_{k=1}^\infty \Phi_k x_{t-k}$ are truncated at lag $T+h+1$, i.e., $\sum_{k=1}^{T+h+1} \Phi_k x_{t-k}$, where $T+h+1 = 179$ in our application.

LASSO-type regularization on the autoregressive coefficients. In contrast, the dynamic factor model achieves regularization by imposing a factor structure on both the autocovariance matrices $\Sigma_x(h)$ and the shock covariance matrix $\mathbb{E}[e_t e_t']$ in a unified way.

The estimated dynamic structure (q_0, m_0) is equal to $(6, 2)$ for all information criteria considered, namely IC_2 , DC_2 , and PC_2 . Accordingly, we set $(q, m) = (6, 2)$ for the subsequent analysis. The VAR lag order for the estimated factors $(\hat{f}_t)$ is selected using the Bayesian Information Criterion (BIC) and is estimated at one month, corresponding to $p = 4$.

With $(q, m) = (6, 2)$, the dynamic factor model explains 99.3% of the variation in U.S. state-level COVID-19 positive cases, indicating that it provides an excellent approximation to the underlying high-dimensional dynamics. The explained variation for each state is displayed in Figure 2.

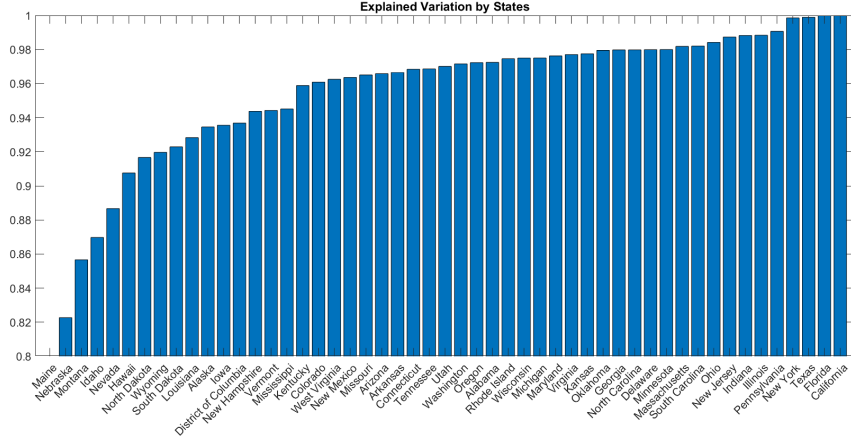


Figure 2: Explained variation of U.S. state-level COVID-19 positive cases under the dynamic factor model with $(q, m) = (6, 2)$.

For each state, we compute the directional connectedness measures to other states, $C_{\bullet \leftarrow i}^h$, and from other states, $C_{i \leftarrow \bullet}^h$. Figure 3 illustrates these measures for Illinois at the one-month horizon ($h = 4$).

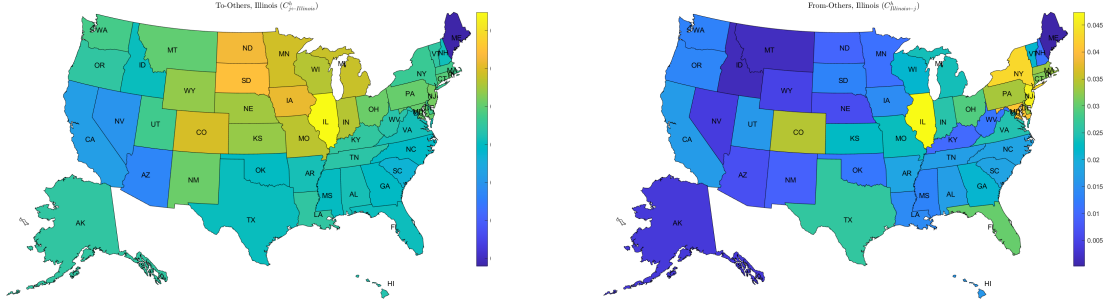


Figure 3: Connectedness from Illinois to other states (left) and from other states to Illinois (right) at the one-month horizon ($h = 4$).

Figure 4 presents the network representation of U.S. state-level COVID-19 connectedness estimated using the dynamic factor model with $(q, m) = (6, 2)$. Following Demirer et al. [2018], node locations are determined using the ForceAtlas2 algorithm of Jacomy et al. [2014], as implemented in Gephi. Node colors indicate net total directional connectedness, with dark green nodes representing net “exporters” of COVID-19 cases and dark brown nodes representing net “importers.” The node size corresponds to the total COVID-19 positive cases, scaled by state population.

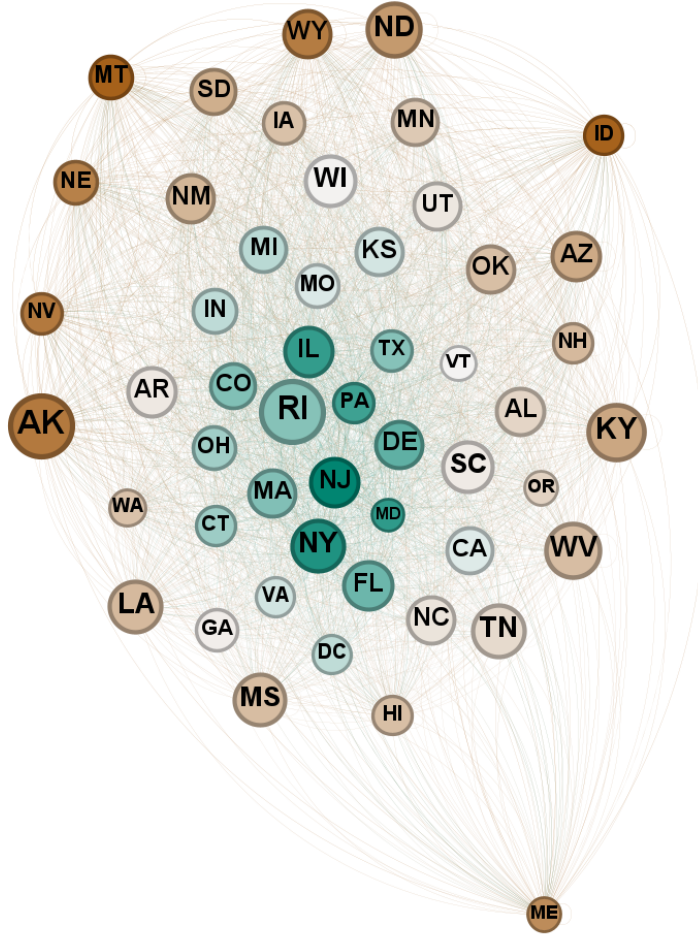


Figure 4: Network connectedness of U.S. state-level COVID-19 positive cases at the one-month horizon ($h = 4$).

The network connectedness framework can be applied at a much finer level of geographic disaggregation. We also estimate network connectedness measures at the county level, focusing on the 988 U.S. counties whose cumulative number of COVID-19 positive cases exceeds 1,000. The estimated dynamic structure (q_0, m_0) equals $(10, 3)$, $(9, 2)$, and $(8, 2)$ under IC_2 , DC_2 , and PC_2 , respectively. We set $(q, m) = (8, 2)$ for the subsequent analysis, accounting for 97.9% of the total variation. The BIC criterion selects $p = 2$. Below is a table of the county-level directional connectedness measures of COVID-19 positive cases, measured at horizons $h = 1$ and $h = 4$. We order the rows with the to-other measure at horizon $h = 1$ and show only 25 rows.

Table 17: County-Level Directional Connectedness Measures at Different Horizons, Ordered by to-others ($h = 1$)

County	To ($h = 1$)	From ($h = 1$)	To ($h = 4$)	From ($h = 4$)
Camden_NewJersey	2.022	0.997	1.827	0.997
Hampden_Massachusetts	1.991	0.997	1.683	0.998
Gloucester_NewJersey	1.953	0.997	1.775	0.997
Lehigh_Pennsylvania	1.947	0.997	1.734	0.998
Frederick_Maryland	1.939	0.997	1.797	0.997
Delaware_Pennsylvania	1.926	0.997	1.743	0.997
Franklin_Ohio	1.924	0.997	1.870	0.997
VirginiaBeach_Virginia	1.919	0.997	1.763	0.997
Hamilton_Indiana	1.912	0.997	1.831	0.997
Marion_Indiana	1.902	0.997	1.744	0.997
Bristol_Massachusetts	1.891	0.997	1.650	0.997
LosAngeles_California	1.887	0.997	1.588	0.997
Essex_Massachusetts	1.868	0.997	1.568	0.997
Putnam_NewYork	1.852	0.997	1.554	0.997
Dutchess_NewYork	1.851	0.997	1.466	0.997
Hamilton_Ohio	1.846	0.997	1.787	0.997
Worcester_Massachusetts	1.843	0.997	1.574	0.997
Burlington_NewJersey	1.841	0.997	1.648	0.997
Fairfield_Ohio	1.836	0.997	1.840	0.997
Mecklenburg_NorthCarolina	1.827	0.997	1.614	0.998
Montgomery_Pennsylvania	1.827	0.997	1.619	0.997
Plymouth_Massachusetts	1.818	0.997	1.527	0.997
St_Clair_Illinois	1.799	0.996	1.899	0.997
SanDiego_California	1.799	0.997	1.539	0.998
Delaware_Ohio	1.789	0.996	1.707	0.997

7 Conclusion

This paper proposes two procedures for jointly selecting the number of dynamic factors and the filter length in dynamic factor models. Consistent estimation is achieved by exploiting the consistency of the estimated dynamic factors. Our approach differs from Bai and Ng [2007] and Amengual and Watson [2007] in that we directly estimate the dynamic factors and, in particular, do not rely on VAR representations of static factors with low-rank innovations.

We illustrate the practical usefulness of the proposed framework through two empirical applications. First, we estimate impulse responses to monetary policy shocks identified using a large-dimensional system. Second, we construct directional connectedness measures for U.S. state-level and county-level COVID-19 positive cases.

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8 Appendix

8.1 Lemmas related to Section 3.1

Lemma 1. *Let Assumption 1 hold. Then, we have*

$$\left\| \left(\sum_{j,k=-m_1}^{m_2} (P_{j-k} \otimes \frac{1}{N} \lambda'_j \lambda_k) \right)^{-1} \right\| = O_p(1)$$

Proof. It suffices to show that the smallest eigenvalue of the matrix S defined as

$$S = \sum_{j,k=-m_1}^{m_2} \left(P_{j-k} \otimes \frac{1}{N} \lambda'_j \lambda_k \right)$$

is bounded away from zero. Defining the lag- h block as

$$B_h = \frac{1}{N} \sum_{k=-m_1}^{m_2} \lambda'_{k+h} \lambda_k, \quad h = j - k,$$

we may write

$$S = \sum_{h=-m_1-m_2}^{m_1+m_2} (P_h \otimes B_h)$$

with the convention that $\lambda_k = 0$ for $k < -m_1$ and $k > m_2$. Let $a = (a'_{1-m_2}, \dots, a'_{T+m_1})'$ be $(T + m_1 + m_2)q$ dimensional vector with q dimensional blocks given by (a_t) . Consider the t -th block of the product Sa . For a single Kronecker term, we have

$$((P_h \otimes B_h)a)_t = \sum_{s=1-m_2}^{T+m_1} (P_h)_{t,s} B_h a_s.$$

Here, P_h is the cyclic shift by $-h$, meaning $(P_h)_{t,s} = 1$ if and only if $s = t + h$ modulo $T + m_1 + m_2$. Therefore,

$$((P_h \otimes B_h)a)_t = B_h a_{t+h}$$

Here and after, the time indices on (a_t) are defined with modulo $T + m_1 + m_2$. Summing over h gives

$$(Sa)_t = \sum_{h=-m_1-m_2}^{m_1+m_2} B_h a_{t+h}$$

This is a convolution in the time index t , where each output block at time t is given by a convolution of the sequence (B_{-h}) with the sequence (a_t) .

We may define the blockwise discrete Fourier transform for frequencies $\omega_l = 2\pi l / (T + m_1 + m_2)$, $l = 1 - m_2, \dots, T + m_1$,

$$a(\omega_l) = \sum_{t=1-m_2}^{T+m_1} a_t e^{-i\omega_l t},$$

which give a q -vector for each $l = 1 - m_2, \dots, T + m_1$. Applying the same transform to the output sequence $b_t = (Sa)_t$, we obtain

$$b(\omega_l) = \sum_{t=1-m_2}^{T+m_1} b_t e^{-i\omega_l t} = \sum_{t=1-m_2}^{T+m_1} \left(\sum_{h=-m_1-m_2}^{m_1+m_2} B_h a_{t+h} \right) e^{-i\omega_l t}.$$

Changing summation order and the time index $s = t + h$, we obtain

$$\begin{aligned} b(\omega_l) &= \sum_{h=-m_1-m_2}^{m_1+m_2} B_h \sum_{s=1-m_2}^{T+m_1} a_s e^{-i\omega_l (s-h)} \\ &= \sum_{h=-m_1-m_2}^{m_1+m_2} B_h e^{i\omega_l h} \sum_{s=1-m_2}^{T+m_1} a_s e^{-i\omega_l s} \\ &= \left(\sum_{h=-m_1-m_2}^{m_1+m_2} B_h e^{i\omega_l h} \right) a(\omega_l) \end{aligned}$$

Thus,

$$b(\omega_l) = B(e^{i\omega_l})a(\omega_l)$$

with

$$B(e^{i\omega}) = \sum_{h=-m_1-m_2}^{m_1+m_2} B_h e^{i\omega h}$$

This implies that the DFT diagonalizes the matrix multiplications where the action of S is reduced to frequency-by-frequency multiplication by a $q \times q$ matrix $B(e^{i\omega})$. Let a phase vector $v(\omega)$ be defined as

$$v(\omega) := (e^{-i\omega(-m_1)}, \dots, e^{-i\omega m_2})' \in \mathbb{C}^m.$$

Note that we have

$$\begin{aligned} B(e^{i\omega}) &= \sum_{h=-m_1-m_2}^{m_1+m_2} B_h e^{i\omega h} \\ &= \frac{1}{N} \sum_{h=-m_1-m_2}^{m_1+m_2} \sum_{k=-m_1}^{m_2} \lambda'_{k+h} \lambda_k e^{i\omega h} \\ &= \frac{1}{N} \sum_{j,k=-m_1}^{m_2} \lambda'_j \lambda_k e^{i\omega(j-k)} \\ &= (v(\omega) \otimes I_q)' \left(\frac{1}{N} \Gamma' \Gamma \right) (v(\omega) \otimes I_q) \end{aligned}$$

By Assumption 1,

$$\frac{1}{N} \Gamma' \Gamma \rightarrow_p M_\gamma > 0$$

Let the smallest eigenvalue of M_γ be given by $c_0 > 0$. For every ω , we have

$$B(e^{i\omega}) = (v(\omega) \otimes I_q)' \left(\frac{1}{N} \Gamma' \Gamma \right) (v(\omega) \otimes I_q) \geq c_0 (v(\omega) \otimes I_q)' (v(\omega) \otimes I_q).$$

for N, T large, where $E \geq F$ is used to denote $E - F$ is positive semi-definite. Since each entry of $v(\omega)$ has modulus 1 and there are m entries,

$$(v(\omega) \otimes I_q)' (v(\omega) \otimes I_q) = m I_q.$$

Therefore,

$$B(e^{i\omega}) \geq c_0 m I_q > 0 \quad \text{for all } \omega.$$

Finally, because the DFT is unitary, the smallest eigenvalue of S is bounded below by $c_0 m$.

To see this, note that

$$\begin{aligned}
\|Sa\|^2 &= \sum_{l=1-m_2}^{T+m_1} \|b(\omega_l)\|^2 \\
&= \sum_{l=1-m_2}^{T+m_1} \|B(e^{i\omega_l})a(\omega_l)\|^2 \\
&\geq c_0 m \sum_{l=1-m_2}^{T+m_1} \|a(\omega_l)\|^2 \\
&= c_0 m \|a\|^2.
\end{aligned}$$

□

8.2 Lemmas related to Section 4

Lemma 2. Let $\hat{X}_{q,m}$ be defined as

$$\hat{X}_{q,m} = \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)}$$

where $(\hat{F}_{-k}^{(q)}, \hat{\lambda}_k^{(q)})_{k=1}^m$ are a solution to the best rank q approximation problem in the frobenius norm. We have

$$\frac{1}{\sqrt{N+T}} \left\| \hat{X}_{q,m} - \sum_{k=1}^{m_0} F_{-k} \lambda'_{-k} \right\| = O_p(1)$$

for the three cases

$$\begin{cases} q = \lceil \frac{q_0 m_0}{m} \rceil, \dots, q_{max} & \text{if } 1 \leq m < m_0 \\ q = q_0, \dots, q_{max} & \text{if } m_0 \leq m \leq m_{max} \\ m = \lceil \frac{q_0 m_0}{q} \rceil, \dots, m_{max} & \text{if } q_0 \leq q \leq q_{max} \end{cases}$$

Proof. Here, we denote G_0 and Γ_0 as the true values, e.g., $X = G_0 \Gamma'_0 + E$, to distinguish them from the argument of the objective function. Note that we have

$$\begin{aligned}
\frac{1}{\sqrt{N+T}} \left\| \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)} - \sum_{k=1}^{m_0} F_{-k} \lambda'_{-k} \right\| &= \frac{1}{\sqrt{N+T}} \left\| \hat{G} \hat{\Gamma}' - G_0 \Gamma'_0 \right\| \\
&= \frac{1}{\sqrt{N+T}} \left\| \hat{G} (\hat{G}' \hat{G})^{-1} \hat{G}' X - G_0 \Gamma'_0 \right\| \\
&= \frac{1}{\sqrt{N+T}} \left\| \hat{G} (\hat{G}' \hat{G})^{-1} \hat{G}' G_0 \Gamma'_0 - G_0 \Gamma'_0 + \hat{G} (\hat{G}' \hat{G})^{-1} \hat{G}' E \right\| \\
&\leq \frac{1}{\sqrt{N+T}} \left\| (I_T - \hat{G} (\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \Gamma'_0 \right\| + \frac{1}{\sqrt{N+T}} \left\| \hat{G} (\hat{G}' \hat{G})^{-1} \hat{G}' E \right\|
\end{aligned}$$

Here, we have

$$\frac{1}{\sqrt{N+T}} \left\| \hat{G} (\hat{G}' \hat{G})^{-1} \hat{G}' E \right\| \leq \frac{1}{\sqrt{N+T}} \left\| \hat{G} (\hat{G}' \hat{G})^{-1} \hat{G}' \right\| \|E\| = O_p(1)$$

and

$$\frac{1}{\sqrt{N+T}} \left\| (I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0\Gamma'_0 \right\| \leq \left(\frac{1}{\sqrt{T}} \left\| (I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0 \right\| \right) \left(\frac{1}{\sqrt{N}} \left\| \Gamma'_0 \right\| \right) \left(\frac{\sqrt{NT}}{\sqrt{N+T}} \right)$$

so that it suffices to show

$$\frac{1}{\sqrt{T}} \left\| (I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0 \right\| = O_p \left(\frac{\sqrt{N+T}}{\sqrt{NT}} \right)$$

Consider the concentrated least squares problem

$$\begin{aligned} \min_{(G,\Gamma)} \frac{1}{NT} \left\| X - G\Gamma' \right\|^2 &= \min_G \frac{1}{NT} \left\| X - G(G'G)^{-1}G'X \right\|^2 \\ &= \min_G \frac{1}{NT} \left\| (I_T - G(G'G)^{-1}G')X \right\|^2 \end{aligned}$$

Note that $\hat{G}$ is the minimizer of the concentrated objective function. We have

$$\frac{1}{NT} \left\| (I_T - G_0(G'_0G_0)^{-1}G'_0)X \right\|^2 \geq \frac{1}{NT} \left\| (I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')X \right\|^2$$

for the three cases

$$\begin{cases} q = \left\lceil \frac{q_0 m_0}{m} \right\rceil, \dots, q_{max} & \text{if } 1 \leq m < m_0 \\ q = q_0, \dots, q_{max} & \text{if } m_0 \leq m \leq m_{max} \\ m = \left\lceil \frac{q_0 m_0}{q} \right\rceil, \dots, m_{max} & \text{if } q_0 \leq q \leq q_{max} \end{cases}$$

This is because on the right-hand side, we minimize the objective function over a larger space. The second case is trivial. Note that for the first and second cases, we have both $m_0 q_0 \leq m q$ and $m_0 \geq m$. The rest of the proof follows from the proof of Proposition 1.

The objective function can be written as

$$\begin{aligned} \frac{1}{NT} \left\| (I_T - G(G'G)^{-1}G')X \right\|^2 &= \frac{1}{NT} \left\| (I_T - G(G'G)^{-1}G')(G_0\Gamma'_0 + E) \right\|^2 \\ &= \frac{1}{NT} \left\| (I_T - G(G'G)^{-1}G')G_0\Gamma'_0 \right\|^2 \\ &\quad + \frac{1}{NT} \left\| (I_T - G(G'G)^{-1}G')E \right\|^2 \\ &\quad + \frac{2}{NT} \text{tr} \left((I_T - G(G'G)^{-1}G')G_0\Gamma'_0 E' \right) \end{aligned}$$

so that we have

$$\begin{aligned}
0 &\leq \frac{1}{NT} \left\| (I_T - G_0(G'_0 G_0)^{-1} G'_0) X \right\|^2 - \frac{1}{NT} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') X \right\|^2 \\
&= \frac{1}{NT} \left\| (I_T - G_0(G'_0 G_0)^{-1} G'_0) G_0 \Gamma'_0 \right\|^2 - \frac{1}{NT} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \Gamma'_0 \right\|^2 \\
&+ \frac{1}{NT} \left\| (I_T - G_0(G'_0 G_0)^{-1} G'_0) E \right\|^2 - \frac{1}{NT} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') E \right\|^2 \\
&+ \frac{2}{NT} \text{tr} \left((I_T - G_0(G'_0 G_0)^{-1} G'_0) G_0 \Gamma'_0 E' \right) - \frac{2}{NT} \text{tr} \left((I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \Gamma'_0 E' \right) \quad (8) \\
&= -\frac{1}{NT} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \Gamma'_0 \right\|^2 \\
&+ \frac{1}{NT} \left\| (I_T - G_0(G'_0 G_0)^{-1} G'_0) E \right\|^2 - \frac{1}{NT} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') E \right\|^2 \\
&- \frac{2}{NT} \text{tr} \left((I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \Gamma'_0 E' \right)
\end{aligned}$$

Using the same logic as in the proof of Proposition 1,

$$\begin{aligned}
0 &\leq -\lambda_{\min} \left(\frac{1}{N} \Gamma'_0 \Gamma_0 \right) \text{tr} \left(\left(\frac{1}{T} G'_0 (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \right) \right) \\
&- \frac{2}{NT} \text{tr} \left((I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \Gamma'_0 E' \right) + O_p \left(\frac{1}{N} + \frac{1}{T} \right)
\end{aligned}$$

Here,

$$\begin{aligned}
\frac{2}{NT} \text{tr} \left((I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \Gamma'_0 E' \right) &\leq \frac{2}{NT} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \right\| \left\| \Gamma'_0 \right\| \|E\| \\
&= O_p \left(\sqrt{\frac{N+T}{NT}} \right) \frac{1}{\sqrt{T}} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \right\|
\end{aligned}$$

Also, we have

$$\text{tr} \left(\frac{1}{T} G'_0 (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \right) = \frac{1}{T} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \right\|^2$$

so that

$$\begin{aligned}
0 &\leq -\lambda_{\min} \left(\frac{1}{N} \Gamma'_0 \Gamma_0 \right) \left(\frac{1}{\sqrt{T}} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \right\| \right)^2 \\
&+ O_p \left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}} \right) \frac{1}{\sqrt{T}} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \right\| + O_p \left(\frac{1}{N} + \frac{1}{T} \right)
\end{aligned}$$

This implies

$$\frac{1}{\sqrt{T}} (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 = O_p \left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}} \right)$$

□

Lemma 3. Let $\hat{X}_{q,m}$ be defined as in Lemma 2. Then, we have

$$\|X - \hat{X}_{q,m}\| \geq \delta_{q_0+q+1, \max(m_0, m)}(E)$$

for any q and m .

Proof. Note that we have

$$\delta_{q,m}(E) \leq \left\| E - \sum_{k=1}^m F_{-k}^{(q-1)} \lambda_k^{(q-1)} \right\|$$

for any matrices $(F_{-k}^{(q-1)}, \lambda_k^{(q-1)})$ of conformable dimensions with $q-1$ number of columns. Thus, we have

$$\begin{aligned} \|X - \hat{X}_{q,m}\| &= \left\| X - \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)} \right\| \\ &= \left\| E + \sum_{k=1}^{m_0} F_{-k} \lambda_k - \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)} \right\| \\ &\geq \delta_{q_0+q+1, \max(m, m_0)}(E), \end{aligned}$$

as desired. $\square$

Lemma 4. Suppose Assumption 2 hold. Let $\hat{X}_{q,m}$ be defined as in Lemma 2. We have

$$\frac{1}{\sqrt{N+T}} \|X - \hat{X}_{q,m}\| = O_p(1)$$

for the three cases

$$\begin{cases} q = \lceil \frac{q_0 m_0}{m} \rceil, \dots, q_{max} & \text{if } 1 \leq m < m_0 \\ q = q_0, \dots, q_{max} & \text{if } m_0 \leq m \leq m_{max} \\ m = \lceil \frac{q_0 m_0}{q} \rceil, \dots, m_{max} & \text{if } q_0 \leq q \leq q_{max} \end{cases}$$

Proof. We have

$$\begin{aligned} \frac{1}{\sqrt{N+T}} \|X - \hat{X}_{q,m}\| &= \frac{1}{\sqrt{N+T}} \left\| E + \sum_{k=1}^{m_0} F_{-k} \lambda'_{-k} - \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)} \right\| \\ &\leq \frac{1}{\sqrt{N+T}} \left(\|E\| + \left\| \sum_{k=1}^{m_0} F_{-k} \lambda'_{-k} - \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)} \right\| \right) \end{aligned}$$

Due to Assumption 2, it suffices to show

$$\frac{1}{\sqrt{N+T}} \left\| \sum_{k=1}^{m_0} F_{-k} \lambda'_{-k} - \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)} \right\| = O_p(1),$$

which follows from Lemma 2. $\square$

Lemma 5. Suppose Assumption 2 and Assumption 3 hold with $(q_{max} + q_0 + 1)m_{max} \leq k_{max}$. Let $\hat{X}_{q,m}$ be defined as in Lemma 2. Suppose one of the three cases holds:

$$\begin{cases} q = \lceil \frac{q_0 m_0}{m} \rceil, \dots, q_{max} & \text{if } 1 \leq m < m_0 \\ q = q_0, \dots, q_{max} & \text{if } m_0 \leq m \leq m_{max} \\ m = \lceil \frac{q_0 m_0}{q} \rceil, \dots, m_{max} & \text{if } q_0 \leq q \leq q_{max} \end{cases}$$

Then, for all $\epsilon > 0$, there exists $c_0, c_1 > 0$ such that

$$\mathbb{P} \left\{ c_0 \leq \frac{1}{\sqrt{N+T}} \|X - \hat{X}_{q,m}\| \leq c_1 \right\} \geq 1 - \epsilon$$

for all $N + T$ large.

Proof. Using Lemma 4, we have $\|X - \hat{X}_{q,m}\| = O_p(\sqrt{N+T})$. Thus, for all $\epsilon > 0$, there exists $c_1 > 0$ such that

$$\mathbb{P} \left\{ \frac{1}{\sqrt{N+T}} \|X - \hat{X}_{q,m}\| > c_1 \right\} \leq \epsilon/2$$

for all $N + T$ large. Now, using Lemma 3 and Assumption 3, for all $\epsilon > 0$, there exists $c_0 > 0$ such that

$$\mathbb{P} \left\{ c_0 > \frac{1}{\sqrt{N+T}} \|X - \hat{X}_{q,m}\| \right\} \leq \epsilon/2$$

for all q, m such that $(q_0 + q + 1) \max(m_0, m) \leq k_{max}$ and for all $N + T$ large. This implies

$$\mathbb{P} \left\{ c_0 \leq \frac{1}{\sqrt{N+T}} \|X - \hat{X}_{q,m}\| \leq c_1 \right\} \geq 1 - \epsilon,$$

because $k_{max} \geq (q_{max} + q_0 + 1)m_{max}$. □

Lemma 6. Suppose Assumption 1, Assumption 2, and 4 hold. Let $\hat{X}_{q,m}$ be defined as in Lemma 2. Suppose one of the four cases holds:

$$\begin{cases} q = 0, \dots, \lceil \frac{q_0 m_0}{m} \rceil - 1 & \text{if } 1 \leq m < m_0 \\ q = 0, \dots, q_0 - 1 & \text{if } m_0 \leq m \leq m_{max} \\ m = 1, \dots, m_{max} & \text{if } 0 \leq q < q_0 \\ m = 1, \dots, \lceil \frac{q_0 m_0}{q} \rceil - 1 & \text{if } 1 \leq q_0 \leq q \leq q_{max} \end{cases}$$

Then, for all $\epsilon > 0$, there exists $c_2, c_3 > 0$ such that

$$\mathbb{P} \left\{ c_2 \leq \frac{1}{\sqrt{NT}} \|X - \hat{X}_{q,m}\| \leq c_3 \right\} \geq 1 - \epsilon$$

for all NT large with $\frac{N+T}{NT} \rightarrow 0$.

Proof. Note that

$$\begin{aligned}
\frac{1}{\sqrt{NT}} \|X - \hat{X}_{q,m}\| &= \frac{1}{\sqrt{NT}} \left\| E + \sum_{k=1}^{m_0} F_{-k} \lambda'_k - \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)'} \right\| \\
&\leq \frac{1}{\sqrt{NT}} \left(\|E\| + \left\| \sum_{k=1}^{m_0} F_{-k} \lambda'_k \right\| + \left\| \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)'} \right\| \right) \\
&= O_p(1)
\end{aligned}$$

due to Assumption 1. This implies that for all $\epsilon > 0$, there exists $c_3 > 0$ such that

$$\mathbb{P} \left\{ \frac{1}{\sqrt{NT}} \|X - \hat{X}_{q,m}\| > c_3 \right\} \leq \epsilon/2$$

for all NT large. Also, note that we have

$$\begin{aligned}
\frac{1}{\sqrt{NT}} \|X - \hat{X}_{q,m}\| &= \frac{1}{\sqrt{NT}} \left\| X - \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)'} \right\| \\
&\geq \frac{1}{\sqrt{NT}} \left\| \sum_{k=1}^{m_0} F_{-k} \lambda'_k - \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)'} \right\| - \frac{1}{\sqrt{NT}} \|E\| \\
&\geq \frac{1}{\sqrt{NT}} \delta_{q+1,m} \left(\sum_{k=1}^{m_0} F_{-k} \lambda'_k \right) + O_p \left(\sqrt{\frac{N+T}{NT}} \right)
\end{aligned}$$

Here, we've used reverse triangle inequality on the first inequality. The second inequality follows from the definition of $\delta_{q+1,m}(\cdot)$ and Assumption 2. Suppose $\frac{N+T}{NT} \rightarrow 0$ so that $O_p \left(\sqrt{\frac{N+T}{NT}} \right) = o_p(1)$. From Assumption 4, for all $\epsilon > 0$, we have

$$\mathbb{P} \left\{ c_2 > \frac{1}{\sqrt{NT}} \|X - \hat{X}_{q,m}\| \right\} \leq \epsilon/2$$

for all $q = 1, \dots, q_0 - 1, m \leq m_{max}$ for all NT large with $\frac{N+T}{NT} \rightarrow 0$. Thus, we have

$$\mathbb{P} \left\{ c_2 \leq \frac{1}{\sqrt{NT}} \|X - \hat{X}_{q,m}\| \leq c_3 \right\} \geq 1 - \epsilon$$

for all $q = 0, \dots, q_0 - 1, m \leq m_{max}$ and for all NT large with $\frac{N+T}{NT} \rightarrow 0$. This proves the result for the second and third cases. For other cases, note that

$$\frac{1}{\sqrt{NT}} \|X - \hat{X}_{q,m}\| \geq \frac{1}{\sqrt{NT}} \delta_{qm+1,1}(G\Gamma') + O_p \left(\sqrt{\frac{N+T}{NT}} \right).$$

It suffices to show that we have $qm+1 \leq q_0 m_0$ because we have $\frac{1}{\sqrt{NT}} \delta_{q_0 m_0,1}(G\Gamma') > c \geq 0$ in the limit due to Assumption 1. Consider the first case: $q = 0, \dots, \lceil \frac{q_0 m_0}{m} \rceil - 1, 1 \leq m < m_0$. Note that

$$qm + 1 \leq m \left(\left\lceil \frac{q_0 m_0}{m} \right\rceil - 1 \right) + 1$$

Here, if $\lceil \frac{q_0 m_0}{m} \rceil$ is an integer, we have

$$m \left(\left\lceil \frac{q_0 m_0}{m} \right\rceil - 1 \right) + 1 = q_0 m_0 - m + 1 \leq q_0 m_0$$

for $m \geq 1$. If $\lceil \frac{q_0 m_0}{m} \rceil$ is not an integer, we have

$$m \left(\left\lceil \frac{q_0 m_0}{m} \right\rceil - 1 \right) + 1 = m \left\lfloor \frac{q_0 m_0}{m} \right\rfloor + 1 < q_0 m_0 + 1$$

so that we have $m \left(\left\lceil \frac{q_0 m_0}{m} \right\rceil - 1 \right) + 1 \leq q_0 m_0$. The same can be shown in a similar way with the fourth case. $\square$

Lemma 7. *We have*

$$V_{q,m}(X) - V_{q_0,m_0}(X) = O_p \left(\frac{N+T}{NT} \right)$$

for the three cases

$$\begin{cases} q = \lceil \frac{q_0 m_0}{m} \rceil, \dots, q_{max} & \text{if } 1 \leq m < m_0 \\ q = q_0, \dots, q_{max} & \text{if } m_0 \leq m \leq m_{max} \\ m = \left\lceil \frac{q_0 m_0}{q} \right\rceil, \dots, m_{max} & \text{if } q_0 \leq q \leq q_{max} \end{cases}$$

Also, for all $\epsilon > 0$, there exists $c > 0$ such that

$$\mathbb{P} \{ V_{q,m}(X) - V_{q_0,m_0}(X) \geq c \} \geq 1 - \epsilon$$

for all N, T large with $\frac{N+T}{NT} \rightarrow 0$ for the four cases

$$\begin{cases} q = 0, \dots, \left\lceil \frac{q_0 m_0}{m} \right\rceil - 1 & \text{if } 1 \leq m < m_0 \\ q = 0, \dots, q_0 - 1 & \text{if } m_0 \leq m \leq m_{max} \\ m = 1, \dots, m_{max} & \text{if } 0 \leq q < q_0 \\ m = 1, \dots, \left\lceil \frac{q_0 m_0}{q} \right\rceil - 1 & \text{if } 1 \leq q_0 \leq q \leq q_{max} \end{cases}$$

Proof. Suppose we have either of the three cases

$$\begin{cases} q = \lceil \frac{q_0 m_0}{m} \rceil, \dots, q_{max} & \text{if } 1 \leq m < m_0 \\ q = q_0, \dots, q_{max} & \text{if } m_0 \leq m \leq m_{max} \\ m = \left\lceil \frac{q_0 m_0}{q} \right\rceil, \dots, m_{max} & \text{if } q_0 \leq q \leq q_{max} \end{cases}$$

In the proof of Lemma 2, we have already seen that

$$V_{q,m}(X) - \frac{1}{NT} \left\| (I_T - G(G'G)^{-1}G')X \right\|^2 = O_p \left(\frac{N+T}{NT} \right)$$

for the three cases. Thus, it suffices to show

$$\frac{1}{NT} \left\| (I_T - G(G'G)^{-1}G')X \right\|^2 - V_{q_0,m_0}(X) = O_p \left(\frac{N+T}{NT} \right)$$

Note that

$$\begin{aligned}
\frac{1}{NT} \left\| (I_T - G(G'G)^{-1}G')X \right\|^2 - V_{q_0, m_0}(X) &= \frac{1}{NT} \left\| (I_T - G(G'G)^{-1}G')X \right\|^2 - \frac{1}{NT} \|E\|^2 \\
&= \frac{1}{NT} \text{tr} \left(E'(I_T - G(G'G)^{-1}G')E \right) - \frac{1}{NT} \|E\|^2 \\
&= -\frac{1}{NT} \text{tr} \left(E'G(G'G)^{-1}G'E \right) \\
&= -\frac{1}{NT} \left\| (G'G)^{-1/2}G'E \right\|^2 \\
&\leq \frac{1}{NT} \left\| (G'G)^{-1/2}G' \right\|^2 \|E\|^2 \\
&= O_p \left(\frac{N+T}{NT} \right)
\end{aligned}$$

Suppose we have either of the four cases

$$\begin{cases} q = 0, \dots, \left\lceil \frac{q_0 m_0}{m} \right\rceil - 1 & \text{if } 1 \leq m < m_0 \\ q = 0, \dots, q_0 - 1 & \text{if } m_0 \leq m \leq m_{max} \\ m = 1, \dots, m_{max} & \text{if } 0 \leq q < q_0 \\ m = 1, \dots, \left\lceil \frac{q_0 m_0}{q} \right\rceil - 1 & \text{if } 1 \leq q_0 \leq q \leq q_{max} \end{cases}$$

Note that we have

$$\begin{aligned}
V_{q, m}(X) - V_{q_0, m_0}(X) &= \frac{1}{NT} \left\| X - \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)'} \right\|^2 - \frac{1}{NT} \|E\|^2 \\
&= \frac{1}{NT} \left\| G\Gamma' - \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)'} \right\|^2 + \frac{2}{NT} \text{tr} \left(E' (G\Gamma' - \hat{G}\hat{\Gamma}') \right)
\end{aligned}$$

Here,

$$\begin{aligned}
\frac{1}{NT} \text{tr} \left(E' (G\Gamma' - \hat{G}\hat{\Gamma}') \right) &\leq \frac{1}{NT} \text{tr} (E'G\Gamma') + \frac{1}{NT} \text{tr} (E'\hat{G}\hat{\Gamma}') \\
&\leq \frac{1}{NT} \|E\| (\|G\Gamma'\| + \|\hat{G}\hat{\Gamma}'\|) \\
&= O_p \left(\sqrt{\frac{N+T}{NT}} \right)
\end{aligned}$$

This implies

$$\begin{aligned}
V_{q, m}(X) - V_{q_0, m_0}(X) &= \frac{1}{NT} \left\| G\Gamma' - \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)'} \right\|^2 + O_p \left(\sqrt{\frac{N+T}{NT}} \right) \\
&\geq \frac{1}{NT} \left\| G\Gamma' - \sum_{k=1}^m \hat{F}_{-k}^{(q)} \hat{\lambda}_k^{(q)'} \right\|^2 + O_p \left(\sqrt{\frac{N+T}{NT}} \right) \\
&\geq \frac{1}{NT} \delta_{mq+1, 1}
\end{aligned}$$

□

8.3 Derivation of FOC

The first order conditions for (λ_k) are trivial. The first order conditions for f_t are given by

$$\sum_{j=\max(-m_1, 1-t)}^{\min(m_2, T-t)} \lambda'_j \left(x_{t+j} - \sum_{k=-m_1}^{m_2} \lambda_k f_{t+j-k} \right) = 0$$

for $t = 1 - m_2, \dots, T + m_1$. This means that the first order conditions are

$$\sum_{j=-m_1}^{m_2} \lambda'_j \left(x_{t+j} - \sum_{k=-m_1}^{m_2} \lambda_k f_{t+j-k} \right) = 0$$

for the $t = m_1 + 1, \dots, T - m_2$. For the boundary values, we have the

$$\sum_{j=1-t}^{m_2} \lambda'_j \left(x_{t+j} - \sum_{k=-m_1}^{m_2} \lambda_k f_{t+j-k} \right) = 0$$

for $t = 1 - m_2, \dots, m_1$ and

$$\sum_{j=-m_1}^{T-t} \lambda'_j \left(x_{t+j} - \sum_{k=-m_1}^{m_2} \lambda_k f_{t+j-k} \right) = 0$$

for $t = T - m_2 + 1, \dots, T + m_1$. The first-order conditions for the boundary values become different due to missing observations at the beginning and end of the sample. Let $(T + m_1 + m_2) \times N$ dimensional matrix $\tilde{X}$ and $(T + m_1 + m_2) \times q$ matrix $\tilde{F}_{-k}$ be given as

$$\tilde{X} = \begin{pmatrix} 0_{m_2 \times N} \\ X \\ 0_{m_1 \times N} \end{pmatrix}, \tilde{F}_{-k} = \begin{pmatrix} 0_{m_2 \times q} \\ F_{-k} \\ 0_{m_1 \times q} \end{pmatrix},$$

i.e., matrices X and F_{-k} that have been augmented with m_2 zero vectors on top and m_1 zero vectors on bottom. Given $a = (a_1, \dots, a_{T+m_1+m_2})' \in \mathbb{R}^{T+m_1+m_2}$, we define P as the one-step upward cyclic permutation that acts as

$$(Pa)_t = a_{t+1 \pmod{T+m_1+m_2}},$$

i.e., P sends $(a_1, \dots, a_{T+m_1+m_2})$ to $(a_2, \dots, a_{T+m_1+m_2-1}, a_1)$. More concretely, P is $(T + m_1 + m_2) \times (T + m_1 + m_2)$ dimensional zero matrix with ones on 1-th diagonal and $-(T + m_1 + m_2)$ -th diagonal. Let $P_j = P^j$. Then, the first order conditions for (f_t) may be summarized as

$$\sum_{j=-m_1}^{m_2} P_j \left(\sum_{k=-m_1}^{m_2} \hat{\tilde{F}}_{-k} \hat{\lambda}'_k \right) \hat{\lambda}_j = \sum_{j=-m_1}^{m_2} P_j \tilde{X} \hat{\lambda}_j,$$

where $\hat{\tilde{F}}_{-k}$ is defined using $\hat{F}_{-k}$. If we restrict the boundary values of f_t to zero, i.e., $f_t = 0$ for $t = 1 - m_2, \dots, m_1$ and $t = T - m_2 + 1, \dots, T + m_1$, we have

$$\sum_{k,j=-m_1}^{m_2} P_{j-k} \hat{\tilde{F}} \hat{\lambda}'_k \hat{\lambda}_j = \sum_{j=-m_1}^{m_2} P_j \tilde{X} \hat{\lambda}_j.$$

8.4 Obtaining the generalized impulse responses

Let $\mathbb{E}[x_t|\Omega_{t-1}] = \sum_{k=1}^{T+h+1} \Phi_k x_{t-k}$. Note that, we have

$$\begin{aligned} s_{\bullet,j}^h &= \mathbb{E}[x_{i,t+h}|e_{j,t} = \delta_j, \Omega_{t-1}] - \mathbb{E}[x_{i,t-1+h}|e_{j,t} = 0, \Omega_{t-1}] \\ &= \mathbb{E}[x_{i,t+h}|e_{j,t} = \delta_j] - \mathbb{E}[x_{i,t-1+h}|e_{j,t} = 0], \end{aligned}$$

because of orthogonality between e_t and Ω_{t-1} . Also,

$$\begin{aligned} s_{\bullet,j}^h &= \mathbb{E}[x_{i,t+h}|e_{j,t} = \delta_j] - \mathbb{E}[x_{i,t-1+h}|e_{j,t} = 0] \\ &= \text{cov}(x_{t+h}, e_{j,t}) \text{var}(e_{j,t})^{-1} \delta_j \\ &= \text{cov}\left(x_{t+h}, x_{j,t} - \sum_{k=1}^{T+h+1} \Phi_{j,k} x_{t-k}\right) \text{var}(e_{j,t})^{-1} \delta_j \\ &= \left([\Sigma_x(h)]_{\bullet,j} - \sum_{k=1}^{T+h+1} \Sigma_x(k+h) \Phi'_{j,k}\right) \text{var}(e_{j,t})^{-1} \delta_j, \end{aligned}$$

where $\Phi_{j,k}$ is N dimensional row vector given by j -th row of Φ_k . Given $\delta_j = \sqrt{\text{var}(e_{j,t})}$, we have

$$s^h = \left(\Sigma_x(h) - \sum_{k=1}^{T+h+1} \Sigma_x(k+h) \Phi'_k\right) \text{Diag}(\text{var}(e_{\bullet,t})^{-1/2})$$

8.5 Simulation, Consistency

We conduct Monte Carlo simulations of dynamic factor models to evaluate the finite-sample performance of estimates of dynamic factors. We use the same simulation model as in Section 5. Specifically, we simulate the model

$$x_t = \sum_{k=-m_1}^{m_2} \lambda_k f_{t-k} + \varepsilon_t,$$

where the idiosyncratic errors satisfy

$$\varepsilon_{it} = \sqrt{\frac{\theta(1-\rho^2)}{1+2J\beta^2}} e_{it}, \quad e_{it} = \rho e_{i,t-1} + v_{it} + \sum_{1 \leq |j| \leq J} \beta v_{i-j,t}.$$

The innovations v_{it} and the factor loadings λ_k are drawn independently from a standard normal distribution. The common factors f_t follow a VARMA(1,1) process,

$$f_t = A f_{t-1} + u_t + \Theta u_{t-1},$$

under the following four specifications:

1. $\rho = \beta = J = 0$ and $A = \Theta = 0$.
2. $\rho = 0.5$, $\beta = 0.2$, $J = 10$, and $A = \Theta = 0$.
3. $\rho = \beta = J = 0$, $A = \text{Diag}(0.7, 0.5, 0.3)$, and $\Theta = 0$.

4. $\rho = \beta = J = 0$, $A = 0$, and $\Theta = \text{Diag}(0.7, 0.5, 0.3)$.

By construction, the variance of ε_{it} equals θ . The variance of the common component $\sum_{k=-m_1}^{m_2} \lambda_k f_{t-k}$ is given by $m \cdot \text{tr}(\Sigma_f)$, where $\Sigma_f = \mathbb{E}[f_t f_t']$. We therefore set $\theta = m \cdot \text{tr}(\Sigma_f)$, ensuring that the signal-to-noise ratio equals one in all specifications.

	$N = 50$	$N = 100$	$N = 150$	$N = 200$
$T = 50$	0.7901	0.8869	0.9139	0.9265
$T = 100$	0.8861	0.9348	0.9514	0.9635
$T = 150$	0.9036	0.9441	0.9624	0.9673
$T = 200$	0.9130	0.9568	0.9658	0.9750

Table 18: R-squared, First Specification

	$N = 50$	$N = 100$	$N = 150$	$N = 200$
$T = 50$	0.5793	0.6521	0.6929	0.7297
$T = 100$	0.6574	0.7783	0.8574	0.8966
$T = 150$	0.6791	0.8227	0.9007	0.9368
$T = 200$	0.6965	0.8541	0.9230	0.9495

Table 19: R-squared, Second Specification

	$N = 50$	$N = 100$	$N = 150$	$N = 200$
$T = 50$	0.5506	0.5873	0.6184	0.6143
$T = 100$	0.6664	0.8017	0.8372	0.8306
$T = 150$	0.7389	0.8786	0.8892	0.8928
$T = 200$	0.7883	0.8975	0.9282	0.9061

Table 20: R-squared, Third Specification

	$N = 50$	$N = 100$	$N = 150$	$N = 200$
$T = 50$	0.5710	0.6605	0.6943	0.7179
$T = 100$	0.7441	0.8451	0.9075	0.9356
$T = 150$	0.7592	0.9088	0.9438	0.9576
$T = 200$	0.7914	0.9322	0.9585	0.9670

Table 21: R-squared, Fourth Specification

8.6 Monetary policy shock identification, extended sample

We estimate the impulse responses to monetary policy shock using the extended sample, March 1973–December 2019.

The estimated dynamic structure (q_0, m_0) is given by $(2, 3), (5, 2), (3, 2)$ for PC_2 , DC_2 , and IC_2 respectively. We use $(q, m) = (3, 2)$ and $(q, m) = (4, 2)$ for the subsequent analysis.

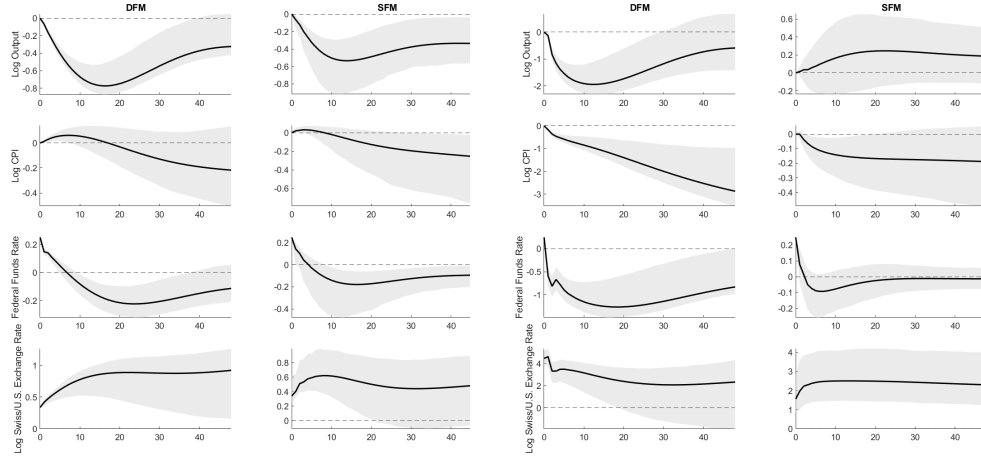


Figure 5: Estimated impulse responses of log output, log CPI, the federal funds rate, and the log exchange rate to a monetary policy shock, based on the $(q, m) = (3, 2)$ (left) and $(q, m) = (4, 2)$ (right).

We see in Figure 5 that the estimated responses of log output to monetary policy shock become unstable for the static factor model when $\hat{q} = 4$, whereas the estimated responses are more robust with the dynamic factor model.

A Proofs for Section 3 (Estimation)

Proposition 1. *Let Assumptions 1 and 2 hold. Suppose both (f_t) and (λ_k) are unknown, and we estimate them by minimizing the objective function (5). Then*

$$\begin{aligned}\frac{1}{\sqrt{N}} \left\| \hat{\Gamma} - \Gamma H'_\gamma \right\| &= O_p\left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}}\right), \\ \frac{1}{\sqrt{T}} \left\| \hat{G} - G H_\gamma^{-1} \right\| &= O_p\left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}}\right),\end{aligned}$$

where $H_\gamma \in \mathbb{R}^{mq \times mq}$ is defined by

$$H_\gamma = (\hat{G}' \hat{G})^{-1} \hat{G}' G.$$

Consequently, the estimated common component satisfies

$$\frac{1}{\sqrt{NT}} \left\| \sum_{k=-m_1}^{m_2} \hat{F}_{-k} \hat{\lambda}'_k - \sum_{k=-m_1}^{m_2} F_{-k} \lambda'_k \right\| = \frac{1}{\sqrt{NT}} \left\| \hat{G} \hat{\Gamma}' - G \Gamma' \right\| = O_p\left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}}\right).$$

Proof. Here, we denote G_0 and Γ_0 as the true values, e.g., $X = G_0 \Gamma'_0 + E$, to distinguish them from the argument of the objective function. We first show that the minimizer $\hat{G}$ of the objective function (5) satisfies

$$\frac{1}{T} G'_0 \hat{G} \left(\frac{1}{T} \hat{G}' \hat{G} \right)^{-1} \frac{1}{T} \hat{G}' G_0 \rightarrow_p M_g > 0$$

where M_g is the positive definite matrix that $\frac{1}{T} G'_0 G_0$ limits to. Consider the concentrated problem

$$\begin{aligned}\min_{(G, \Gamma)} \frac{1}{NT} \left\| X - G \Gamma' \right\|^2 &= \min_G \frac{1}{NT} \left\| X - G(G'G)^{-1} G' X \right\|^2 \\ &= \min_G \frac{1}{NT} \left\| (I_T - G(G'G)^{-1} G') X \right\|^2\end{aligned}$$

The objective function can be written as

$$\begin{aligned}\frac{1}{NT} \left\| (I_T - G(G'G)^{-1} G') X \right\|^2 &= \frac{1}{NT} \left\| (I_T - G(G'G)^{-1} G') (G_0 \Gamma'_0 + E) \right\|^2 \\ &= \frac{1}{NT} \left\| (I_T - G(G'G)^{-1} G') G_0 \Gamma'_0 \right\|^2 \\ &\quad + \frac{1}{NT} \left\| (I_T - G(G'G)^{-1} G') E \right\|^2 \\ &\quad + \frac{2}{NT} \text{tr} \left((I_T - G(G'G)^{-1} G') G_0 \Gamma'_0 E' \right)\end{aligned}$$

Because $\hat{G}$ is the minimizer of the concentrated objective function, we have

$$\begin{aligned}
0 &\leq \frac{1}{NT} \left\| (I_T - G_0(G'_0 G_0)^{-1} G'_0) X \right\|^2 - \frac{1}{NT} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') X \right\|^2 \\
&= \frac{1}{NT} \left\| (I_T - G_0(G'_0 G_0)^{-1} G'_0) G_0 \Gamma'_0 \right\|^2 - \frac{1}{NT} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \Gamma'_0 \right\|^2 \\
&+ \frac{1}{NT} \left\| (I_T - G_0(G'_0 G_0)^{-1} G'_0) E \right\|^2 - \frac{1}{NT} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') E \right\|^2 \\
&+ \frac{2}{NT} \text{tr} \left((I_T - G_0(G'_0 G_0)^{-1} G'_0) G_0 \Gamma'_0 E' \right) - \frac{2}{NT} \text{tr} \left((I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \Gamma'_0 E' \right) \quad (9) \\
&= -\frac{1}{NT} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \Gamma'_0 \right\|^2 \\
&+ \frac{1}{NT} \left\| (I_T - G_0(G'_0 G_0)^{-1} G'_0) E \right\|^2 - \frac{1}{NT} \left\| (I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') E \right\|^2 \\
&- \frac{2}{NT} \text{tr} \left((I_T - \hat{G}(\hat{G}' \hat{G})^{-1} \hat{G}') G_0 \Gamma'_0 E' \right)
\end{aligned}$$

where we've used

$$\begin{aligned}
\frac{1}{NT} \left\| (I_T - G_0(G'_0 G_0)^{-1} G'_0) G_0 \Gamma'_0 \right\|^2 &= 0 \\
\frac{2}{NT} \text{tr} \left((I_T - G_0(G'_0 G_0)^{-1} G'_0) G_0 \Gamma'_0 E' \right) &= 0
\end{aligned}$$

in the final equality. Now, we show that the term

$$\frac{1}{NT} \left\| (I_T - G_0(G'_0 G_0)^{-1} G'_0) E \right\|^2 - \frac{1}{NT} \left\| (I_T - G(G' G)^{-1} G') E \right\|^2 \quad (10)$$

as a function of G is $O_p\left(\frac{1}{N} + \frac{1}{T}\right)$ uniformly over G . Similarly, the term

$$\frac{2}{NT} \text{tr} \left((I_T - G(G' G)^{-1} G') G_0 \Gamma'_0 E' \right) \quad (11)$$

as a function of G is $O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right)$ uniformly over G . To show the rate on (10), note that

$$\begin{aligned}
&\frac{1}{NT} \left\| (I_T - G_0(G'_0 G_0)^{-1} G'_0) E \right\|^2 - \frac{1}{NT} \left\| (I_T - G(G' G)^{-1} G') E \right\|^2 \\
&= \frac{1}{NT} \left\| (G'_0 G_0)^{-1/2} G'_0 E \right\|^2 - \frac{1}{NT} \left\| (G' G)^{-1/2} G' E \right\|^2 \\
&\leq \frac{1}{NT} \left\| (G'_0 G_0)^{-1/2} G'_0 \right\|^2 \|E\|^2 - \frac{1}{NT} \left\| (G' G)^{-1/2} G' \right\|^2 \|E\|^2 \\
&= O_p\left(\frac{1}{N} + \frac{1}{T}\right)
\end{aligned}$$

To show the rate on (11), note that

$$\begin{aligned}
\frac{2}{NT} \text{tr} \left((I_T - G(G' G)^{-1} G') G_0 \Gamma'_0 E' \right) &\leq \frac{2}{NT} \text{tr} (G_0 \Gamma'_0 E') \\
&\leq \frac{2}{NT} \|G_0 \Gamma'_0 E\| \\
&\leq \frac{2}{NT} \|G_0 \Gamma'_0\| \|E\| \\
&= O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right)
\end{aligned}$$

where the first inequality follows due to Von Neumann's trace inequality. This implies that

$$0 \leq -\frac{1}{NT} \left\| (I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0\Gamma'_0 \right\|^2 + o_p(1)$$

Since the term on the right hand side is not positive, it must converge to zero. Thus, we have

$$\begin{aligned} & \frac{1}{NT} \left\| (I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0\Gamma'_0 \right\|^2 \\ &= \frac{1}{NT} \text{tr} \left(\Gamma'_0\Gamma_0G'_0(I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0 \right) \\ &= \text{tr} \left(\left(\frac{1}{N}\Gamma'_0\Gamma_0 \right) \left(\frac{1}{T}G'_0(I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0 \right) \right) \\ &= o_p(1) \end{aligned}$$

Since $\frac{1}{N}\Gamma'_0\Gamma_0$ is positive definite in the limit and $\frac{1}{T}G'_0(I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0$ is positive semi-definite for any N and T , we have

$$\frac{1}{T}G'_0(I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0 = o_p(1)$$

Since $\frac{1}{T}G'_0G_0 \rightarrow_p M_g$, we have $\frac{1}{T}G'_0\hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}'G_0 \rightarrow_p M_g$. Using the first order condition for $\hat{\Gamma}$, $\hat{\Gamma} = X'\hat{G}(\hat{G}'\hat{G})^{-1}$, we have

$$\hat{\Gamma} = \Gamma_0G'_0\hat{G}(\hat{G}'\hat{G})^{-1} + E'\hat{G}(\hat{G}'\hat{G})^{-1}$$

This implies

$$\frac{1}{NT}\Gamma'_0\hat{\Gamma}\hat{G}'G_0 = \frac{1}{NT}\Gamma'_0\Gamma_0G'_0\hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}'G_0 + \frac{1}{NT}\Gamma'_0E'\hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}'G_0$$

Here, the first term on the right hand side converges in probability to $M_\gamma M_g$. The second term is $o_p(1)$ because

$$\begin{aligned} \frac{1}{NT} \left\| \Gamma'_0E'\hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}'G_0 \right\| &\leq \frac{1}{NT} \left\| \Gamma'_0E' \right\| \left\| \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}' \right\| \left\| G_0 \right\| \\ &\leq \frac{1}{NT} \left\| \Gamma'_0 \right\| \left\| E' \right\| \left\| \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}' \right\| \left\| G_0 \right\| \\ &= O_p \left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}} \right) \end{aligned}$$

Thus,

$$\frac{1}{NT}\Gamma'_0\hat{\Gamma}\hat{G}'G_0 \rightarrow_p M_\gamma M_g \tag{12}$$

Let $\delta_k(\cdot)$ be k -th singular value of $(\cdot)$. Note that for any $K \times K$ dimensional matrices G_1 and G_2 , we have

$$\delta_K(G_1G_2) \leq \delta_1(G_1)\delta_K(G_2) \tag{13}$$

$$\delta_1(G_1)\delta_K(G_2) \leq \delta_1(G_1G_2) \tag{14}$$

Let $G_1 = \frac{1}{N}\Gamma'_0\hat{\Gamma}$, and $G_2 = \frac{1}{N}\hat{G}'G_0$. Then, using (13), we have

$$\delta_{mq} \left(\frac{1}{NT}\Gamma'_0\hat{\Gamma}\hat{G}'G_0 \right) \leq \delta_1 \left(\frac{1}{N}\Gamma'_0\hat{\Gamma} \right) \delta_{mq} \left(\frac{1}{N}\hat{G}'G_0 \right)$$

Using continuity of singular values, $\delta_{mq} \left(\frac{1}{NT}\Gamma'_0\hat{\Gamma}\hat{G}'G_0 \right)$ is strictly positive for N, T large. Because $\delta_1 \left(\frac{1}{N}\Gamma'_0\hat{\Gamma} \right)$ is finite for all N and T , we must have that $\delta_{mq} \left(\frac{1}{T}\hat{G}'G_0 \right)$ is strictly positive for N, T large. Similarly, $\delta_{mq} \left(\frac{1}{N}\Gamma'_0\hat{\Gamma} \right)$ is strictly positive for N, T large.

Similarly, let $G_1 = G'_0\hat{G}(\hat{G}'\hat{G})^{-1}$, and $G_2 = \frac{1}{T}\hat{G}'G_0$. Then, using (14), we have

$$\delta_1 \left(G'_0\hat{G}(\hat{G}'\hat{G})^{-1} \right) \delta_{mq} \left(\frac{1}{T}\hat{G}'G_0 \right) \leq \delta_1 \left(\frac{1}{T}G'_0\hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}'G_0 \right)$$

Using continuity of singular values, $\delta_1 \left(\frac{1}{T}G'_0\hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}'G_0 \right)$ is finite for N, T large. Since $\delta_{mq} \left(\frac{1}{T}\hat{G}'G_0 \right)$ is strictly positive, this implies $\delta_1 \left(G'_0\hat{G}(\hat{G}'\hat{G})^{-1} \right)$ is finite. Going through the argument one more time with

$$\delta_1 \left(\left(\frac{1}{T}\hat{G}'\hat{G} \right)^{-1} \right) \delta_{mq} \left(\frac{1}{T}G'_0\hat{G} \right) \leq \delta_1 \left((\hat{G}'\hat{G})^{-1}\hat{G}'G_0 \right)$$

we have

$$\left(\frac{1}{T}\hat{G}'\hat{G} \right)^{-1} = O_p(1)$$

Considering (9) again, we have

$$\begin{aligned} 0 &\leq -\text{tr} \left(\left(\frac{1}{N}\Gamma'_0\Gamma_0 \right) \left(\frac{1}{T}G'_0(I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0 \right) \right) \\ &\quad - \frac{2}{NT} \text{tr} \left((I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0\Gamma'_0E' \right) + O_p \left(\frac{1}{N} + \frac{1}{T} \right) \\ &\leq -\lambda_{\min} \left(\frac{1}{N}\Gamma'_0\Gamma_0 \right) \text{tr} \left(\frac{1}{T}G'_0(I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0 \right) \\ &\quad - \frac{2}{NT} \text{tr} \left((I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0\Gamma'_0E' \right) + O_p \left(\frac{1}{N} + \frac{1}{T} \right) \end{aligned}$$

Note that we have

$$\begin{aligned} \frac{2}{NT} \text{tr} \left((I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0\Gamma'_0E' \right) &\leq \frac{2}{NT} \left\| (I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0 \right\| \left\| \Gamma_0 \right\| \|E\| \\ &= O_p \left(\sqrt{\frac{N+T}{NT}} \right) \frac{1}{\sqrt{T}} \left\| (I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}')G_0 \right\| \\ &= O_p \left(\sqrt{\frac{N+T}{NT}} \right) \frac{1}{\sqrt{T}} \left\| (\hat{G} - GH_\gamma^{-1})H_\gamma \right\| \end{aligned} \tag{15}$$

where $H_\gamma = (\hat{G}'\hat{G})^{-1}\hat{G}'G$. Also, we have

$$\begin{aligned}\text{tr} \left(\frac{1}{T} G_0' (I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}') G_0 \right) &= \frac{1}{T} \left\| (I_T - \hat{G}(\hat{G}'\hat{G})^{-1}\hat{G}') G_0 \right\|^2 \\ &= \left(\frac{1}{\sqrt{T}} \left\| (\hat{G} - GH_\gamma^{-1}) H_\gamma \right\| \right)^2\end{aligned}$$

so that

$$\begin{aligned}0 &\leq -\lambda_{\min} \left(\frac{1}{N} \Gamma_0' \Gamma_0 \right) \left(\frac{1}{\sqrt{T}} \left\| (\hat{G} - GH_\gamma^{-1}) H_\gamma \right\| \right)^2 \\ &\quad + O_p \left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}} \right) \frac{1}{\sqrt{T}} \left\| (\hat{G} - GH_\gamma^{-1}) H_\gamma \right\| + O_p \left(\frac{1}{N} + \frac{1}{T} \right)\end{aligned}$$

Let $x = \frac{1}{\sqrt{T}} \left\| (\hat{G} - GH_\gamma^{-1}) H_\gamma \right\|$. Since $\lambda_{\min} \left(\frac{1}{N} \Gamma_0' \Gamma_0 \right)$ is strictly positive, the above equation implies

$$x^2 - O_p \left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}} \right) x + O_p \left(\frac{1}{N} + \frac{1}{T} \right) \leq 0$$

Consider the second order polynomial $f(x) = x^2 - bx + c$. This is made negative only if $b^2 - 4c > 0$ and $\frac{1}{2}(b - \sqrt{b^2 - 4c}) \leq x \leq \frac{1}{2}(b + \sqrt{b^2 - 4c})$. This implies that

$$\frac{1}{\sqrt{T}} \left\| (\hat{G} - GH_\gamma^{-1}) H_\gamma \right\| = O_p \left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}} \right)$$

Note that the smallest singular value of H_γ is strictly bounded away from zero. This implies

$$\frac{1}{\sqrt{T}} \left\| \hat{G} - GH_\gamma^{-1} \right\| = O_p \left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}} \right)$$

Considering the first order condition for $\hat{\Gamma}$, we have

$$\begin{aligned}\hat{\Gamma} &= X' \hat{G} (\hat{G}' \hat{G})^{-1} \\ &= \Gamma_0 G_0' \hat{G} (\hat{G}' \hat{G})^{-1} + E' \hat{G} (\hat{G}' \hat{G})^{-1}\end{aligned}$$

so that

$$\hat{\Gamma} - \Gamma_0 H_\gamma' = E' \hat{G} (\hat{G}' \hat{G})^{-1}$$

and thus,

$$\begin{aligned}\frac{1}{\sqrt{N}} \left\| \hat{\Gamma} - \Gamma_0 H_\gamma' \right\| &= \frac{1}{N} \left\| E \hat{G} (\hat{G}' \hat{G})^{-1} \right\| \\ &\leq \frac{1}{\sqrt{N}} \left\| E \right\| \left\| \hat{G} (\hat{G}' \hat{G})^{-1} \right\| \\ &\leq O_p \left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}} \right)\end{aligned}$$

Finally,

$$\begin{aligned}
\frac{1}{\sqrt{NT}} \left\| \hat{G}\hat{\Gamma}' - G\Gamma' \right\| &= \frac{1}{\sqrt{NT}} \left\| (\hat{G} - GH_\gamma^{-1})\hat{\Gamma}' + GH_\gamma^{-1}(\hat{\Gamma} - \Gamma H_\gamma)' \right\| \\
&\leq \frac{1}{\sqrt{NT}} \left\| (\hat{G} - GH_\gamma^{-1})\hat{\Gamma}' \right\| + \frac{1}{\sqrt{NT}} \left\| GH_\gamma^{-1}(\hat{\Gamma} - \Gamma H_\gamma)' \right\| \\
&= O_p \left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}} \right)
\end{aligned}$$

□

Theorem 1. *Let Assumptions 1 and 2 hold. Suppose both (f_t) and (λ_k) are unknown, and we estimate them by minimizing the objective function (5). Then, for $k = -m_1, \dots, m_2$, we have*

$$\begin{aligned}
\frac{1}{\sqrt{T}} \left\| \hat{F} - FH_f' \right\| &= O_p \left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}} \right) \\
\frac{1}{\sqrt{N}} \left\| \hat{\lambda}_k - \lambda_k H_f^{-1} \right\| &= O_p \left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}} \right),
\end{aligned}$$

where $H_f \in \mathbb{R}^{q \times q}$ is defined as

$$H_f = \left(\sum_{k=-m_1}^{m_2} \lambda_k' \hat{\lambda}_k \right) \left(\sum_{k=-m_1}^{m_2} \hat{\lambda}_k' \hat{\lambda}_k \right)^{-1}.$$

Consequently, each component of the common component satisfies

$$\frac{1}{\sqrt{NT}} \left\| \hat{F}_{-k} \hat{\lambda}_k' - F_{-k} \lambda_k' \right\| = O_p \left(\frac{1}{\sqrt{N}} + \frac{1}{\sqrt{T}} \right), \quad k = -m_1, \dots, m_2.$$

Proof. We first show that H_γ^{-1} is a block diagonal matrix in the limit. Note that

$$\begin{aligned}
\frac{1}{T} \left\| F_j' (\hat{G} - GH_\gamma^{-1}) \right\| &\leq \left(\frac{1}{\sqrt{T}} \left\| F_j \right\| \right) \left(\frac{1}{\sqrt{T}} \left\| \hat{G} - GH_\gamma^{-1} \right\| \right) \\
&= O_p \left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}} \right)
\end{aligned}$$

for $j = -m_2 - 1, \dots, m_1 + 1$. Let $q \times q$ blocks of H_γ^{-1} be denoted by

$$H_\gamma^{-1} = \begin{pmatrix} H_{1,1} & \dots & H_{1,m} \\ \vdots & \ddots & \vdots \\ H_{m,1} & \dots & H_{m,m} \end{pmatrix}$$

Subtracting the first block of $\frac{1}{T} F_{j+1}' (\hat{G} - GH_\gamma^{-1})$ from the second block of $\frac{1}{T} F_j' (\hat{G} - GH_\gamma^{-1})$, we have

$$\frac{1}{T} F_j' (\hat{G} - GH_\gamma^{-1}) \begin{pmatrix} 0 \\ I_q \\ \vdots \\ 0 \end{pmatrix} - \frac{1}{T} F_{j+1}' (\hat{G} - GH_\gamma^{-1}) \begin{pmatrix} I_q \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

Both of the terms above are $O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right)$. Thus, we have

$$\begin{aligned}
O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right) &= \frac{1}{T}F'_j(\hat{G} - GH_\gamma^{-1}) \begin{pmatrix} 0 \\ I_q \\ \vdots \\ 0 \end{pmatrix} - \frac{1}{T}F'_{j+1}(\hat{G} - GH_\gamma^{-1}) \begin{pmatrix} I_q \\ 0 \\ \vdots \\ 0 \end{pmatrix} \\
&= \frac{1}{T}F'_{j+1}GH_\gamma^{-1} \begin{pmatrix} I_q \\ 0 \\ \vdots \\ 0 \end{pmatrix} - \frac{1}{T}F'_jGH_\gamma^{-1} \begin{pmatrix} 0 \\ I_q \\ \vdots \\ 0 \end{pmatrix} \\
&= \frac{1}{T}F'_{j+1} \sum_{k=-m_1}^{m_2} F_{-k}H_{m_1+1+k,1} - \frac{1}{T}F'_j \sum_{k=-m_1}^{m_2} F_{-k}H_{m_1+1+k,2} \\
&= -M_{j-m_1}H_{1,2} + \sum_{k=-m_1}^{m_2-1} M_{j+1+k}(H_{m_1+1+k,1} - H_{m_1+2+k,2}) + M_{j+1+m_2}H_{m,1}
\end{aligned}$$

Collecting the above expression for $j = m_1, \dots, -m_2 - 1$ in a matrix, we may write

$$O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right) = \begin{pmatrix} M_0 & \dots & M_m \\ \vdots & \ddots & \vdots \\ M_{-m} & \dots & M_0 \end{pmatrix} \begin{pmatrix} -H_{1,2} \\ H_{1,1} - H_{2,2} \\ \vdots \\ H_{m-1,1} - H_{m,2} \\ H_{m,1} \end{pmatrix}$$

Similarly, considering the $(k-1)$ -th block of $\frac{1}{T}F'_j(\hat{G} - GH_\gamma^{-1})$ minus the k -th block of $\frac{1}{T}F'_{j+1}(\hat{G} - GH_\gamma^{-1})$ for $j = m_1, \dots, -m_2 - 1$ and for $k = 2, \dots, m$, we have

$$\begin{pmatrix} M_0 & \dots & M_m \\ \vdots & \ddots & \vdots \\ M_{-m} & \dots & M_0 \end{pmatrix} \begin{pmatrix} -H_{1,2} & -H_{1,3} & \dots & -H_{1,m} \\ H_{1,1} - H_{2,2} & H_{1,2} - H_{2,3} & \dots & H_{1,m-1} - H_{2,m} \\ \vdots & \vdots & \ddots & \vdots \\ H_{m-1,1} - H_{m,2} & H_{m-1,2} - H_{m,3} & \dots & H_{m-1,m-1} - H_{m,m} \\ H_{m,1} & H_{m,2} & \dots & H_{m,m-1} \end{pmatrix} = O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right)$$

Since $\begin{pmatrix} M_0 & \dots & M_m \\ \vdots & \ddots & \vdots \\ M_{-m} & \dots & M_0 \end{pmatrix}$ is positive definite in the limit, we have

$$\begin{pmatrix} -H_{1,2} & -H_{1,3} & \dots & -H_{1,m} \\ H_{1,1} - H_{2,2} & H_{1,2} - H_{2,3} & \dots & H_{1,m-1} - H_{2,m} \\ \vdots & \vdots & \ddots & \vdots \\ H_{m-1,1} - H_{m,2} & H_{m-1,2} - H_{m,3} & \dots & H_{m-1,m-1} - H_{m,m} \\ H_{m,1} & H_{m,2} & \dots & H_{m,m-1} \end{pmatrix} = O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right)$$

This implies

$$H_{j,k} = O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right), \quad H_{j,j} - H_{k,k} = O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right)$$

for all $j \neq k$. Thus, there exists a sequence of $q \times q$ dimensional matrices (H_{NT}) such that

$$\|H_\gamma^{-1} - (I_m \otimes H'_{NT})\| = O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right)$$

or equivalently,

$$\|H'_\gamma - (I_m \otimes H_{NT}^{-1})\| = O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right)$$

because H'_γ is positive definite in the limit. This implies

$$\begin{aligned} \frac{1}{\sqrt{N}} \|\hat{\Gamma} - \Gamma(I_m \otimes H_{NT}^{-1})\| &= \frac{1}{\sqrt{N}} \|(\hat{\Gamma} - \Gamma H'_\gamma) - \Gamma((I_m \otimes H_{NT}^{-1}) - H'_\gamma)\| \\ &\leq \frac{1}{\sqrt{N}} \|\hat{\Gamma} - \Gamma H'_\gamma\| + \frac{1}{\sqrt{N}} \|\Gamma((I_m \otimes H_{NT}^{-1}) - H'_\gamma)\| \end{aligned}$$

so that we have

$$\frac{1}{\sqrt{N}} \|\hat{\Gamma} - \Gamma(I_m \otimes H_{NT}^{-1})\| = O_p\left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}}\right) \quad (16)$$

Now, using the first order condition for (f_t) , we have

$$\sum_{j,k=-m_1}^{m_2} P_j \left(\hat{\tilde{F}}_{-k} \hat{\lambda}'_k - \tilde{F}_{-k} \lambda'_k \right) \hat{\lambda}_j = \sum_{j=-m_1}^{m_2} P_j \tilde{E} \hat{\lambda}_j$$

This can be written as

$$\sum_{j,k=-m_1}^{m_2} P_j \left(\hat{\tilde{F}}_{-k} (\lambda_k H_{NT}^{-1})' - \tilde{F}_{-k} H'_{NT} (\lambda_k H_{NT}^{-1})' + \hat{\tilde{F}}_{-k} (\hat{\lambda}_k - \lambda_k H_{NT}^{-1})' \right) \hat{\lambda}_j = \sum_{j=-m_1}^{m_2} P_j \tilde{E} \hat{\lambda}_j$$

so that we have

$$\sum_{j,k=-m_1}^{m_2} P_j \left(\hat{\tilde{F}}_{-k} - \tilde{F}_{-k} H'_{NT} \right) H_{NT}^{-1'} \lambda'_k \hat{\lambda}_j = - \sum_{j,k=-m_1}^{m_2} P_j \hat{\tilde{F}}_{-k} (\hat{\lambda}_k - \lambda_k H_{NT}^{-1})' \hat{\lambda}_j + \sum_{j=-m_1}^{m_2} P_j \tilde{E} \hat{\lambda}_j$$

Here, note that the matrix $\hat{\tilde{F}}_0$ permuted by P_{-k} , $k = -m_1, \dots, m_2$, i.e.,

$$P_{-k} \hat{\tilde{F}}_0 = \begin{pmatrix} 0_{(m_2+k) \times q} \\ \hat{\tilde{F}}_0 \\ 0_{(m_1-k) \times q} \end{pmatrix},$$

and the matrix $\hat{\tilde{F}}_{-k}$ shares the same element on the interior. Specifically, we have

$$\hat{\tilde{F}}_{-k} - P_{-k}\hat{\tilde{F}}_0 = \begin{pmatrix} 0_{m_2 \times q} \\ \hat{\tilde{F}}_{-k} \\ 0_{m_1 \times q} \end{pmatrix} - \begin{pmatrix} 0_{(m_2+k) \times q} \\ \hat{\tilde{F}}_0 \\ 0_{(m_1-k) \times q} \end{pmatrix} = \begin{cases} \begin{pmatrix} 0_{m_2 \times q} \\ \hat{f}'_{1-k} \\ \vdots \\ \hat{f}'_0 \\ 0_{(T-k) \times q} \\ -f'_{T-k+1} \\ \vdots \\ -f'_T \\ 0_{(m_1-k) \times q} \end{pmatrix} & k > 0 \\ \begin{pmatrix} 0_{m_2 \times q} \\ -\hat{f}'_1 \\ \vdots \\ -\hat{f}'_{-k} \\ 0_{(T-k) \times q} \\ f'_{T+1} \\ \vdots \\ f'_{T-k} \\ 0_{(m_1-k) \times q} \end{pmatrix} & k < 0 \end{cases}$$

so that $\hat{\tilde{F}}_{-k} - P_{-k}\hat{\tilde{F}}_0$ only has $2k$ number of nonzero rows. Similarly, $\tilde{\tilde{F}}_{-k} - P_{-k}\tilde{\tilde{F}}_0$ only has $2k$ number of nonzero rows. Based on this observation, we may write

$$\hat{\tilde{F}}_{-k} - \tilde{\tilde{F}}_{-k}H'_{NT} = (P_{-k}\hat{\tilde{F}}_0 - P_{-k}\tilde{\tilde{F}}_0H'_{NT}) + (\hat{\tilde{F}}_{-k} - P_{-k}\hat{\tilde{F}}_0) - (\tilde{\tilde{F}}_{-k}H'_{NT} - P_{-k}\tilde{\tilde{F}}_0H'_{NT}),$$

so that we have

$$\begin{aligned} \sum_{j,k=-m_1}^{m_2} P_{j-k} \left(\hat{\tilde{F}}_0 - \tilde{\tilde{F}}_0H'_{NT} \right) H_{NT}^{-1'} \lambda'_k \hat{\lambda}_j &= - \sum_{j,k=-m_1}^{m_2} P_j (\hat{\tilde{F}}_{-k} - P_{-k}\hat{\tilde{F}}_0) H_{NT}^{-1'} \lambda'_k \hat{\lambda}_j \\ &+ \sum_{j,k=-m_1}^{m_2} P_j (\tilde{\tilde{F}}_{-k} - P_{-k}\tilde{\tilde{F}}_0) \lambda'_k \hat{\lambda}_j \\ &- \sum_{j,k=-m_1}^{m_2} P_j \hat{\tilde{F}}_{-k} (\hat{\lambda}_k - \lambda_k H_{NT}^{-1})' \hat{\lambda}_j + \sum_{j=-m_1}^{m_2} P_j \tilde{\tilde{F}} \hat{\lambda}_j \end{aligned}$$

Vectorizing, we have

$$\begin{aligned}
\sum_{j,k=-m_1}^{m_2} \left(P_{j-k} \otimes \hat{\lambda}'_j \lambda_k H_{NT}^{-1} \right) \text{vec}(\hat{\tilde{F}}_0 - \tilde{F}_0 H'_{NT}) &= -\text{vec} \left(\sum_{j,k=-m_1}^{m_2} P_j (\hat{\tilde{F}}_{-k} - P_{-k} \hat{\tilde{F}}_0) H_{NT}^{-1'} \lambda'_k \hat{\lambda}_j \right) \\
&+ \text{vec} \left(\sum_{j,k=-m_1}^{m_2} P_j (\tilde{F}_{-k} - P_{-k} \tilde{F}_0) \lambda'_k \hat{\lambda}_j \right) \\
&- \text{vec} \left(\sum_{j,k=-m_1}^{m_2} P_j \hat{\tilde{F}}_{-k} (\hat{\lambda}_k - \lambda_k H_{NT}^{-1})' \hat{\lambda}_j \right) \\
&+ \text{vec} \left(\sum_{j=-m_1}^{m_2} P_j \tilde{E} \hat{\lambda}_j \right)
\end{aligned}$$

so that we have

$$\begin{aligned}
\text{vec}(\hat{\tilde{F}}_0 - \tilde{F}_0 H'_{NT}) &= - \left(\sum_{j,k=-m_1}^{m_2} \left(P_{j-k} \otimes \hat{\lambda}'_j \lambda_k H_{NT}^{-1} \right) \right)^{-1} \text{vec} \left(\sum_{j,k=-m_1}^{m_2} P_j (\hat{\tilde{F}}_{-k} - P_{-k} \hat{\tilde{F}}_0) H_{NT}^{-1'} \lambda'_k \hat{\lambda}_j \right) \\
&+ \left(\sum_{j,k=-m_1}^{m_2} \left(P_{j-k} \otimes \hat{\lambda}'_j \lambda_k H_{NT}^{-1} \right) \right)^{-1} \text{vec} \left(\sum_{j,k=-m_1}^{m_2} P_j (\tilde{F}_{-k} - P_{-k} \tilde{F}_0) \lambda'_k \hat{\lambda}_j \right) \\
&- \left(\sum_{j,k=-m_1}^{m_2} \left(P_{j-k} \otimes \hat{\lambda}'_j \lambda_k H_{NT}^{-1} \right) \right)^{-1} \text{vec} \left(\sum_{j,k=-m_1}^{m_2} P_j \hat{\tilde{F}}_{-k} (\hat{\lambda}_k - \lambda_k H_{NT}^{-1})' \hat{\lambda}_j \right) \\
&+ \left(\sum_{j,k=-m_1}^{m_2} \left(P_{j-k} \otimes \hat{\lambda}'_j \lambda_k H_{NT}^{-1} \right) \right)^{-1} \text{vec} \left(\sum_{j=-m_1}^{m_2} P_j \tilde{E} \hat{\lambda}_j \right)
\end{aligned}$$

This implies

$$\begin{aligned}
\frac{1}{\sqrt{T}} \left\| \hat{\tilde{F}}_0 - \tilde{F}_0 H'_{NT} \right\| &\leq \left\| \left(\sum_{j,k=-m_1}^{m_2} \left(P_{j-k} \otimes \frac{1}{N} \hat{\lambda}'_j \lambda_k H_{NT}^{-1} \right) \right)^{-1} \right\| \left\| \frac{1}{N\sqrt{T}} \left\| \sum_{j,k=-m_1}^{m_2} P_j (\hat{\tilde{F}}_{-k} - P_{-k} \hat{\tilde{F}}_0) H_{NT}^{-1'} \lambda'_k \hat{\lambda}_j \right\| \right\| \\
&+ \left\| \left(\sum_{j,k=-m_1}^{m_2} \left(P_{j-k} \otimes \frac{1}{N} \hat{\lambda}'_j \lambda_k H_{NT}^{-1} \right) \right)^{-1} \right\| \left\| \frac{1}{N\sqrt{T}} \left\| \sum_{j,k=-m_1}^{m_2} P_j (\tilde{F}_{-k} - P_{-k} \tilde{F}_0) \lambda'_k \hat{\lambda}_j \right\| \right\| \\
&+ \left\| \left(\sum_{j,k=-m_1}^{m_2} \left(P_{j-k} \otimes \frac{1}{N} \hat{\lambda}'_j \lambda_k H_{NT}^{-1} \right) \right)^{-1} \right\| \left\| \frac{1}{N\sqrt{T}} \left\| \sum_{j,k=-m_1}^{m_2} P_j \hat{\tilde{F}}_{-k} (\hat{\lambda}_k - \lambda_k H_{NT}^{-1})' \hat{\lambda}_j \right\| \right\| \\
&+ \left\| \left(\sum_{j,k=-m_1}^{m_2} \left(P_{j-k} \otimes \frac{1}{N} \hat{\lambda}'_j \lambda_k H_{NT}^{-1} \right) \right)^{-1} \right\| \left\| \frac{1}{N\sqrt{T}} \left\| \sum_{j=-m_1}^{m_2} P_j \tilde{E} \hat{\lambda}_j \right\| \right\|
\end{aligned}$$

so that we have

$$\frac{1}{\sqrt{T}} \left\| \hat{F} - FH'_{NT} \right\| = O_p \left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}} \right) \quad (17)$$

as long as

$$\left\| \left(\sum_{j,k=-m_1}^{m_2} \left(P_{j-k} \otimes \frac{1}{N} \hat{\lambda}'_j \lambda_k H_{NT}^{-1} \right) \right)^{-1} \right\| = O_p(1).$$

We may use Lemma 1 to show that the above rate is satisfied because $\frac{1}{N} \hat{\Gamma}' \Gamma$ is strictly positive definite in the limit. The two equations (16) and (17) implies

$$\frac{1}{\sqrt{NT}} \left\| F_{-k} \lambda'_k - \hat{F}_{-k} \hat{\lambda}'_k \right\| = O_p \left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}} \right) \quad (18)$$

for all $k = -m_1, \dots, m_2$. Note that the first order condition for (f_t) can be rewritten as

$$\begin{aligned} \sum_{k,j=-m_1}^{m_2} P_{j-k} \hat{F}_0 \hat{\lambda}'_k \hat{\lambda}_j - \sum_{k,j=-m_1}^{m_2} P_{j-k} \tilde{F}_0 \lambda'_k \hat{\lambda}_j = & - \sum_{j=-m_1}^{m_2} P_j \sum_{k=-m_1}^{m_2} \left(\hat{F}_{-k} - P_{-k} \hat{F}_0 \right) \hat{\lambda}'_k \hat{\lambda}_j \\ & + \sum_{j=-m_1}^{m_2} P_j \sum_{k=-m_1}^{m_2} \left(\tilde{F}_{-k} - P_{-k} \tilde{F}_0 \right) \lambda'_k \hat{\lambda}_j + \sum_{j=-m_1}^{m_2} P_j \tilde{E} \hat{\lambda}_j \end{aligned}$$

Subtracting terms on the left hand side with $k \neq j$ on both sides, we have

$$\begin{aligned} \hat{F}_0 \left(\sum_{k=-m_1}^{m_2} \hat{\lambda}'_k \hat{\lambda}_k \right) - \tilde{F}_0 \left(\sum_{k=-m_1}^{m_2} \lambda'_k \hat{\lambda}_k \right) = & - \sum_{k \neq j} P_j \left(\hat{F}_{-k} \hat{\lambda}'_k - \tilde{F}_{-k} \lambda'_k \right) \hat{\lambda}_j \\ & - \sum_{j=-m_1}^{m_2} P_j \sum_{k=-m_1}^{m_2} \left(\hat{F}_{-k} - P_{-k} \hat{F}_0 \right) \hat{\lambda}'_k \hat{\lambda}_j \\ & + \sum_{j=-m_1}^{m_2} P_j \sum_{k=-m_1}^{m_2} \left(\tilde{F}_{-k} - P_{-k} \tilde{F}_0 \right) \lambda'_k \hat{\lambda}_j + \sum_{j=-m_1}^{m_2} P_j \tilde{E} \hat{\lambda}_j \end{aligned}$$

Defining $H_f = \left(\sum_{k=-m_1}^{m_2} \hat{\lambda}'_k \hat{\lambda}_k \right)^{-1} \left(\sum_{k=-m_1}^{m_2} \lambda'_k \hat{\lambda}_k \right)$, we have

$$\begin{aligned} \hat{F}_0 - \tilde{F}_0 H'_f = & - \left(\sum_{k \neq j} P_j \left(\hat{F}_{-k} \hat{\lambda}'_k - \tilde{F}_{-k} \lambda'_k \right) \hat{\lambda}_j \right) \left(\sum_{k=-m_1}^{m_2} \hat{\lambda}'_k \hat{\lambda}_k \right)^{-1} \\ & - \left(\sum_{j=-m_1}^{m_2} P_j \sum_{k=-m_1}^{m_2} \left(\hat{F}_{-k} - P_{-k} \hat{F}_0 \right) \hat{\lambda}'_k \hat{\lambda}_j \right) \left(\sum_{k=-m_1}^{m_2} \hat{\lambda}'_k \hat{\lambda}_k \right)^{-1} \\ & - \left(\sum_{k \neq j} P_j \left(\hat{F}_{-k} \hat{\lambda}'_k - \tilde{F}_{-k} \lambda'_k \right) \hat{\lambda}_j \right) \left(\sum_{k=-m_1}^{m_2} \hat{\lambda}'_k \hat{\lambda}_k \right)^{-1} + \left(\sum_{j=-m_1}^{m_2} P_j \tilde{E} \hat{\lambda}_j \right) \left(\sum_{k=-m_1}^{m_2} \hat{\lambda}'_k \hat{\lambda}_k \right)^{-1} \end{aligned}$$

so that

$$\begin{aligned} \frac{1}{\sqrt{T}} \left\| \hat{F} - F H_f' \right\| &\leq O_p \left(\frac{1}{\sqrt{T}} \right) + O_p(1) \frac{1}{\sqrt{NT}} \sum_{k \neq j} P_j \left\| F_{-k} \lambda_k' - \hat{F}_{-k} \hat{\lambda}_k' \right\| + O_p \left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}} \right) \\ &= O_p \left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}} \right) \end{aligned} \tag{19}$$

where we've used equation (18). From (17) and (19), we may deduce that

$$H_f - H_{NT} = O_p \left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}} \right)$$

so that we have

$$\frac{1}{\sqrt{N}} \left\| \hat{\Gamma} - \Gamma(I_m \otimes H_f^{-1}) \right\| = O_p \left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}} \right),$$

from which we may easily deduce

$$\frac{1}{\sqrt{N}} \left\| \hat{\lambda}_k - \lambda_k H_\lambda^{-1} \right\| = O_p \left(\frac{1}{\sqrt{T}} + \frac{1}{\sqrt{N}} \right)$$

for $k = -m_1, \dots, m_2$. □

B Proofs for Section 4 (Testing for Number of Dynamic Factors)

Proposition 2. *Let (q_0, m_0) denote the true number of dynamic factors and the true filter length. Then the dynamic factor model with (q_0, m_0) also admits equivalent representations with*

$$(q, m) = \left(\left\lceil \frac{q_0 m_0}{m} \right\rceil, m \right), \quad 1 \leq m < m_0,$$

or

$$(q, m) = \left(q, \left\lceil \frac{q_0 m_0}{q} \right\rceil \right), \quad q > q_0.$$

In other words, the common component $\sum_{k=1}^{m_0} \lambda_k f_{t-k}$, with $f_t \in \mathbb{R}^{q_0}$, can also be represented as $\sum_{k=1}^m \Gamma_k g_{t-k}$ with the factor number and filter length given above.

Proof. We first consider the representations with the number of factors and filter length given by $(\lceil \frac{q_0 m_0}{m} \rceil, m)$ for $1 \leq m < m_0$.

Suppose $\frac{q}{q_0}$ is an integer. The common component of the original representation is given by $\sum_{k=1}^{m_0} \lambda_k f_{t-k}$. Let $\sum_{k=1}^m \Gamma_k g_{t-k}$ be a candidate representation with $\Gamma_k \in \mathbb{R}^{N \times q}$ and $g_t \in \mathbb{R}^q$ where $q = \lceil \frac{q_0 m_0}{m} \rceil$. We may construct g_t and Γ_k as

$$g_t = \begin{pmatrix} f_t \\ f_{t-m} \\ \vdots \\ f_{t-(\frac{q}{q_0}-1)m} \end{pmatrix}, \quad \Gamma_k = \begin{pmatrix} \lambda_k & \lambda_{k+m} & \dots & \lambda_{k+(\frac{q}{q_0}-1)m} \end{pmatrix},$$

where we define $\lambda_k = 0$ for $k > m_0$. The vector g_t collects $\frac{q}{q_0}$ number of f_t so that $g_t \in \mathbb{R}^q$. Note that because $m < m_0$, $\frac{q}{q_0}$ is an integer that is at least 2. Also, we have $\frac{q_0 m_0}{m} \leq q$ so that $m + (\frac{q}{q_0} - 1)m \geq m_0$. By constructing g_t and Γ_k in this way, we see that

$$\begin{aligned} \sum_{k=1}^m \Gamma_k g_{t-k} &= \sum_{k=1}^m (\lambda_k f_{t-k} + \lambda_{k+m} f_{t-k-m} + \dots + \lambda_{k+m_0-m} f_{t-k-m_0+m}) \\ &= \sum_{k=1}^{m_0} \lambda_k f_{t-k} \end{aligned}$$

Now, we consider the case where $\frac{q}{q_0}$ is not an integer. Let $\sum_{k=1}^m \Gamma_k g_{t-k}$ be a candidate representation with $\Gamma_k \in \mathbb{R}^{N \times q}$ and $g_t \in \mathbb{R}^q$ where $q = \lceil \frac{q_0 m_0}{m} \rceil$. We first consider g_t constructed as

$$g_t = \begin{pmatrix} f_t \\ f_{t-m} \\ \vdots \\ f_{t-(\lceil \frac{q}{q_0} \rceil - 2)m} \\ 0_{(q-(\lceil \frac{q}{q_0} \rceil - 1)q_0) \times 1} \end{pmatrix} + \begin{pmatrix} 0_{(q-q_0) \times 1} \\ f_{t-m_0+m} \end{pmatrix}$$

Note that by constructing g_t in this way all the elements in f_{t-k} , $k = 1, \dots, m_0$ appear in g_{t-k} , $k = 1, \dots, m$. To see this, note that $\lceil \frac{q}{q_0} \rceil = \lceil \frac{m_0}{m} \rceil$ so that first term of the right-hand side accounts for f_{t-k} , $k = 1, \dots, (\lceil \frac{m_0}{m} \rceil - 1)m$. The second term accounts for f_{t-k} , $k = 1 + m_0 - m, \dots, m_0$. Here, we have $(\lceil \frac{m_0}{m} \rceil - 1)m > m_0 - m$ so that all of f_{t-k} , $k = 1, \dots, m_0$ are accounted for. We have $q > (\lceil \frac{q}{q_0} \rceil - 1)q_0$ so that both terms on the right-hand side of g_t has q elements. Also, $q - (\lceil \frac{q}{q_0} \rceil - 1)q_0 = q - \lfloor \frac{q}{q_0} \rfloor q_0 < q_0$ so that some elements of g_{t-k} consists of sum of two elements in (f_{t-k}) . We first consider the top $(\lceil \frac{q}{q_0} \rceil - 2)$ blocks of g_t , i.e., (f_{t-k}) , $k = 0, m, \dots, (\lceil \frac{q}{q_0} \rceil - 3)$. These elements do not appear as a sum and do not appear as elements in other (g_{t-k}) , $k = 1, \dots, m$. This implies that we may construct the left $(\lceil \frac{q}{q_0} \rceil - 2)$ block of Γ_k as $\begin{pmatrix} \lambda_k & \lambda_{k+m} & \dots & \lambda_{k+(\lceil \frac{q}{q_0} \rceil - 3)m} \end{pmatrix}$ for $k = 1, \dots, m$ so that the top $(\lceil \frac{q}{q_0} \rceil - 2)$ blocks of (g_{t-k}) accounts for $\sum_{k=1}^{(\lceil \frac{q}{q_0} \rceil - 2)m} \lambda_k f_{t-k}$. By constructing the top $(\lceil \frac{q}{q_0} \rceil - 2)$ block of g_t in this way, we see that it suffices to show that the alternative dynamic factor representation $(\lceil \frac{q_0 m_0}{m} \rceil, m)$ holds for (q_0, m_0) where $\lceil \frac{q}{q_0} \rceil = 2$. We first consider the case with $q - q_0 = m_0 - m$. Suppose $q - q_0 = m_0 - m$. We may construct g_t as

$$g_{t-k} = \begin{pmatrix} f_{t-k} \\ 0_{(q-q_0) \times 1} \end{pmatrix} + \begin{pmatrix} 0 \\ f_{t-(k+1)} \\ 0_{(q-q_0-1) \times 1} \end{pmatrix} + \dots + \begin{pmatrix} 0_{(q-q_0) \times 1} \\ f_{t-(m_0-m)} \end{pmatrix},$$

where we sum over $q - q_0$ number of (f_{t-k}) while performing cyclic permutation and adding lags. For example, suppose we want to represent a dynamic factor model with $(q_0, m_0) = (3, 5)$ as a model with $(q, m) = (5, 3)$. We can define g_{t-k} as

$$g_{t-k} = \begin{pmatrix} f_{t-k,1} \\ f_{t-k,2} \\ f_{t-k,3} \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ f_{t-(k+1),1} \\ f_{t-(k+1),2} \\ f_{t-(k+1),3} \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ f_{t-(k+2),1} \\ f_{t-(k+2),2} \\ f_{t-(k+2),3} \end{pmatrix}$$

Note that all of $(f_{t-k}), k = 1, \dots, m_0$ are included in $(g_{t-k}), k = 1, \dots, m$. Also, each elements in $(f_{t-k}), k = 1, \dots, m_0$ exists in (g_{t-k}) either alone, e.g., $f_{t-k,1}$ and $f_{t-(k+2),3}$, or as a sum over other elements in (f_{t-k}) . The key observation here is that each element in $(f_{t-k}), k = 1, \dots, m_0$ appears in (g_{t-k}) as a collection of groups whose number of unique elements match the number of groups in each collection. To illustrate, we have

$$(g_{t-1}, g_{t-2}, g_{t-3}) = \left(\begin{pmatrix} f_{t-1,1} \\ f_{t-1,2} + f_{t-2,1} \\ f_{t-1,3} + f_{t-2,2} + f_{t-3,1} \\ f_{t-2,3} + f_{t-3,2} \\ f_{t-3,3} \end{pmatrix}, \begin{pmatrix} f_{t-2,1} \\ f_{t-2,2} + f_{t-3,1} \\ f_{t-2,3} + f_{t-3,2} + f_{t-4,1} \\ f_{t-3,3} + f_{t-4,2} \\ f_{t-4,3} \end{pmatrix}, \begin{pmatrix} f_{t-3,1} \\ f_{t-3,2} + f_{t-4,1} \\ f_{t-3,3} + f_{t-4,2} + f_{t-5,1} \\ f_{t-4,3} + f_{t-5,2} \\ f_{t-5,3} \end{pmatrix} \right)$$

Here, the elements in (f_{t-k}) are formed into groups consisting of

$$\begin{aligned} & (f_{t-1,1}) \\ & ((f_{t-1,2} + f_{t-2,1}), f_{t-2,1}) \\ & ((f_{t-1,3} + f_{t-2,2} + f_{t-3,1}), (f_{t-2,2} + f_{t-3,1}), f_{t-3,1}) \\ & ((f_{t-2,3}, f_{t-3,2}), (f_{t-2,3} + f_{t-3,2} + f_{t-4,1}), (f_{t-3,2}, f_{t-4,1})) \\ & ((f_{t-3,3}), (f_{t-3,3} + f_{t-4,2}), (f_{t-3,3} + f_{t-4,2} + f_{t-5,1})) \\ & ((f_{t-4,3}), (f_{t-4,3} + f_{t-5,2})) \\ & (f_{t-5,3}), \end{aligned}$$

where within each collection, the number of unique elements of (f_{t-k}) and the number of groups are the same. This implies that we may solve for (Γ_k) such that $\sum_{k=1}^m \Gamma_k g_{t-k} = \sum_{k=1}^{m_0} \lambda_k f_{t-k}$. In this example, we obtain

$$\begin{aligned} \Gamma_1 &= (\lambda_{1,1} \quad \lambda_{1,2} \quad \lambda_{1,3} \quad \lambda_{3,2} - \lambda_{4,1} \quad \lambda_{3,3} - \lambda_{4,2}) \\ \Gamma_2 &= (\lambda_{2,1} - \lambda_{1,2} \quad \lambda_{2,2} - \lambda_{1,3} \quad \lambda_{4,1} - \lambda_{3,2} + \lambda_{2,3} \quad \lambda_{4,2} - \lambda_{5,1} \quad \lambda_{4,3} - \lambda_{5,2}) \\ \Gamma_3 &= (\lambda_{3,1} - \lambda_{2,2} \quad \lambda_{3,2} - \lambda_{2,3} \quad \lambda_{5,1} \quad \lambda_{5,2} \quad \lambda_{5,3}) \end{aligned}$$

This can be done for any (q_0, m_0) and (q, m) as long as $q - q_0 = m_0 - m$. Now, consider the general case where $q - q_0$ may not equal $m_0 - m$. For this general case, we may construct (g_{t-k}) as

$$\begin{aligned} g_{t-k} &= \begin{pmatrix} f_{t-k} \\ 0_{(q-q_0) \times 1} \end{pmatrix} + \begin{pmatrix} 0 \\ f_{t-k} \\ 0_{(q-q_0-1) \times 1} \end{pmatrix} + \dots + \begin{pmatrix} 0_{(q-q_0) \times 1} \\ f_{t-k} \end{pmatrix} \\ &+ \begin{pmatrix} f_{t-(k+1)} \\ 0_{(q-q_0) \times 1} \end{pmatrix} + \begin{pmatrix} 0 \\ f_{t-(k+1)} \\ 0_{(q-q_0-1) \times 1} \end{pmatrix} + \dots + \begin{pmatrix} 0_{(q-q_0) \times 1} \\ f_{t-(k+1)} \end{pmatrix} \\ &\vdots \\ &+ \begin{pmatrix} f_{t-(m_0-m)} \\ 0_{(q-q_0) \times 1} \end{pmatrix} + \begin{pmatrix} 0 \\ f_{t-(m_0-m)} \\ 0_{(q-q_0-1) \times 1} \end{pmatrix} + \dots + \begin{pmatrix} 0_{(q-q_0) \times 1} \\ f_{t-(m_0-m)} \end{pmatrix} \end{aligned}$$

With this choice of (g_{t-k}) , all the elements of (f_{t-k}) are formed into one collection, where the number of unique elements of (f_{t-k}) , i.e., $q_0 m_0$, is not greater than the number of groups,

i.e., qm . This implies that we may solve for (Γ_k) such that $\sum_{k=1}^m \Gamma_k g_{t-k} = \sum_{k=1}^{m_0} \lambda_k f_{t-k}$. The representation is not unique if we have $qm > q_0 m_0$.

The representations with the number of factors and filter length given by $(q, \lceil \frac{q_0 m_0}{q} \rceil)$ for $q > q_0$ can be shown in the same way as in the case with $(\lceil \frac{q_0 m_0}{m} \rceil, m)$ for $1 \leq m < m_0$. $\square$

Theorem 2. Suppose Assumptions 1, 2, 3, and 4 hold. Let $q_{\max}$ and $m_{\max}$ be integers satisfying $(q_{\max} + q_0 + 1)m_{\max} \leq k_{\max}$. Fix $m \geq 1$ and define

$$\hat{q} = \operatorname{argmax}_{1 \leq q \leq q_{\max}} DR_{q,m}(X),$$

where $DR_{q,m}(X) = \frac{\delta_{q,m}(X)}{\delta_{q+1,m}(X)}$. If $m_0 \leq m \leq m_{\max}$, then

$$\mathbb{P}\{\hat{q} = q_0\} \rightarrow 1 \quad \text{as } N, T \rightarrow \infty.$$

If instead $m < m_0$, then

$$\mathbb{P}\left\{\hat{q} = \left\lceil \frac{q_0 m_0}{m} \right\rceil\right\} \rightarrow 1,$$

where $\lceil \cdot \rceil$ denotes the ceiling function.

Proof. We first consider the case where $m_0 \leq m \leq m_{\max}$. For all $\epsilon > 0$, there exists $c_4 > 0$ such that

$$\mathbb{P}\left\{\frac{\frac{1}{\sqrt{NT}}\|X - \hat{X}_{q_0-1,m}\|}{\frac{1}{\sqrt{N+T}}\|X - \hat{X}_{q_0,m}\|} \leq c_4\right\} \leq \epsilon$$

for all $N + T, NT$ large with $\frac{N+T}{NT} \rightarrow 0$. To see this, note that

$$\mathbb{P}\left\{\frac{\frac{1}{\sqrt{NT}}\|\hat{X}_{q_0-1,m}\|}{\frac{1}{\sqrt{N+T}}\|\hat{X}_{q_0,m}\|} \leq \frac{c_2}{c_1}\right\} \leq \mathbb{P}\left\{\frac{1}{\sqrt{NT}}\|\hat{X}_{q_0-1,m}\| \leq c_2\right\} + \mathbb{P}\left\{\frac{1}{\sqrt{N+T}}\|\hat{X}_{q_0,m}\| \geq c_1\right\}$$

because

$$\left\{\frac{1}{\sqrt{NT}}\|\hat{X}_{q_0-1,m}\| > c_2\right\} \cap \left\{\frac{1}{\sqrt{N+T}}\|\hat{X}_{q_0,m}\| < c_1\right\} \subset \left\{\frac{\frac{1}{\sqrt{NT}}\|\hat{X}_{q_0-1,m}\|}{\frac{1}{\sqrt{N+T}}\|\hat{X}_{q_0,m}\|} > \frac{c_2}{c_1}\right\}$$

We use Lemma 5 and Lemma 6 with $\epsilon/2$ and take $c_4 = \frac{c_2}{c_1}$ to finish. This implies $\frac{\frac{1}{\sqrt{NT}}\|X - \hat{X}_{q_0-1,m}\|}{\frac{1}{\sqrt{N+T}}\|X - \hat{X}_{q_0,m}\|}$ is strictly positive with probability approaching one as $N, T \rightarrow \infty$. Note that

$$\begin{aligned} DR_{q_0,m}(X) &= \frac{\|X - \hat{X}_{q_0-1,m}\|}{\|X - \hat{X}_{q_0,m}\|} \\ &= \left(\frac{\sqrt{NT}}{\sqrt{N+T}}\right) \frac{\frac{1}{\sqrt{NT}}\|X - \hat{X}_{q_0-1,m}\|}{\frac{1}{\sqrt{N+T}}\|X - \hat{X}_{q_0,m}\|} \end{aligned}$$

Thus, for all $\epsilon > 0$, there exists $c_4 > 0$ such that

$$\mathbb{P} \left\{ DR_{q_0, m}(X) > \left(\frac{\sqrt{NT}}{\sqrt{N+T}} \right) c_4 \right\} \geq 1 - \epsilon$$

for all N, T large. This implies that $DR_{q_0, m}(X)$ is stochastically unbounded as $N, T \rightarrow \infty$. Also, using Lemma 5, for all $q = q_0 + 1, \dots, q_{\max}$ and for all $\epsilon > 0$, there exists c_0 and c_1 such that

$$\mathbb{P} \left\{ DR_{q, m}(X) \geq \frac{c_1}{c_0} \right\} \leq \epsilon$$

for all $N + T$ large. To see this, note that

$$\left\{ \frac{1}{\sqrt{N+T}} \|X - \hat{X}_{q-1, m}\| < c_1 \right\} \cap \left\{ \frac{1}{\sqrt{N+T}} \|X - \hat{X}_{q, m}\| > c_0 \right\} \subset \left\{ \frac{\frac{1}{\sqrt{N+T}} \|X - \hat{X}_{q-1, m}\|}{\frac{1}{\sqrt{N+T}} \|X - \hat{X}_{q, m}\|} < \frac{c_1}{c_0} \right\}$$

This implies $DR_{q, m}(X)$ is stochastically bounded for all $q = q_0 + 1, 2, \dots, q_{\max}$ as $N + T \rightarrow \infty$. Similarly, using Lemma 6, we may show that $DR_{q, m}(X)$ is stochastically bounded for all $q = 1, \dots, q_0 - 1$ as $N, T \rightarrow \infty$. This implies that

$$\mathbb{P} \{ \hat{q} = q_0 \} \rightarrow 1$$

as $N, T \rightarrow \infty$.

Now we consider the case $m < m_0$. Using Lemma 5 and Lemma 6, $\frac{1}{\sqrt{N+T}} \|X - \hat{X}_{q, m}\|$ is asymptotically strictly positive and bounded for $q = \lceil \frac{q_0 m_0}{m} \rceil, \dots, q_{\max}$ and $\frac{1}{\sqrt{NT}} \|X - \hat{X}_{q, m}\|$ is asymptotically strictly positive and bounded for $q = 1, \dots, \lceil \frac{q_0 m_0}{m} \rceil - 1$. These implies that the ratio $DR_{q, m}(X) = \frac{\|X - \hat{X}_{q-1, m}\|}{\|X - \hat{X}_{q, m}\|}$ is stochastically bounded for all $q = 1, \dots, \lceil \frac{q_0 m_0}{m} \rceil - 1$ and $q = \lceil \frac{q_0 m_0}{m} \rceil + 1, \dots, q_{\max}$. The ratio $DR_{q, m}(X) = \frac{\|X - \hat{X}_{q-1, m}\|}{\|X - \hat{X}_{q, m}\|}$ diverges with large probability only at $q = \lceil \frac{q_0 m_0}{m} \rceil$. $\square$

Theorem 3. Suppose Assumptions 1, 2, 3, and 4 hold. Let $q_{\max}$ and $m_{\max}$ satisfy $(q_{\max} + q_0 + 1)m_{\max} \leq k_{\max}$. Fix q and define

$$\hat{m} = \operatorname{argmax}_{1 \leq m \leq m_{\max}} DR_{q, m}(X),$$

where $DR_{q, m}(X) = \frac{\delta_{q+1, m-1}(X)}{\delta_{q+1, m}(X)}$. If $q = q_0 \geq 1$, then

$$\mathbb{P} \{ \hat{m} = m_0 \} \rightarrow 1$$

as $N, T \rightarrow \infty$. If $q > q_0$, then

$$\mathbb{P} \left\{ \hat{m} = \left\lceil \frac{q_0 m_0}{q} \right\rceil \right\} \rightarrow 1,$$

where $\lceil \cdot \rceil$ denotes the ceiling operator.

Proof. Note that

$$DR_{q,m}(X) = \frac{\|X - \hat{X}_{q,m-1}\|}{\|X - \hat{X}_{q,m}\|}$$

Suppose $q = q_0$. Using Lemma 5, $\frac{1}{\sqrt{N+T}}\|\hat{X}_{q_0,m}\|$ is asymptotically strictly positive and bounded for all $m_0 \leq m \leq m_{max}$. It suffices to show that $\frac{1}{\sqrt{NT}}\|X - \hat{X}_{q_0,m}\|$ is strictly positive for $m \leq m_0 - 1$. This is true using fourth case of Lemma 6.

Now suppose $q > q_0$. Note that $\frac{1}{\sqrt{N+T}}\|X - \hat{X}_{q,m}\|$ is asymptotically strictly positive and bounded for $m = \left\lceil \frac{q_0 m_0}{q} \right\rceil, \dots, m_{max}$ due to Lemma 5, third case. Also, $\frac{1}{\sqrt{NT}}\|X - \hat{X}_{q,m}\|$ is asymptotically strictly positive and bounded for $m = 1, \dots, \left\lceil \frac{q_0 m_0}{q} \right\rceil - 1$ due to Lemma 6 fourth case. $\square$

Theorem 4. Suppose Assumptions 1, 2, 3, and 4 hold. Let $\hat{q}$ and $\hat{m}$ be given by

$$\hat{q}, \hat{m} = \underset{0 \leq q \leq q_{max}, 0 \leq m \leq m_{max}}{\operatorname{argmin}} PC(q, m)$$

or

$$\hat{q}, \hat{m} = \underset{0 \leq q \leq q_{max}, 0 \leq m \leq m_{max}}{\operatorname{argmin}} DC(q, m).$$

If $g(N, T) \rightarrow 0$ and $\left(\frac{N+T}{NT}\right)g(N, T) \rightarrow \infty$ as $N, T \rightarrow \infty$, we have

$$P(\hat{q} = q_0) \rightarrow 1, \quad \text{and} \quad P(\hat{m} = m_0) \rightarrow 1$$

as $N, T \rightarrow \infty$.

Proof. We show that

$$\mathbb{P}\{PC(q, m) < PC(q_0, m_0)\} \rightarrow 0$$

as $N, T \rightarrow \infty$ for all $q \leq q_{max}$ and $m \leq m_{max}$ such that both $q \neq q_0$ and $m \neq m_0$. Using Lemma 7, for all $\epsilon > 0$, there exists $c > 0$ such that

$$\mathbb{P}\{V_{q,m}(X) - V_{q_0,m_0}(X) \geq c\} \geq 1 - \epsilon$$

for all N, T large with $\frac{N+T}{NT} \rightarrow 0$ for the four cases

$$\begin{cases} q = 0, \dots, \left\lceil \frac{q_0 m_0}{m} \right\rceil - 1 & \text{if } 1 \leq m < m_0 \\ q = 0, \dots, q_0 - 1 & \text{if } m_0 \leq m \leq m_{max} \\ m = 1, \dots, m_{max} & \text{if } 0 \leq q < q_0 \\ m = 1, \dots, \left\lceil \frac{q_0 m_0}{q} \right\rceil - 1 & \text{if } 1 \leq q_0 \leq q \leq q_{max} \end{cases} \quad (20)$$

Note that we have

$$PC(q, m) - PC(q_0, m_0) = V_{q,m}(X) - V_{q_0,m_0}(X) - (q_0(m_0 + 1) - q(m + 1))g(N, T),$$

where the first term $V_{q,m}(X) - V_{q_0,m_0}(X)$ is strictly positive in the limit, whereas the second term $(q_0(m_0 + 1) - q(m + 1))g(N, T)$ vanishes. Thus, we have $\mathbb{P}\{PC(q, m) < PC(q_0, m_0)\} \rightarrow 0$ for the four cases in 20.

Also, using Lemma 7, we have

$$V_{q_0, m_0}(X) - V_{q, m}(X) = O_p\left(\frac{N+T}{NT}\right)$$

for the three cases

$$\begin{cases} q = \lceil \frac{q_0 m_0}{m} \rceil, \dots, q_{max} & \text{if } 1 \leq m < m_0 \\ q = q_0, \dots, q_{max} & \text{if } m_0 \leq m \leq m_{max} \\ m = \lceil \frac{q_0 m_0}{q} \rceil, \dots, m_{max} & \text{if } q_0 \leq q \leq q_{max} \end{cases} \quad (21)$$

Note that we may write

$$\mathbb{P}\{PC(q, m) < PC(q_0, m_0)\} = \mathbb{P}\{V_{q_0, m_0}(X) - V_{q, m}(X) > (q(m+1) - q_0(m_0+1))g(N, T)\}.$$

For all three cases in 21 such that either $q \neq q_0$ or $m \neq m_0$, we have

$$(q(m+1) - q_0(m_0+1))g(N, T) \geq g(N, T),$$

which converges to zero at a slower rate than $O_p\left(\frac{N+T}{NT}\right)$. Thus, for these three cases in 21 such that either $q \neq q_0$ or $m \neq m_0$, we have $\mathbb{P}\{PC(q, m) < PC(q_0, m_0)\} \rightarrow 0$ as $N, T \rightarrow \infty$. The prove the result for $PC(q, m)$.

For $DC(q, m)$, note that we have

$$\frac{1}{NT} \hat{\delta}_{q+1, m}^2(X) = O_p\left(\frac{N+T}{NT}\right)$$

for the three cases in 21 due to Lemma 5 and for all $\epsilon > 0$, there exists $c > 0$ such that

$$\mathbb{P}\left\{\frac{1}{\sqrt{NT}}\|X - \hat{X}_{q, m}\| \geq c\right\} \geq 1 - \epsilon$$

for the four cases in 20 due to Lemma 6. We use the same argument as in the case of $PC(q, m)$ to finish. $\square$